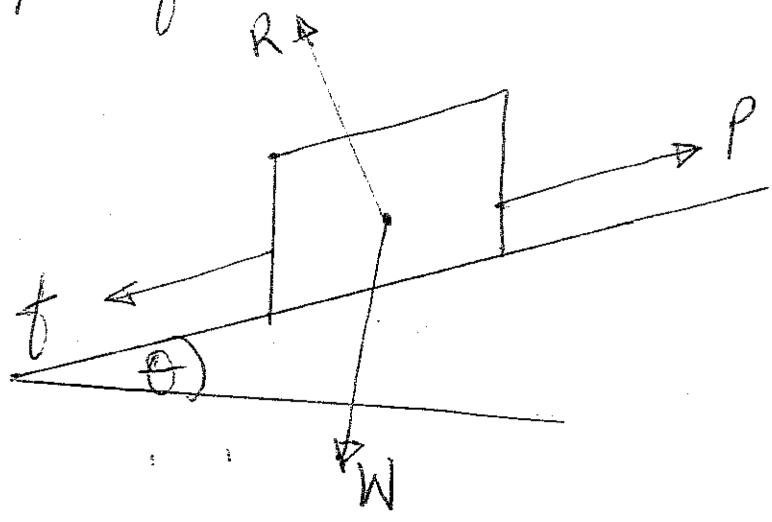
MEC 315 **CONCEPT OF FRICTION**

When one body slides on another, a certain degree of interlocking of minutely projecting particles takes place. This interlocking properties of projecting particles oppose any tendency of the body to move. The resisting force acts in the direction opposite to that of the motion of the upper body and is called friction. Frictional force may be defined as the opposing force which is called into play in between the surfaces of contact of two bodies, when one body moves over the surface of another body. It can also be defined as the opposing force which acts in the opposite direction of the movement of the block.

FACTORS WHICH INFLUENCE FRICTION

The factors which influence friction are
i. The frictional force (f)
ii. Normal reaction (R)
iii. The weight of the body (W)
iv. The angle of inclination (θ) and
v. The applied force (P).

diag.



- NORMAL REACTION (R): This is an equal and opposite reacting force acting vertically upwards.
- WEIGHT OF THE BODY (W): It is the product of mass and acceleration due to gravity. $W = mg$.
- ANGLE OF FRICTION [θ]: This is the angle which the normal reaction makes with vertical.
- APPLIED FORCE [P]: This is the horizontal force tangential to the surfaces which acted on the body.

The ratio of friction force to normal force is termed the coefficient of friction and is denoted by μ. It is constant for any given pair of surfaces and is independent of the area of contact, the sliding velocity and the intensity of pressure.

The force required to start a body moving is slightly more than that necessary to keep the body moving at uniform speed, which leads to concepts of the coefficient of static and kinetic frictions.

- STATIC FRICTION: It is the friction experienced by a body when it is at rest or when the body tends to move or friction offered by the surfaces subjected to external forces until there is no motion between them. It is always higher than the friction produced once the object has started to move e.g vehicles or pistons in the engine cylinders. Considerable force is required to move them, but once they move the resistance is less.

- KINETIC OR DYNAMIC FRICTION: It is a friction experienced by a body when it is in motion or it is the friction which exists after movement has started e.g when vehicles or pistons are moving.

- DRY FRICTION :- Is the friction that exist between two unlubricated surfaces. Or in the complet absence lubricant, the friction, rate of wear and heat genera is high. This type of friction is used for brakes and clutches. It is of two types:

i, Sliding friction and

ii, Rolling friction.

- SLIDING FRICTION: Is the friction that exists when one surface slides over the other.

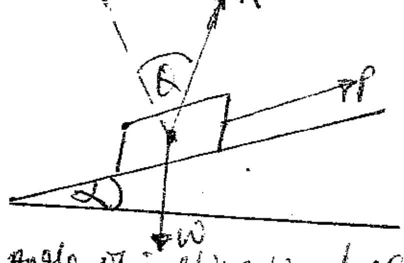
- ROLLING FRICTION: It is the friction experienced by a body when it rolls over the other or when balls or rollers are interposed between the two ~~sources~~. surfaces.

- FLUID FRICTION :- This is the complete seperation of surfaces by a film of lubricant. It is the ideal condition and reduces friction, rate of wear and heat generated to a minimum.

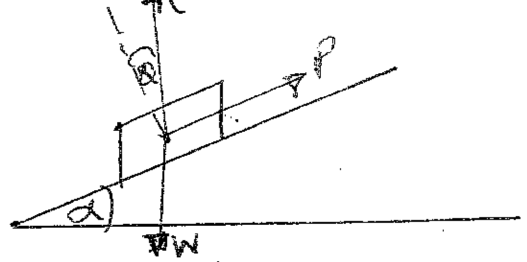
- BOUNDARY FRICTION :- The film of lubricant is not comp and the surfaces are only partially seperated, the friction rate of wear and heat generated is less than in the condition of dry friction. This type of friction exists on pistons, rings and valve stems.

EQUILIBRIUM OF A BODY LYING ON A ROUGH INCLINED PLANE

diag



(a) Angle of inclination less than angle of friction



(b) Angle of inclination greater than angle of friction.

(4)

Consider a body of weight W, lying on a ~~rough~~ rough plane inclined at an angle α with the horizontal as shown above. In "a" above, the body will be automatically be in equilibrium, therefore it requires a corresponding force to move the body upwards or downwards. In "b" the body move down and an upward force P will be required to resist the body from moving down the plane. The following forces are very important in this topic.

i, Force acting along the inclined plane.
ii, Force acting horizontally and
iii, Force acting at some angle with the inclined plane.

EQUILIBRIUM OF A BODY ON A ROUGH INCLINED PLANE SUBJECTED TO A FORCE ACTING ALONG THE INCLINED PLANE

Consider a body lying on a rough inclined plane subjected to a force acting along the inclined plane, which keeps it in equilibrium. let

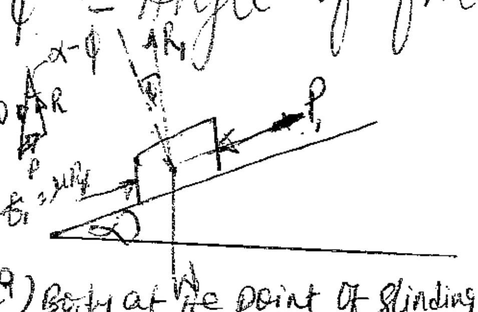
W = weight of the body

α = Angle which the inclined plane makes with the horizontal.

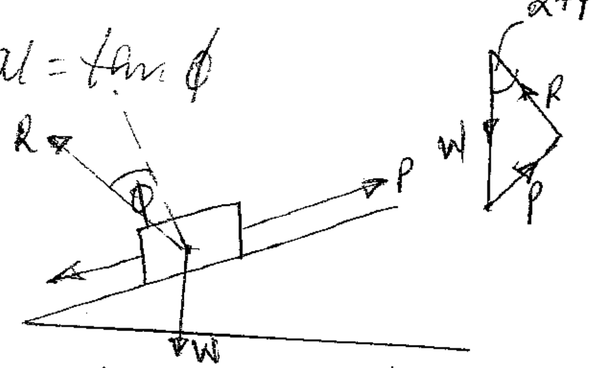
R = Normal reaction

μ = Co-efficient of friction between the body and the inclined plane.

ϕ = Angle of friction such that $\mu = \tan\phi$



(a) Body at the point of sliding downwards



(b) Body at the point of sliding upwards.

In "a" above, minimum force P_1 will keep the body in equilibrium when it is at the point of sliding downwards. In this case the force of friction ($F_1 = \mu R_1$) will act upwards, as the body is at the point of sliding downwards.

Now resolving the forces along the plane

$$P_1 = W\sin\alpha - \mu R_1 \quad \text{------------ (1)}$$

And now resolving the forces perpendicular to the plane

$$R_1 = W\cos\alpha \quad \text{------------ (2)}$$

Substituting the value of R_1 in eqn (1)

$$P_1 = W\sin\alpha - \mu W\cos\alpha$$
$$= W[\sin\alpha - \mu\cos\alpha]$$

Substitute the value of $\mu = \tan\phi$

$$P_1 = W[\sin\alpha - \tan\phi\cos\alpha]$$

multiplying both sides by $\cos\phi$

$$P_1\cos\phi = W[\sin\alpha\cos\phi - \sin\phi\cos\alpha]$$
$$= \cancel{W[} W\sin[\alpha - \phi]$$

$$\therefore P_1 = W \times \frac{\sin[\alpha - \phi]}{\cos\phi}$$

In "b" above, minimum force P_2 which will keep the body in equilibrium when it is at the point of sliding upwards. In this case, the force of friction ($F_2 = \mu R_2$) will act downwards as the body is at the point of sliding upwards.

Now resolving the forces along the plane

$$P_2 = W\sin\alpha + \mu R_2 \quad \text{------------} \quad (1)$$

Resolving the forces perpendicular to the plane

$$R_2 = W\cos\alpha \quad \text{------------} \quad (2)$$

Substituting the value of R_2 in eqn (1)

$$P_2 = W\sin\alpha + \mu W\cos\alpha$$
$$= W[\sin\alpha + \mu\cos\alpha]$$

Substituting the value of $\mu = \tan\phi$

$$P_2 = W(\sin\alpha + \tan\phi\cos\alpha]$$

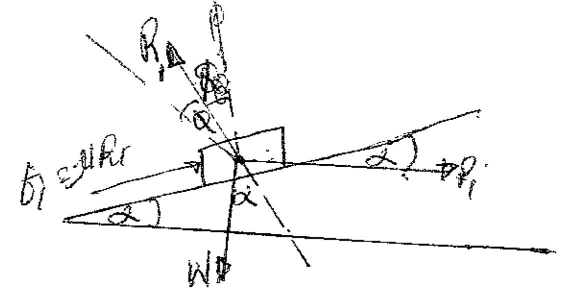
Multiplying both sides by $\cos\phi$

$$P_2\cos\phi = W(\sin\alpha\cos\phi + \sin\phi\cos\alpha]$$
$$= W\sin[\alpha + \phi]$$
$$\therefore P_2 = W\frac{\sin[\alpha+\phi]}{\cos\phi}$$

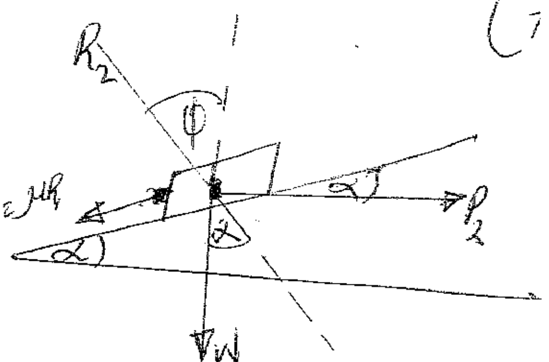
EQUILIBRIUM OF A BODY ON A ROUGH INCLINED PLANE SUBJECTED TO A FORCE ACTING HORIZONTALLY

Consider a body lying on a rough inclined plane subjected to a force acting horizontally, which keep it in an equilibrium as shown in "a" and "b" below.

Fig.



(a) Body at a point of Sliding downwards.



(b) Body at the point of sliding upwards.

Let

W = Weight of the body

α = Angle which the inclined plane makes with the horizontal

R = Normal reaction

μ = co-efficient of friction between the body and inclined plane

ϕ = Angle of friction such that $\mu = \tan\phi = \frac{\sin\phi}{\cos\phi}$

In "a" to find the minimum force (P_1) which will keep the body in equilibrium, when it is at the point of sliding downwards i.e the force of friction $F_1 = \mu R_1$ will act upwards as the body is at the point of sliding downwards.

Resolving forces along the plane

$$P_1 \cos\alpha = W\sin\alpha - \mu R_1 \quad \text{------------- (1)}$$

Resolving forces perpendicular to the plane

$$R_1 = W\cos\alpha + P_1 \sin\alpha \quad \text{------------- (2)}$$

Substituting the value of R_1 in eqn (1)

$$P_1 \cos\alpha = W\sin\alpha - \mu(W\cos\alpha + P_1\sin\alpha)$$

$$= W\sin\alpha - \mu W\cos\alpha - \mu P_1 \sin\alpha$$

$$P_1\cos\alpha + \mu P_1 \sin\alpha = W\sin\alpha - \mu W\cos\alpha$$

$$P_1(\cos\alpha + \mu\sin\alpha) = W(\sin\alpha - \mu\cos\alpha)$$

$$P_1 = W[\sin\alpha - \mu\cos\alpha] / \cos\alpha + \mu\sin\alpha$$

Now, Substituting the value of $\mu = \tan\phi$

$$P_1 = \frac{W[\sin\alpha - \tan\phi\cos\alpha]}{\cos\alpha + \tan\phi\sin\alpha}$$

Multiply by $\frac{\cos\phi}{\cos\phi}$

$$P_1 = W \times \frac{\sin\alpha\cos\phi - \sin\phi\cos\alpha}{\cos\alpha\cos\phi + \sin\alpha\sin\phi}$$

$$= W \times \frac{\sin(\alpha - \phi)}{\cos(\alpha - \phi)}$$

$= W\tan(\alpha - \phi)$ — when $\alpha > \phi$

$= W\tan(\phi - \alpha)$ — When $\phi > \alpha$

In "b" to find the minimum force P_2 which will keep the body in equilibrium, when it is at the point of sliding upwards. The force of friction will act downwards ($F_2 = \mu R_2$) as the body is at the point of sliding upwards.

Resolving the forces at right angles to the plane (horizontally)

$P_2\cos\alpha = W\sin\alpha + \mu R_2$ ---------- (1)

Resolving forces perpendicular to the plane

$R_2 = W\cos\alpha + P_2\sin\alpha$ ---------- (2)

Substituting the value of R_2 in eqn(1)

$P_2\cos\alpha = W\sin\alpha + \mu(W\cos\alpha + P_2\sin\alpha)$

$= W\sin\alpha + \mu W\cos\alpha + \mu P_2\sin\alpha$

$P_2\cos\alpha - \mu P_2\sin\alpha = W\sin\alpha + \mu W\cos\alpha$

P.T.

$$P_2(\cos\alpha - \mu\sin\alpha) = W(\sin\alpha + \mu\cos\alpha)$$

$$P_2 = \frac{W(\sin\alpha + \mu\cos\alpha)}{\cos\alpha - \mu\sin\alpha}$$

Substitute for $\mu = \tan\phi$

$$P_2 = \frac{W(\sin\alpha + \tan\phi\cos\alpha)}{\cos\alpha - \tan\phi\sin\alpha}$$

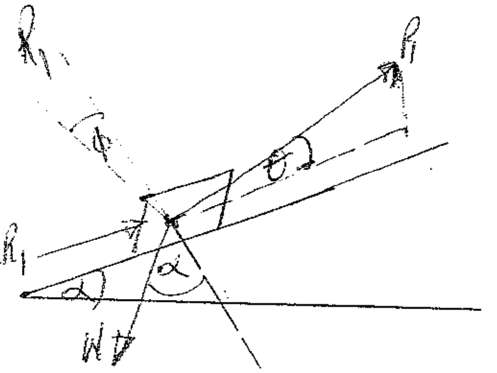
multiply both sides by $\frac{\cos\phi}{\cos\phi}$

$$P_2 = W\frac{\sin\alpha\cos\phi + \sin\phi\cos\alpha}{\cos\alpha\cos\phi - \sin\phi\sin\alpha}$$

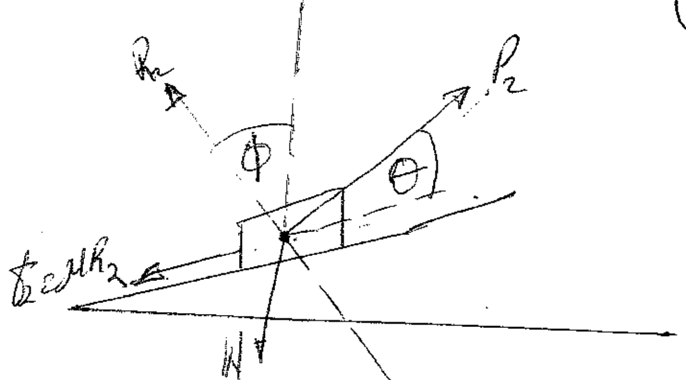
$$= W \times \frac{\sin(\alpha+\phi)}{\cos(\alpha+\phi)} = W\tan(\alpha+\phi)$$

EQUILIBRIUM OF A BODY ON A ROUGH INCLINED PLANE SUBJECTED TO A FORCE ACTING AT SOME ANGLE WITH THE INCLINED PLANE

Consider a body lying on a rough inclined plane subjected to a force acting at some angle with the inclined plane, which keeps it in equilibrium as shown below.



(a) body at the point of sliding downwards.



(b) Body at the point of sliding upwards

To find the minimum force P_1 which will keep the body in equilibrium when it is at the point of sliding downwards. The force of friction $f_1 = \mu R_1$ will act upwards as the body is at the point of sliding downwards.

Resolving forces along the plane

$$P_1 \cos\theta = W\sin\alpha - \mu R_1 \quad \text{------------ (1)}$$

Resolving forces perpendicular to the plane.

$$R_1 = W\cos\alpha - P_1 \sin\theta$$

Substitute for R_1 above in Eqn (1)

$$P_1 \cos\theta = W\sin\alpha - \mu\left[W\cos\alpha - P_1\sin\theta\right]$$

$$= W\sin\alpha - \mu W\cos\alpha + \mu P_1 \sin\theta$$

$$P_1\cos\theta - \mu P_1 \sin\theta = W\sin\alpha - \mu W\cos\alpha$$

$$P_1\left[\cos\theta - \mu\sin\theta\right] = W\sin\alpha - \mu W\cos\alpha$$

$$P_1 = \frac{\cancel{W\sin\alpha}\, W\left[\sin\alpha - \mu\cos\alpha\right]}{\cos\theta - \mu\sin\theta}$$

Substitute for the value of $\mu = \tan\phi = \frac{\sin\phi}{\cos\phi}$

$$P_1 = \frac{W(\sin\alpha - \tan\phi\cos\alpha)}{\cos\theta - \tan\phi\sin\theta}$$

Multiply both side by $\frac{\cos\phi}{\cos\phi}$

$$P_1 = W \frac{(\sin\alpha\cos\phi - \sin\phi\cos\alpha)}{\cos\theta\cos\phi - \sin\phi\sin\theta}$$

$$\therefore P_1 = W \times \frac{\sin(\alpha - \phi]}{\cos(\theta + \phi)}$$

To find the maximum force P_2 which will keep the body in equilibrium when it is at a point of sliding upwards. The force of friction $f_2 = \mu R_1$ will act downwards as the body is at the point of sliding upw

Resolving the forces along the plane

$$P_2\cos\theta = W\sin\alpha + \mu R_2 \quad \text{------ (1)}$$

Resolving forces perpendicular to the plane

$$R_2 = W\cos\alpha \mp P_2\sin\theta \quad \text{------ (2)}$$

Substitute R_2 in eqn (1)

$$P_2\cos\theta = W\sin\alpha + \mu[W\cos\alpha - P_2\sin\theta]$$

$$= W\sin\alpha + \mu W\cos\alpha - \mu P_2\sin\theta$$

$$P_2\cos\theta + \mu P_2\sin\theta = W\sin\alpha + \mu W\cos\alpha$$

$$P_2[\cos\theta + \mu\sin\theta] = W\sin\alpha + \mu W\cos\alpha$$

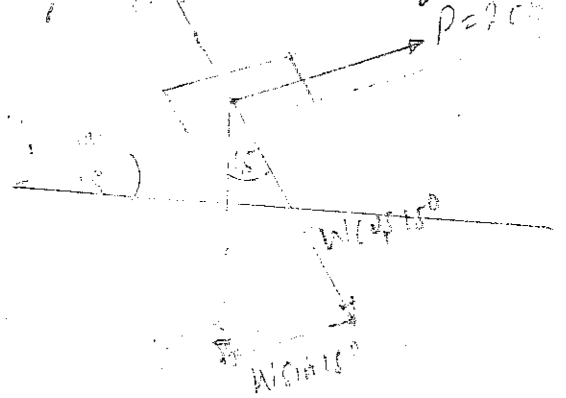
$$P_2 = W \times \frac{\sin\alpha + \mu\cos\alpha}{\cos\theta + \mu\sin\theta}$$ Substitute for $\mu = \tan\phi$

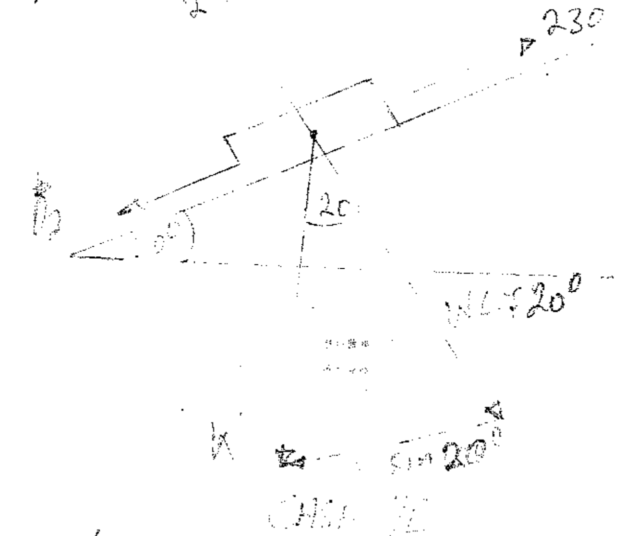
$$P_2 = W \times \frac{\sin\alpha + \tan\phi\cos\alpha}{\cos\theta + \tan\phi\sin\theta}$$ multiply both sides by $\frac{\cos\phi}{\cos\phi}$

$$P_2 = W \times \frac{\sin\alpha\cos\phi + \sin\phi\cos\alpha}{\cos\theta\cos\phi + \sin\phi\sin\theta} = W \times \frac{\sin(\alpha + \phi)}{\cos(\theta - \phi)}$$

Example I

An effort of 200N is required just to move a certain body up an inclined plane of angle 15° the force acting parallel to the plane. If the angle of inclination of the plane is made 20°, the effort required again applied parallel to the plane is found to be 230N. Find the weight of the body and the co-efficient of friction.





Case I. Resolving forces along the plane

$200 = F_1 + W\sin 15^\circ = \mu R_1 + W\sin 15^\circ$

Resolving forces perpendicular to the plane.

$R_1 = W\cos 15^\circ$ Substitute R_1 in above.

$200 = \mu[W\cos 15^\circ] + W\sin 15^\circ$

$= \mu W\cos 15^\circ + W\sin 15^\circ$

$= W[\mu\cos 15^\circ + \sin 15^\circ]$ - - - - - - - - (1)

Case II. Resolving forces along the plane

$230 = F_2 + W\sin 20^\circ = \mu R_2 + W\sin 20^\circ$

Resolving forces perpendicular to the plane

$R_2 = W\cos 20^\circ$, Substitute R_2 in above

$230 = \mu W\cos 20^\circ + W\sin 20^\circ$

$= W[\mu\cos 20^\circ + \sin 20^\circ]$ - - - - - - - - (2)

Dividing equation (2) by (2)

$$\frac{280}{230} = \frac{W(\mu \cos 25^\circ + \sin 25^\circ)}{W(\mu \cos 20^\circ + \sin 20^\circ)}$$

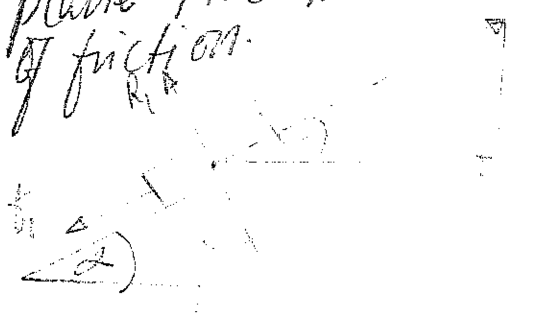
$\mu = 0.259$, Substitute μ in eqn (1)

$$200 = W(0.259 \cos 15^\circ + \sin 15^\circ)$$

$$W = \frac{200}{1.509} = 392.9N.$$

Example II

A load of 1.5 KN resting on an inclined rough plane can be move up the plane by a force of 2 kN applied horizontally or by a force of 1.25 kN applied parallel to the plane. Find the inclination of the plane and the co-efficient of friction.



Remember that when the force is applied horizontally, the magnitude of the force which can move the load up the plane is

$$P = W \tan[\alpha + \phi]$$

$$2 = 1.5 \tan(\alpha + \phi]$$

$$\therefore \tan(\alpha + \phi] = \frac{2}{1.5} = 1.333$$

$$\therefore \alpha + \phi = 53^\circ 8'$$

But, when the force is applied parallel to the plane, the magnitude of the force which can move up the plane

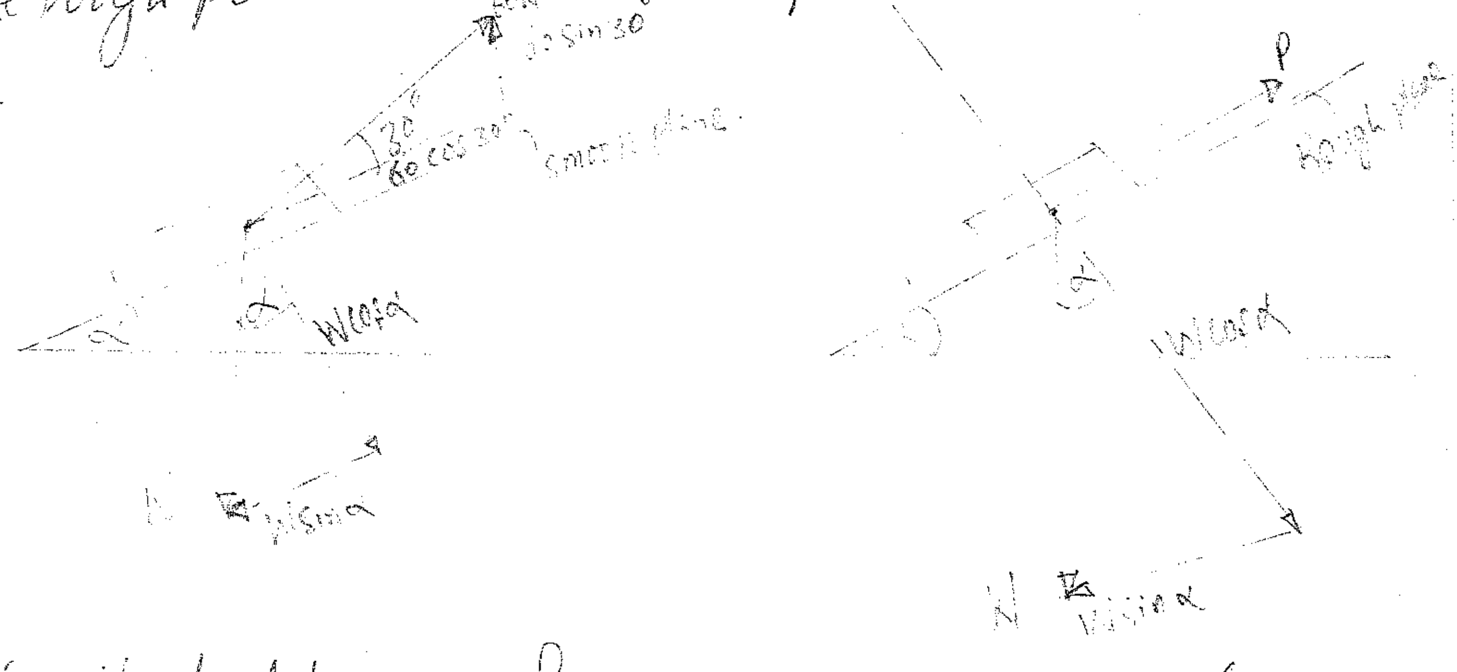
$$P = W \times \frac{\sin(\alpha + \phi)}{\cos\phi}$$

$$1.25 = 1.5 \times \frac{\sin 53^\circ 8'}{\cos\phi} \quad \therefore \phi = 16^\circ 16'$$

$$\therefore \alpha = 53^\circ 8' - 16^\circ 16' = 36^\circ 52'.$$

Example III

Find the force required to move a load of 300N up a rough plane, the force being applied parallel to plane. The inclination of the plane is such that, when the same load is kept on a perfectly smooth plane inclined at the same angle, a force of 60N applied at an inclination of 30° to the plane, keeps the same load in equilibrium. Assume the co-efficient of friction between the rough plane and the load is equal to 0.3.



Consider load lying on smooth plane, $\mu = 0$, $\phi = 0$ (because of smooth plane surface)

Remember, the force (P) required when the load is at the point of sliding downward

$$60 = W \times \frac{\sin(\alpha + \phi)}{\cos(\theta - \phi)} = 300 \times \frac{\sin\alpha}{\cos 30^\circ} \qquad \phi = 0$$

$$= 300 \times \frac{\sin\alpha}{0.866}$$

$$60 = \frac{300 \sin\alpha}{0.866}$$

$$\sin\alpha = \frac{60 \times 0.866}{300} = 0.1732$$

$$\therefore \alpha = 9^\circ 58' \text{ or } 9.97^\circ$$

Consider the load lying on the rough plane inclined at an angle of $9^\circ 58'$ with the horizontal.

Given $\mu = 0.3 = \tan\phi$

$\therefore \phi = 16^\circ 42'$ or 16.7°

Remember the force required to move the load up the plane

$$P = W \times \frac{\sin(\alpha+\phi)}{\cos\phi} = 300 \times \frac{\sin(9^\circ 58' + 16^\circ 42')}{\cos 16^\circ 42'}$$

$$= 300 \times \frac{\sin 26^\circ 40'}{\cos 16^\circ 42} = 300 \times \frac{0.4488}{0.9578} = 140.6N$$

OR

Resolving forces parallel to the plane

$W \sin\alpha = 60 \cos 30^\circ$

$300 \sin\alpha = 60 \cos 30^\circ$

$\sin\alpha = \frac{60 \cos 30^\circ}{300} = 0.1732 \therefore \alpha \doteq 9.97^\circ$ or $9^\circ 58'$

Now, when the force is applied parallel to the rough plane

Resolving forces parallel to the plane,

$P = F + W \sin\alpha$

$P = \mu R + W \sin\alpha$ - - - - - - - - - - - (i)

Resolving perpendicular to the plane

$R = W \cos\alpha$ - - - - - - - - - - - (ii)

Substitute the value of R in eqn. (i)

$P = \mu W \cos\alpha + W \sin\alpha$

$= W[\mu \cos\alpha + \sin\alpha]$

$$P = W\left(\frac{\sin\phi}{\cos\phi} \times \cos\alpha + \sin\alpha\right] \qquad \mu = \tan\phi = \frac{\sin\phi}{\cos\phi}$$

$$= \frac{W \sin(\alpha + \phi)}{\cos\phi}$$

Given

$W = 300N$

$\mu = 0.3$

$\tan\phi = 0.3$

$\phi = 16.7^\circ$

$$P = \frac{300 \sin(9.97^\circ + 16.7^\circ)}{\cos 16.7^\circ} = \frac{300 \times 0.5023}{0.9578} = 15.73N$$

THE SCREW FRICTION

The Screw threads are mainly V-threads and Square thread. The V-threads are stronger and offer more frictional resistance to motion than Square threads. Square threads are used for lifting heavy loads. The following terms are important

i, HELIX: It is the curve traced by a particle while describing a circular path at ~~an~~ a uniform speed and advancing in the axial direction at a uniform rate.

ii, PITCH: It is the distance from a point of a screw to a corresponding point on the next thread. It is measured parallel to the axis of the screw.

iii, LEAD: It is the distance through which a screw thread advances axially in one turn.

(iv) DEPTH OF THREAD: It is the distance between the top and bottom surfaces of a thread, also known as the crest and root of thread.

V, SINGLE-THREADED SCREW: If the lead of a screw is equal to its pitch, it is known as single threaded screw

Vi, MULTI-THREADED SCREW: If more than one threads are cut in one lead distance of a screw.

Lead = pitch × number of threads.

(vii) SLOPE OF THREAD :- It is the inclination of the thread with horizontal.

$$\tan\alpha = \frac{\text{Lead of Screw}}{\text{Circumference of Screw}}$$

$$\tan\alpha = \frac{p}{\pi d} \quad \text{(in a Single-threaded Screw)}$$

$$\text{"} \quad = \frac{np}{\pi d} \quad \text{(in a Multi-threaded Screw)}$$

Example I

The Screw of a Jack is square threaded with two threads to a Centimeter, the outer diameter of the Screw is 5cm. If the co-efficient of friction is 0.1. Calculate the force required to be applied at the end of the lever, which is 70cm long. (a) to lift a load of 4kN and (b) to lower it.

Given

$\Delta = 5cm$, $\mu = 0.1$, $L = 70cm$, $n = 2$ (two threaded)

$\therefore p = \frac{1}{2} = 0.5cm$

Internal diameter of the Screw $= 5 - (2 \times 0.5) = 4cm$

Mean diameter of the Screw $= d = \frac{5+4}{2} = 4.5cm$

$$\tan\alpha = \frac{p}{\pi d} = \frac{0.5}{\pi \times 4.5} = 0.0335$$

(a) Force to lift the load P_1

$$P = W\tan(\alpha + \phi) = W \times \frac{\tan\alpha + \tan\phi}{1 - \tan\alpha\tan\phi}$$

$$= 4000 \times \frac{0.0335 + 0.1}{1 - 0.0335 \times 0.1} = 535.8N$$

$$\therefore P_1 \times 70 = P \times d/2 \quad = P_1 \times 70 = 535.8 \times \frac{4.5}{2}$$

Force required at the end of the lever $\quad P_1 = 17.2N$

(b) Force to lower the load P_2

Force required to be applied at the mean radius to lower the load P

$$P = W\tan(\phi - \alpha) = 4000 \times \frac{\tan\phi - \tan\alpha}{1 + \tan\phi\tan\alpha}$$

$$= 4000 \times \frac{0.1 - 0.0335}{1 + 0.1 \times 0.0335}$$

$$P = 265.1\,N$$

$$P_2 \times 70 = p \times d/2, \quad P_2 \times 70 = 265.1 \times \frac{4.85}{2}$$

$$8.5$$

$$P_2 = 9N$$

Example II The mean diameter of a whitworth bolt having V-threads is 25mm. The pitch of the thread is 5mm and the angle of V is 55°. The bolt is tightened by screwing a nut whose mean radius of the bearing surface is 25 mm. If the co-efficient of friction for nut and bolt is 0.1 and for nut and bearing surfaces is 0.16, find the force required at the end of a spanner 0.5m long when the load on the bolt is 10kN.

Given:

$d = 25\,mm$, $P = 5\,mm$, $2\beta = 55^\circ$ $\therefore \beta = 27.5^\circ$, $R = 25\,mm$

$\mu_1 = \tan\phi = 0.1$, $\mu_2 = 0.16$, $L = 0.5\,m$, $W = 10\,kN$.

Virtual co-efficient of friction

β = is a semi-angle of the V-thread.

$$\mu_1 = \tan\phi_1 = \frac{\mu_1}{\cos\beta} = \frac{0.1}{\cos 27.5^\circ} = 0.113$$

Known that $\tan\alpha = \frac{P}{\pi d} = \frac{5}{\pi \times 25} = 0.064$

Force on screw

$$P = W\tan(\alpha + \phi_1) = W\left[\frac{\tan\alpha + \tan\phi_1}{1 - \tan\alpha\tan\phi_1}\right]$$

$$P = 11 \times 10^3 \left[\frac{0.064 + 0.113}{1 - (0.064 \times 0.113)} \right] = 1783N$$

Total torque transmitted

$$T = P \times \frac{d}{2} + \mu_2 WR = 1783 \times \frac{25}{2} + 0.16 \times 10 \times 10^3 \times 25$$

$$= 62300Nmm = 62.3NM$$

P_1 = Force required at the end of a spanner

$\therefore$ Torque required at the end of the spanner T

$$T = P_1 \times L = \cancel{P_1 \times 0.5 = 0.5P_1}$$

$$62.3 = P_1 \times 0.5$$

$$P_1 = \frac{62.3}{0.5} = 124.6N$$

ADVANTAGES AND DISADVANTAGES OF FRICTION

Friction can be used to great advantage in a motor vehi[cle] when applied to such units as tyres, clutches and brakes in which instances the surfaces must be dry and free fro[m] oil and grease. In bearings and other moving parts, fricti[on] is a disadvantage and systems of lubrication must be used to reduce friction to a minimum.

LUBRICANTS AND THEIR PROPERTIES

The must common lubricants are vegetable oils such as castor oil and mineral oils such as those obtained by distilation and refining of crude petroleum.

The two must important properties of lubricating oils are oiliness and viscosity. Oiliness can be described as the adhering property of lubricant to metal surface. Under severe conditions of loading and high rubbing speed seizure between a journal and its bearing would be less likely to occur with the lubricant of greater oiliness.

It concerns the resistance to oil-film breakdown during the application of heavy loads.

The viscosity of a lubricating oil is a measure of its resistance to flow. It concerns the ease with which an oil flows through the oil pipe-lines and spreads over the bearing surfaces.

SOLID LUBRICANTS AND THEIR APPLICATIONS

Solid lubricants includes lime soda base grease, heavy duty soda grease, medium grease and high-melting point grease. They are used to lubricate bearings operating under high pressure and it helps to seal bearings against the entry of water, dust and mud. They are used to lubricate joint, wheel hub, to prevent corrosion and reduce temperature.

EFFECT OF TEMPERATURE ON VISCOSITY OF LUBRICANTS

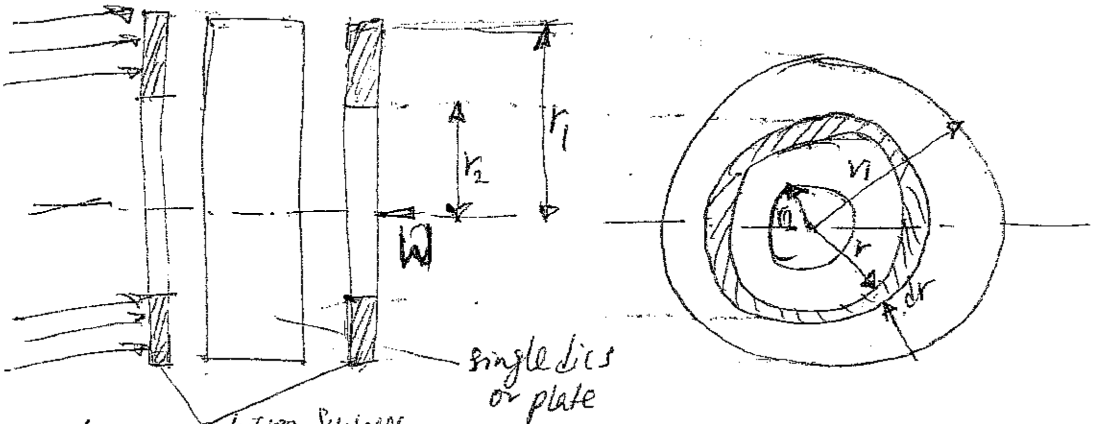
The viscosity of an oil changes with temperature. Increasing temperature causes an oil to thin, thus, the viscosity of the oil is lowered. Falling temperature causes an oil to thicken, thus, viscosity of the oil is raised.

CLUTCHES

The clutch fitted to the transmission system of a motor car is a means of disconnecting and connecting the engine to the gearbox. With frictional ~~clutch~~ plate clutches of the axial type, the frictional contact pressure between the driven and driving member is in the direction parallel to the engine crankshaft. The clutches are usually ~~consider of~~ single plate type and multiplate types. The types are

1, Disc or plate clutches (single or multiple disc clutch)
2, Cone clutches and
3, Centrifugal clutches.

- diag.



Consider friction Surfaces maintained in Contact by an axial thrust W

Let, T = Torque transmitted by the clutch

p = Intensity of the axial pressure with which the contact Surfaces are held together.

r_1 and r_2 = External and internal radii of friction Surface

μ = Co-efficient of friction.

Consider an elementary ring of radius r and thickness dr.

Area of Contact Surface or friction Surface $= 2\pi r\,dr$

δw = pressure x area.

$\therefore$ Normal or axial force on ring $\delta w = p \times 2\pi r\,dr$

Frictional torque acting on the ring $\qquad F_r = \mu \cdot \delta w$

$T_r = F_r \times r = \mu p\, 2\pi r\,dr \cdot r = 2\pi\mu p r^2\,dr$

CASE I, Considering when there is a Uniform pressure, the intensity of pressure, $p = \dfrac{W}{\pi(r_1^2 - r_2^2)}$

W = The axial thrust with which the contact or friction Surfaces are held together. From above.

$T_r = 2\pi\mu p r^2\,dr$, Integrate within the limits from r_2 and r_1

Total frictional torque

$$T = \int_{r_2}^{r_1} 2\pi\mu p r^2\,dr = 2\pi\mu p\left[\frac{r^3}{3}\right]_{r_2}^{r_1} = 2\pi\mu p\left[\frac{r_1^3 - r_2^3}{3}\right]$$

Substituting the value of P

$$T = 2\pi\mu \times \frac{W}{\pi(r_1^2 - r_2^2)} \times \frac{r_1^3 - r_2^3}{3}$$

$$= \frac{2}{3}\mu W\left[\frac{r_1^3 - r_2^3}{r_1^2 \cancel{3} r_2^2}\right] = \mu W R$$

R = mean radius of friction surface.

$$\frac{2}{3}\left[\frac{r_1^3 - r_2^3}{r_1^2 - r_2^2}\right]$$

CASE II, Considering when there is a Uniform Wear,

let p = Normal intensity of pressure at a distance r from the axis of a clutch.

Since the intensity varies inversely with the distance.

$$\therefore\ p \cdot r = C \quad \text{or} \quad p = \frac{C}{r}$$

Normal force on the ring, $\delta W = p \cdot 2\pi r \cdot dr = \frac{C}{r} \times 2\pi r \cdot dr$

$$= 2\pi C \cdot dr.$$

Total force acting on the friction surface

$$W = \int_{r_2}^{r_1} 2\pi C\, dr = 2\pi C\left[r\right]_{r_2}^{r_1} = 2\pi C\left[r_1 - r_2\right]$$

$$C = \frac{W}{2\pi\left[r_1 - r_2\right]}$$

Frictional torque acting on the ring

$$T_r = 2\pi\mu p r^2 dr = 2\pi\mu \times \frac{C}{r} \times r^2 dr = 2\pi\mu C r\, dr.$$

Total frictional torque on the friction surface

$$T = \int_{r_2}^{r_1} 2\pi\mu C r\, dr = 2\pi\mu C\left[\frac{r^2}{2}\right]_{r_2}^{r_1}$$

$$= 2\pi\mu C\left[\frac{r_1^2 - r_2^2}{2}\right]$$

$$= \pi \mu C [r_1^2 - r_2^2] = \pi \mu \times \frac{W}{2\pi [r_1 - r_2]} \times [r_1^2 - r_2^2]$$

$$T = \frac{1}{2} \mu W [r_1 + r_2] = \mu W R$$

Where, R = Mean radius of the friction surface $= \left[\frac{r_1 + r_2}{2}\right]$

Example I

A single plate clutch with both sides effective has outer and inner diameter 300mm and 200mm respectively. The maximum intensity of pressure at any point in the contact surface is not to exceed $0.1 N/mm^2$. If the co-efficient of friction is 0.3, determine the power transmitted by a clutch at a speed 2500rpm.

Given

$d_1 = 300mm$, $d_2 = 200mm$, $p = 0.1 N/mm^2$; $\mu = 0.3$

$N = 2500 rev/min$ or $\omega = \frac{2\pi N}{60} = \frac{2\pi \cdot 2500}{60} = 261.8 rad/s$

Since the intensity of pressure "p" is maximum at the inner radius r_2

$\therefore$ Uniform wear $C = p \cdot r_2 = 0.1 \times 100 = 10 N/mm$

Axial thrust $W = 2\pi C (r_1 - r_2)$

$$= 2\pi \times 10 (150 - 100) = 3142 N$$

Mean radius of friction surfaces for uniform wear

$$R = \frac{r_1 + r_2}{2} = \frac{150 + 100}{2} = 125mm = 0.125m$$

Torque transmitted

$$T = n \mu W R = 2 \times 0.3 \times 3142 \times 0.125 = 235.65 Nm$$

($n = 2$ for both sides of plate effective)

$\therefore$ power transmitted by a clutch

$$P = T\omega = 235.65 \times 261.8 = 61693 W$$

P.T.O

Example ii

A multiple disc clutch has five plates having four pairs of active friction surfaces. If the intensity of pressure is not to exceed $0.127 N/mm^2$. Find the power transmitted at 500 rpm. The outer and inner radii of friction surfaces are 125 mm and 75 mm respectively. Assume uniform wear and take co-efficient of friction $= 0.3$

Given $n_1 + n_2 = 5$, $n = 4$, $p = 0.127 N/mm^2$; $N = 500 rpm$

or $\omega = \frac{2\pi \times 500}{60} = 52.4$ rad/s, $r_1 = 125 mm$, $r_2 = 75 mm$

$\mu = 0.3$

Since the intensity of pressure is maximum at the inner radius r_2

$\therefore p r_2 = C = 0.127 \times 75 = \cancel{96}$ $\therefore C = 9.525 N/mm$

Axial force required to engage the clutch

$W = 2\pi C(r_1 - r_2) = 2\pi \times 9.525(125 - 75) = 2990 N$

Mean radius of the friction surfaces.

$R = \frac{r_1 + r_2}{2} = \frac{125 + 75}{2} = 100 mm = 0.1 m$

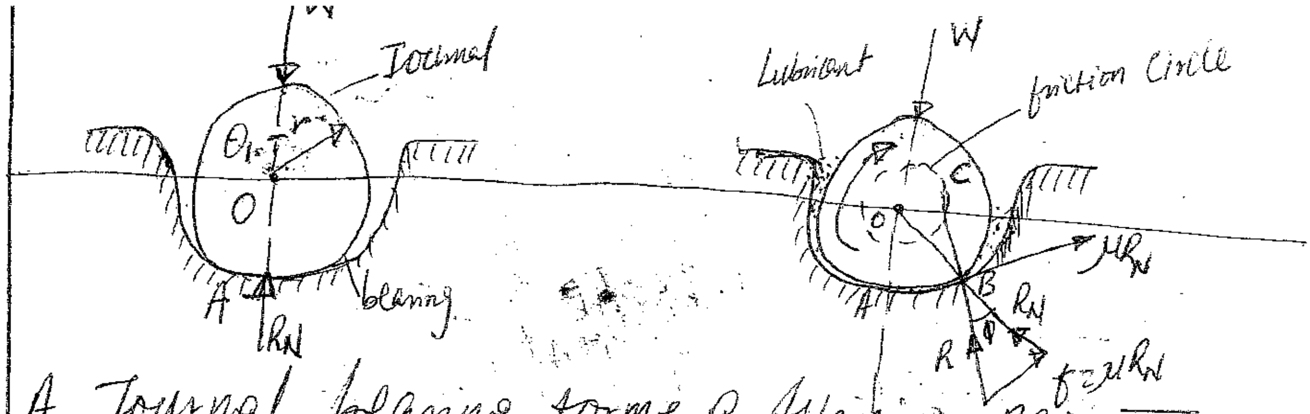
Torque transmitted, $T = n \cdot \mu W R$

$= 4 \times 0.3 \times 2990 \times 0.1 = 358.8 Nm.$

power transmitted, $P = T\omega = 358.8 \times 52.4 = 18800 W = 18.8 kW$

FRICTION IN JOURNAL BEARING

Diag.



A Journal bearing forms a turning pair. The fixed outer element of a turning pair is called a bearing and the portion of the inner element i.e Shaft which fits in the bearing is called a Journal. The Journal is slightly less in diameter than the bearing, in order to permit the free movement of the Journal in a bearing. When the ~~force~~ bearing is not lubricated (or Journal is stationary), there is a line of contact between the two elements. The load W on the Journal and normal reaction R_N (equal to W) of the bearing acts through the centre. The reaction R_N acts vertically upwards at point A. Now consider a shaft rotating inside a bearing in clockwise direction as shown above. The lubricant between the Journal and bearing forms a thin layer which gives rise to a greasy friction. Therefore, the reaction R_N does not act vertically upwards, but acts at another point of pressure B. This is due to the fact that when the shaft rotates, a frictional force $f = \mu R_N$ acts at the circumference of the shaft in opposite direction of the motion and this ~~shaft~~ shifts the point A to point B.

For uniform motion, the resultant force acting on the shaft must be zero and the resultant turning ~~mome~~ moment on the shaft must be zero.

$$\therefore R = W \text{ and } T = W \times OC = W \times OB \sin\phi$$

$$= W \cdot r \cdot \sin\phi$$

Since ϕ is very small, therefore substituting $\sin\phi = \tan\phi$

$T = W \cdot r \cdot \tan\phi = \mu W r$

If the shaft rotates with angular velocity ω rad/s then the power wasted in friction

$P = T.\omega = T \times \frac{2\pi N}{60}$ watts.

Example I A 60mm diameter shaft running in a bearing carries a load of 2000N. If the co-efficient of friction between the shaft and bearing is 0.03. Find the power transmitted when it runs at 1440rpm.

Given, $d = 60mm$, $r = 30mm = 0.03m$, $W = 2000N$, $\mu = 0.03$

$N = 1440rpm$ or $\omega = \frac{2\pi \times 1440}{60} = 150.8 rad/s$.

$T = \mu W r = 0.03 \times 2000 \times 0.03 = 0.18Nm$

$P = T.\omega = 0.18 \times 150.8 = 27.15W$.

Example II The main journal bearings of an engine crankshaft carry a total load of 10kN when the engine speed is 3500 rev/min. If the co-efficient of friction between each bearing and shaft is 0.015 and the diameter of the shaft is 60mm, calculate

(a) The frictional torque on the shaft.

(b) The power absorbed by friction.

(c) The heat energy generated in the bearing per minute.

Given- $W = 10,000N$, $N = 3500rpm$, $r = 0.03m$, $\mu = 0.015$

(a) Frictional torque, $T_f = \mu W r = 0.015 \times 10,000 \times 0.03$
$= 4.5Nm$

(b) Power absorbed by friction $= T_f \times \frac{2\pi N}{60} = \frac{4.5 \times 2\pi \times 3500}{60}$
$= 1650W$

(c) Heat energy generated in bearings per minute due to friction.
$= 1650 \times 60 = 99000J = 99KJ$
[Since $1W = 1 J/s$].

BALANCING

IMPORTANCE OF BALANCING

The high speed of engines and other machines is a common phenomenon nowadays. It is very essential that all the rotating and reciprocating parts should be completely balanced as far as possible. If these parts are not properly balanced, the dynamic forces are set up. These forces not only increase the loads on bearings and stresses in the various members but also produce unpleasant and even dangerous vibrations. Balancing is very important in order to minimise pressure on the main bearing when an engine is running, vibrations, noise-emission and permanent failure of a fatigue nature.

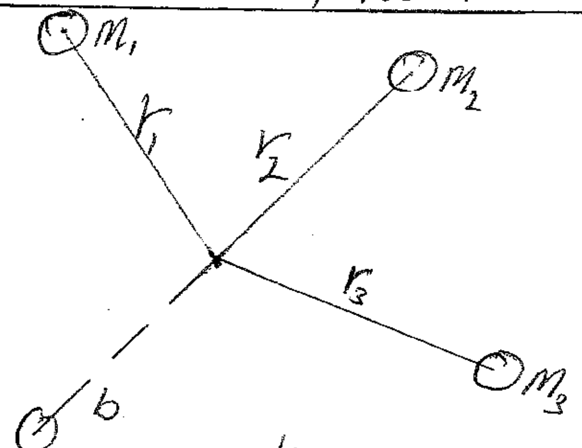
STATIC AND DYNAMIC BALANCE

When a shaft carrying several eccentric masses is in static balance, the centre of mass of the system lies in the axis of the shaft so that the shaft and attached masses remain in any position in which it is placed. When the shaft rotates, the centrifugal forces act upon the masses and if these are not rotating in the same plane, couples also act upon the shaft. Therefore, for complete dynamic balance

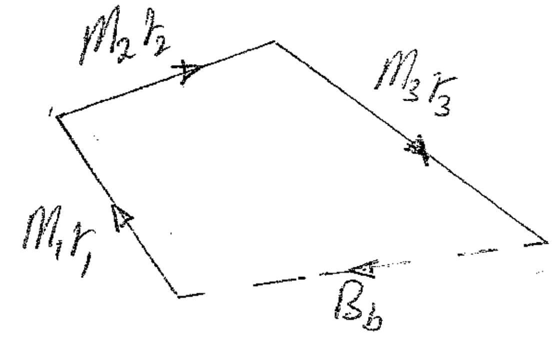
i, The resultant force acting upon the shaft must be zero.

ii, The resultant couple acting upon the shaft must be zero.

BALANCING OF MASSES ROTATING IN THE SAME PLANE



Space diagram



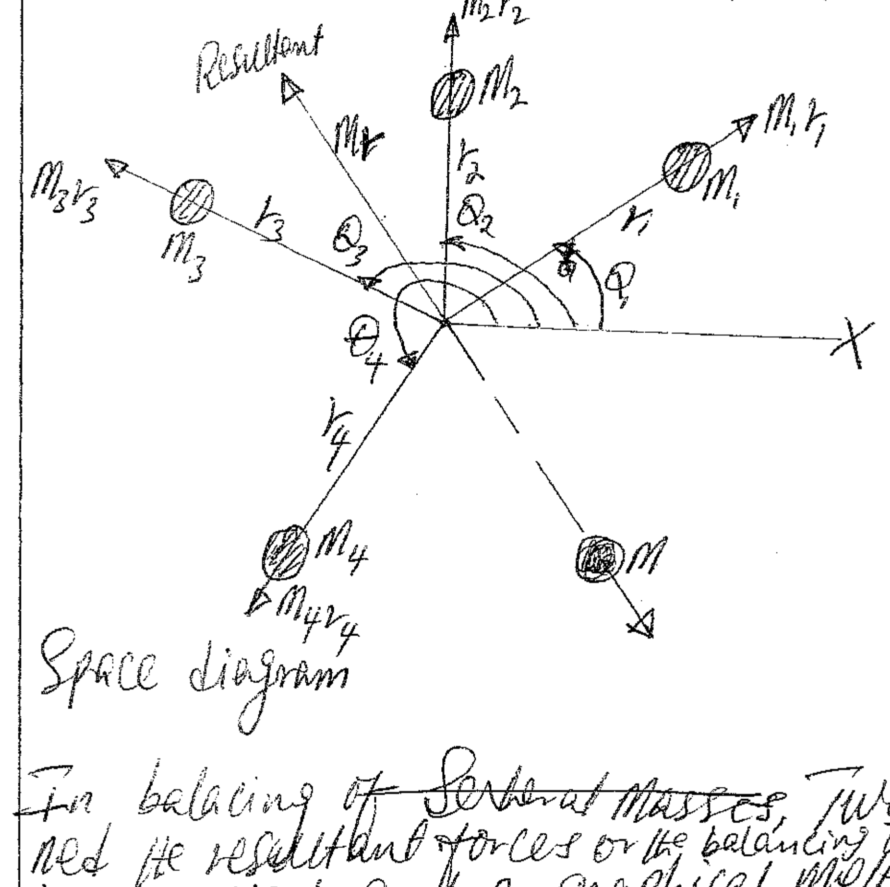
Vector diagram

If M_1, M_2, M_3 etc are the out-of-balance masses and r_1, r_2, r_3 etc are the respective radii of rotation. For dynamic balance, the vector sum of the centrifugal forces must be zero.

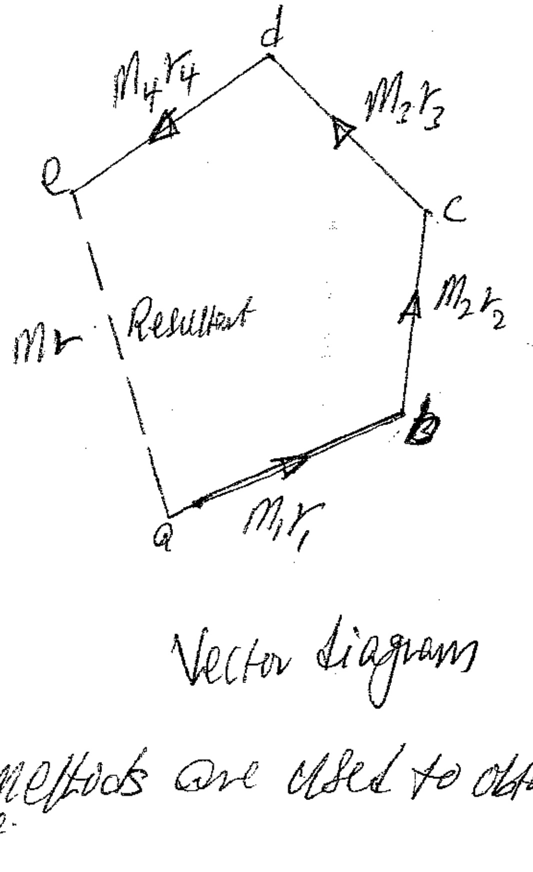
$\sum M\omega^2 r = 0$, where, ω is the angular speed of the shaft

$\sum mr = 0$, since ω^2 is the same for each mass. From the force polygon drawn above, the closing side represents the product of the balance mass B and its radius of rotation b.

~~BALANCING OF SEVERAL MASSES ROTATING IN THE SAME PLANE~~



Space diagram



Vector diagram

In balacing of ~~Serveral masses~~, two methods are used to obtained the resultant forces or the balancing force.

i, Analytical and 2, graphical method.

ANALYTICAL METHOD

i, Find out the centrifugal force (or the product of mass and its radius of rotation) exerted by each mass on the rotating shaft.

ii, Resolve the centrifugal forces horizontally and vertically and find their sum i.e $\sum H$ and $\sum V$. Sum of the horizontal components of the centrifugal forces $\sum H = M_1 r_1 \cos\theta_1 + M_2 r_2 \cos\theta_2 + \text{------}$

The sum of ~~horizontal~~ vertical components of the centrifugal forces, $\Sigma V = m_1 r_1 \sin\theta_1 + m_2 r_2 \sin\theta_2 + \cdots$

iii, Magnitude of the resultant force, $M_r = \sqrt{(\Sigma H)^2 + (\Sigma V)^2}$

iv, If θ is the angle, which the resultant force makes with the horizontal then $\tan\theta = \Sigma V / \Sigma H$

v, The balancing force is then equal to the resultant force, but in opposite direction.

vi, Now, find out the magnitude of the balancing mass such that, Resultant centrifugal force $= m r$.

GRAPHICAL METHOD

i) Draw the space diagram with the positions of the several masses

ii, Find out the centrifugal force (or product of the mass and radius of rotation) exerted by each mass on the rotating shaft.

iii, Now, draw the vector diagram with the obtained centrifugal forces (or the product of the masses and their radii of rotation). such that "ab" represent the centrifugal force exerted by the mass m_1 ($m_1 r_1$) in magnitude and direction drawn to a suitable scale. Similarly draw bc, cd, and de to represent centrifugal forces of other masses m_2, m_3, and m_4 (or $m_2 r_2$, $m_3 r_3$ and $m_4 r_4$).

(IV) Now as per polygon law of forces, the closing side "ae" represents the resultant force in magnitude and direction.

V, ~~Draw~~ The balancing force is equal to the resultant force, but in opposite direction

vi, Now, find out the magnitude of the balancing mass (m) at a given radius of rotation (r) such that, $m\omega^2 r$ = Resultant force

$m \cdot r$ = Resultant of $m_1 r_1$, $m_2 r_2$, $m_3 r_3$ and $m_4 r_4$.

BALANCING OF SEVERAL MASSES ROTATING IN DIFFERENT PLANES

When several masses revolve in different planes, they may be transfered to reference plane (R.P) which may be defined as the plane passing through a point on the axis of rotation and perpendicular to it. The effect of transfering a revolving mass (in one plane) to a reference plane is to cause force of magnitude equal to the centrifugal force of the revolving mass to act in reference

plane, together with the couple of magnitude equal to the product of the force and the distance between the plane of rotation and the reference plane. In order to have a complete balance of the several masses in different plane, these two conditions must be satisfied.

i, The forces in the reference plane must balance i-e the resultant force must be zero.

ii, The couples about the reference plane must balance i-e the resultant couple must be zero.

BALANCING IN RECIPROCATING ENGINES

Apart from the reaction torque produced by the forces causing accelaration of reciprocating parts, there could be unbalanced forces transmitted through bearings equal to the reciprocating and rotating inertia forces given by

$$P = mr\omega^2\left(\cos\theta + \frac{r}{l}\cos 2\theta\right) \quad \text{----- 1}$$

The first term in eqn(1) i-e $m\omega^2 r\cos\theta$ is called the primary inertia force and reaches a maximum value of $m\omega^2 r$ twice per revolution, when $\theta = 0^\circ$ and 180°. The secound term $m\omega^2 r^2 \cos 2\theta / l$ is the secoundary inertia force and reaches a maximum value of $m\omega^2 r^2/l$ four times in a revolution at $\theta = 0^\circ, 90^\circ, 180^\circ$ and 270°. Each term represents a force which is constant in direction but varying in magnitude and therefore cannot be completely balanced by a rotating mass on which the inertia force is varying in direction but constant in magnitude.

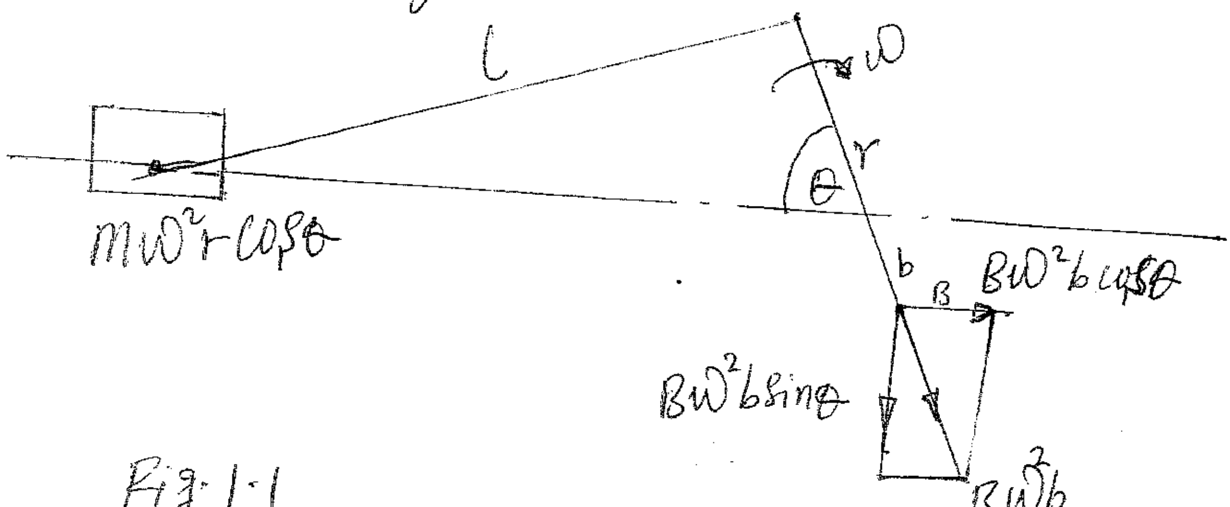


Fig. 1·1

In the case of the primary inertia force, this can be balanced (? by a mass "B", fig 1.1, rotating at a radius "b" such that $Bb = Mr$, the component of inertia on B in the line of stroke being $B\omega^2 b\cos\theta$. But a force $B\omega^2 b\sin\theta$ is then produced in the direction perpendicular to the line of stroke.

A similar balance, partial balance of the secoundary inertia force can only be achieved by a balance mass which rotates such that the radius makes an angle 2θ ~~with the~~, i.e. rotates at twice the crankshaft speed. Due to the practical difficulties of such an arrangement and relatively small magnitude of the secoundary inertial force, such balance is not normally attempted.

With more than one cylinder, however, it will be possible to obtain a degree of balance of the reciprocating inertia forces by suitable choice of relative crank spacing both radially and axially. For complete balance, omitting ω^2,

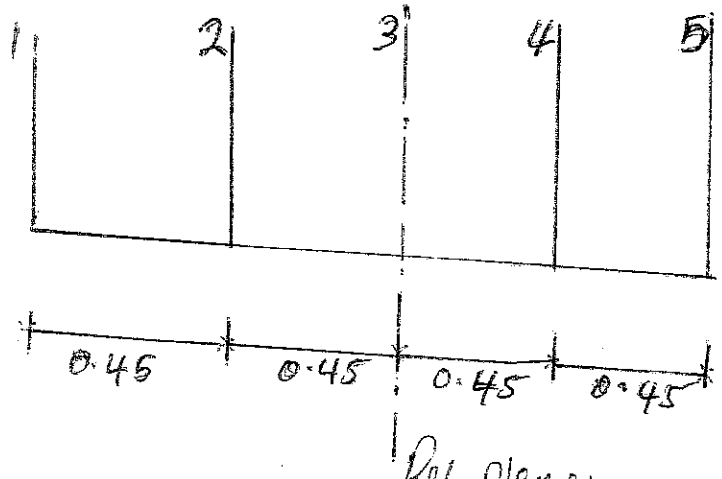
$$\left.\begin{aligned} &\sum mr\cos\theta = 0 \text{ — (a): primary force} \\ &\sum (mr/n)\cos 2\theta = 0 \text{ — (b): secoundary force} \\ &\sum mrx\cos\theta = 0 \text{ — (c): primary moment} \\ &\sum (mrx/n)\cos 2\theta = 0 \text{ — (d): secoundary moment} \end{aligned}\right\} \text{— 1.2}$$

Where $n = l/r$, each of these summations can be represented by vectors of the appropriate lengths inclined at corresponding θ values to the line of stroke.

__Example I__

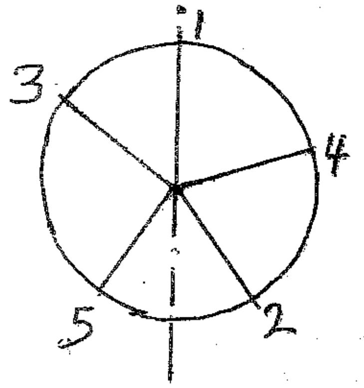
An engine having 5 cylinders inline has successive cranks 144° apart, the distance between cylinder centre line been 450mm. The reciprocating mass for each cylinder is 16kg, the crank radius is 135 mm and the connecting rod length is 540 mm, the engine runs at 600 rpm. Examine the engine for balance of primary and secoundary forces and couples. Determine also the maximum values of these and the position of the central crank at which the maximum value occur.

Soln

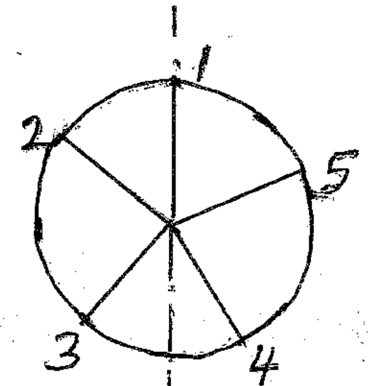


From the data, the table is obtained.

Plane	M(kg)	r(m)	x (m)	Mr (kgm)	Mrx(kgm²)	θ(°)	2θ(°)
1	16	0.135	−0.90	2.16	−1.944	0	0
2	16	0.135	−0.45	2.16	−0.972	144	288
3	16	0.135	0.00	2.16	0.000	288	216
4	16	0.135	+0.45	2.16	+0.972	72	144
5	16	0.135	+0.90	2.16	+1.944	216	72

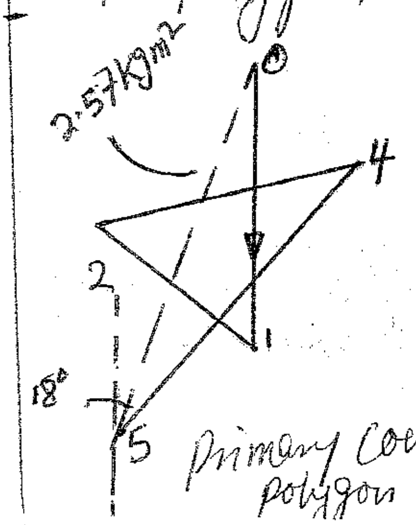


Primary Crank positions

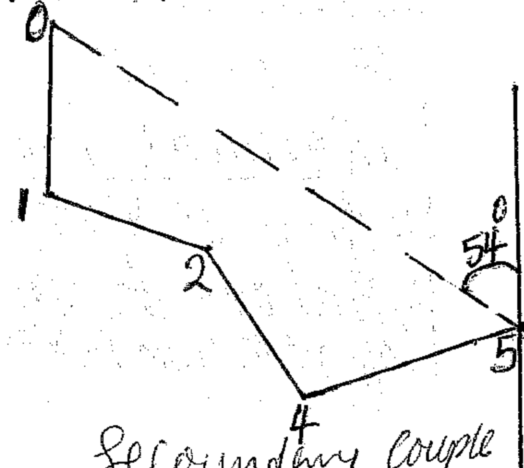


Secoundary Crank position.

Maximum Unbalanced primary couple ~~a~~ From the primary couple polygon, Closing side 05 = 2.57 kgm² by measurement



Primary Couple polygon

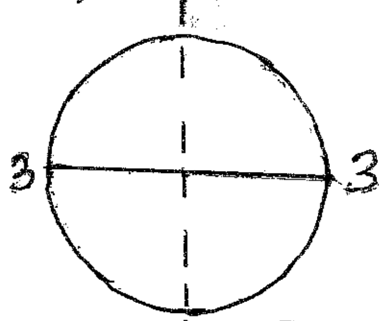


Secoundary Couple polygon

For the maximum unbalanced primary couple the 2.5 kgm² (3 must be multiplied by ω^2 to give the actual unbalanced coupl i.e

$$\text{maximum unbalanced primary couple} = 2.57 \times \left(600 \times \frac{2\pi}{60}\right)^2$$
$$= 10150\,Nm$$

In this case the vector $\vec{OS}$ is parallel with the line of strok when couple polygon has rotated through 18° and 198° antic ckwise from the position shown i.e when crank No. 3 i at 90° on either side of the line of stroke.



or 18° and 198° to T.D.C
i.e when crank No. 3 is at 288−1?
= 90° on either side of t.d.c.

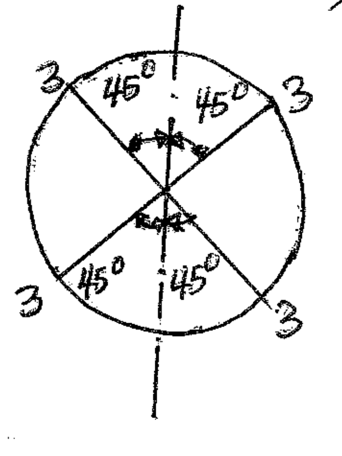
From the secoundary couple polygon, the closing side i = 4.15 kgm² which must be multiply by ω^2/n to give th actual unbalance couples. i.e.

maximum unbalanced ~~couple~~ secoundary couple

$$= 4.15\left(600 \times \frac{2\pi}{60}\right)^2 \times \frac{0.135}{0.54} \quad \text{where } n = \tfrac{l}{r}$$

$$= 4100\,Nm$$

Vector OS is parallel with the line of stroke when the couple polygon turned through 54°, 234°, 414° and 594° clockwise from the position shown. Since the crank angles are twice the actual crank angles measured from the T.D.C, the actual cranks must be turned through 27°, 117°, 207° and 297° clockwise. For maximum secoundary couple to occur i.e crank No. 3 is at 45° on the either side of the line of stroke.



P.T.O.

Example II

A shaft carries four masses A, B, C and D of magnitude 200kg, 300kg, 400kg and 200kg respectively and revolving at radii 80mm, 70mm, 60mm and 80mm in planes measured from A at ~~8~~300mm, 400mm and 700mm. The angles between the cranks measured anticlockwise are A to B 45°, B to C 70° and C to D 120°. The balancing masses are to be placed in planes X and Y. The distance between the planes A and X is 100mm, between X and Y is 400mm and between Y and D is 200mm. If the balancing masses revolve at a radius of 100mm, find their magnitudes and angular positions.

Given $M_A = 200kg$, $M_B = 300kg$, $M_C = 400kg$, $M_D = 200kg$

$r_A = 80mm = 0.08m$, $r_B = 70mm = 0.07m$, $r_C = 60mm = 0.06m$, $r_D = 80mm = 0.08m$

$r_X = r_Y = 100mm = 0.1m$

Let,

M_X = Balancing mass placed in plane X

M_Y = Balancing mass placed in plane Y

The position of planes and angular position of the ~~measured~~ masses (assuming the mass A as horizontal) are shown below. Assuming the plane X as the reference plane. The distance of the planes to the right of plane X are taken as +ve while distances of the planes to the left of plane X are taken as −ve. The data may be tabulated as shown below.

Plane (1)	Mass(m) kg (2)	Radius(r) m (3)	Centrifugal force ÷ ω^2 m.r (kgm) (4)	Distance from plane X (L) m (5)	Couple ÷ ω^2 mrL (kgm²) (6)
A	200	0.08	16	−0.1	−1.6
X (R.P)	MX	0.1	0.1MX	0	0
B	300	0.07	21	0.2	4.2
C	400	0.06	24	0.3	7.2
Y	MY	0.1	0.1MY	0.4	0.04MY
D	200	0.08	16	0.6	9.6

the balancing masses M_x and M_y and their angular positions may be determined graphically as discussed below:

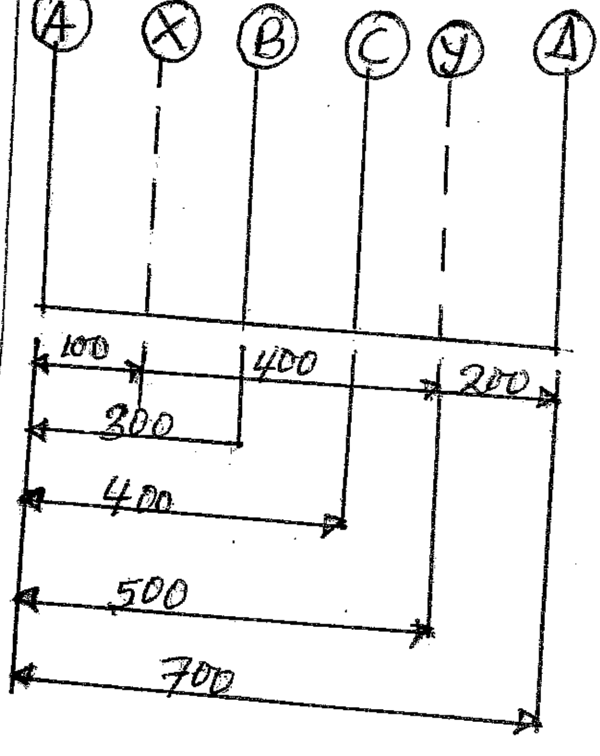
(i) Draw the couple polygon from the data given in column 6 of the table above. Since the balance couple is proportional to $0.04 M_y$, therefore, by measurement

$$0.04 M_y = \text{Vector "do"} = 7.3\ kgm^2 \quad \text{or} \quad M_y = 182.5\ kg$$

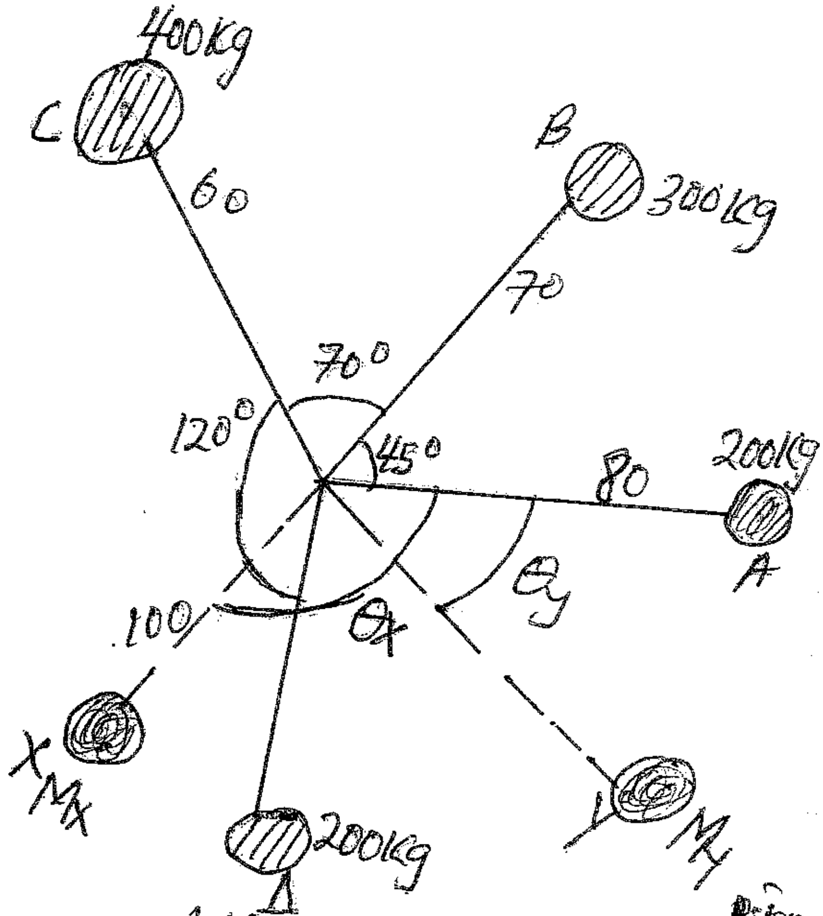
The angular position of the mass M_y is obtained by drawing OM_y parallel to vector "do". By measurement, the angular position of M_y is $\theta_y = 12°$ in the clockwise direction from mass M_A i.e. 200kg.

(ii) Now draw the force polygon from the data given in column 4 from the table. The vector eo represents the balanced force. Since the balanced force is proportional to $0.1 M_x$, therefore by measurement, $0.1 M_x = \text{Vector } eo = 35.5 \therefore M_x = 355\ kg$.

The angular position of the mass M_x is obtained by drawing OM_x parallel to vector eo. By measurement, the angular position of M_x is $\theta_x = 145°$ in the clockwise direction from mass M_A i.e. 200kg.

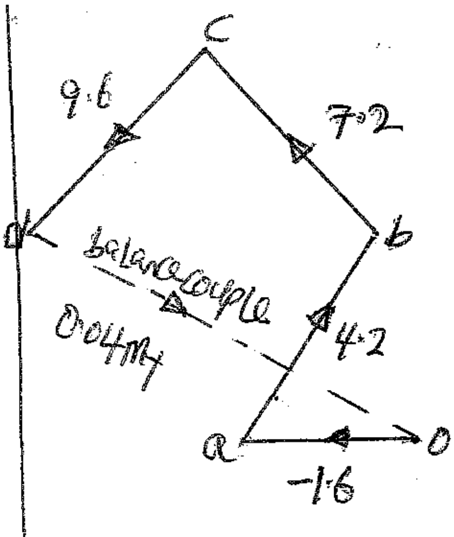


POSITION OF PLANES

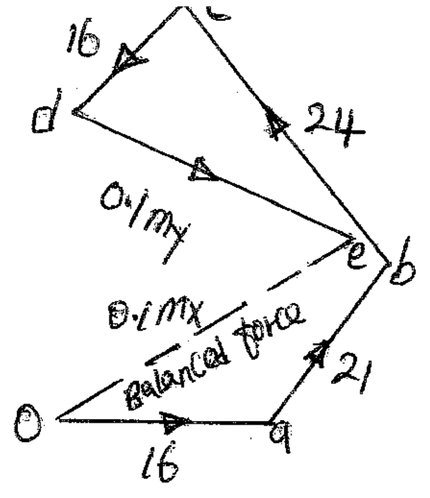


ANGULAR POSITION OF MASSES

p.t.o



Couple polygon

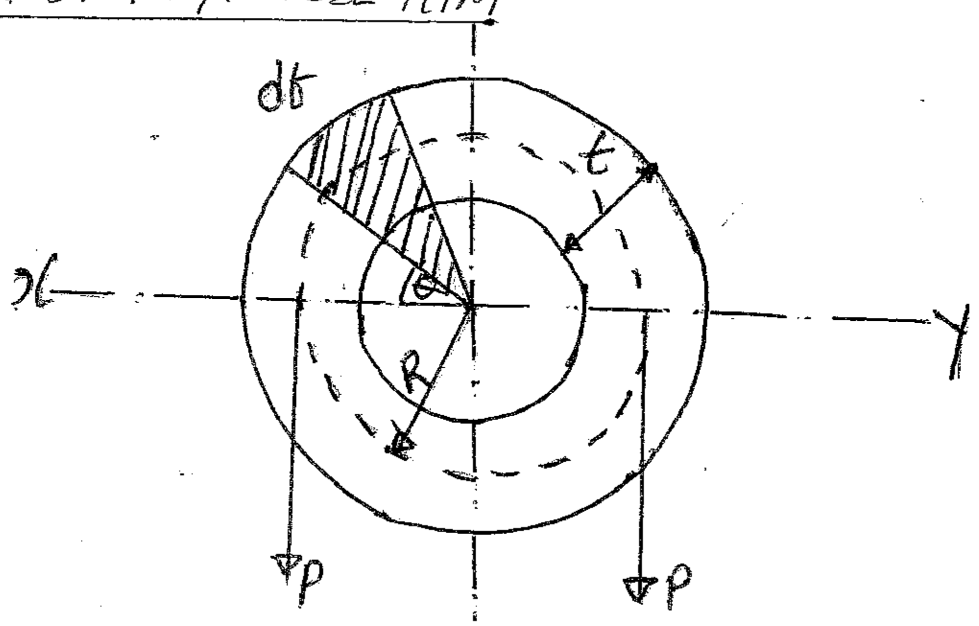


Force polygon

FLYWHEEL

FLYWHEEL :- A flywheel used in machines serves as a reservoir which stores energy during the period when the supply of energy is more than the requirement and releases it during the period when the requirement of energy is more than the supply. A little consideration shows that, when the flywheel absorbs energy its speed increases and when it releases energy, the speed decreases. A flywheel does not maintain a constant speed, it simply reduces the fluctuation of speed. In other words, a flywheel controls the speed variations caused by the fluctuation of engine ~~speed~~ turning moment during each cycle operation.

DIMENSION OF FLYWHEEL RIM



diag.

Consider a rim shown above.

Let, Δ = Mean diameter of the rim in meters

R = Mean radius of rim in m

A = Cross-Sectional area of rim in m^2

ρ = Density of rim material in kg/m^3

ω = Angular velocity of the flywheel in rad/s

V = linear velocity at the mean radius in m/s $= \cancel{\omega \cdot R = \pi \Delta N/60}$

σ = Tensile Stress in N/m^2 due to Centrifugal force.

Consider a Small element of the rim as shown ~~above~~ shaded above. Let it Substends an angle $\delta\theta$ at the Cent of the flywheel.

Volume of the Small element $= A \times R\delta\theta$

Mass of the Small element $= \delta m$ = Density × Volume $= \rho \cdot A \cdot R \cdot \delta\theta$.

Centrifugal force on the element, acting radially outwards,

$$\delta f = \delta m \cdot \omega^2 \cdot R = \rho \cdot A \cdot R^2 \omega^2 \delta\theta$$

Vertical Component of $\delta f = \delta f \sin\theta = \rho \cdot A \cdot R^2 \omega^2 \delta\theta \sin\theta$

$\therefore$ Total Vertical upwards force tending to twist the rim across the diameter XY

$$= \rho \cdot A \cdot R^2 \cdot \omega^2 \int_0^{\pi} \sin\theta\, \delta\theta = \rho \cdot A \cdot R^2 \cdot \omega^2 \left[-\cos\theta\right]_0^{\pi}$$

$$= 2\rho \cdot A \cdot R^2 \cdot \omega^2 \quad \text{------ (1)}$$

The Vertical upwards force will produce tensile Stress (Centrifugal or Circumferential Stresses) and it is resisted by $2P$ such that

$$2P = 2\sigma A \quad \text{------ (2)}$$

Equating eqn (1) & (2)

$$2\rho A R^2 \omega^2 = 2\sigma A$$

$$\sigma = \rho R^2 \omega^2 = \rho V^2 \qquad V = R\omega$$

$$V = \sqrt{\frac{\sigma}{\rho}}$$

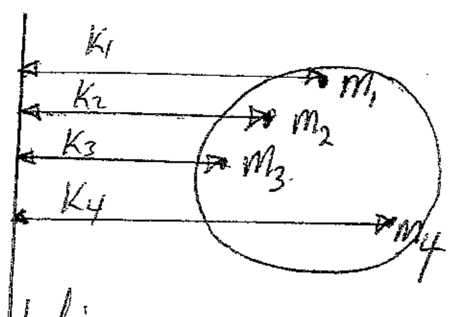
Mass of the rim, m = Volume × density $= \pi \Delta A \rho$

$$A = \frac{m}{\pi \Delta \rho}$$

TERMS RELATING TO FLYWHEELS

1. MASS: It is the amount of matter contained in a given body and does not vary with the change in its position on the earth's surface.

2. MOMENT OF INERTIA: It is the sum of the product of every particle of a body and square of its perpendicular distance from a fixed line



Moment of inertia, $I = m_1 K_1^2 + m_2 K_2^2 + m_3 K_3^2 + m_4 K_4^2$ ------ etc

3, TORQUE: It may be defined as the product of force and the perpendicular distance of its line of action from the given point or axis.

4. TURNING MOMENT DIAGRAM: The turning moment diagram (which is known as crank effort diagram) is the graphical representation of the turning moment or crank effort for various positions of the crank.

5. CO-EFFICIENT OF FLUCTUATION OF SPEED: This is the ratio of the maximum fluctuation of speed to the mean speed. Note that the difference between the maximum and minimum speeds during a cycle is called maximum fluctuation of speed. If N_1 and N_2 = Maximum and minimum speeds in rpm during the cycle.

$$N = \text{Mean speed in rpm} = \frac{N_1 + N_2}{2}$$

∴ Co-efficient of fluctuation of speed,

$$C_s = \frac{N_1 - N_2}{N} = \frac{2[N_1 - N_2]}{N_1 + N_2}$$

$$= \frac{\omega_1 - \omega_2}{\omega} = \frac{2(\omega_1 - \omega_2)}{\omega_1 + \omega_2} \quad \text{— (in angular speeds)}$$

$$= \frac{V_1 - V_2}{V} = \frac{2[V_1 - V_2]}{V_1 + V_2} \quad \text{— [in linear speeds]}$$

6, __CO-EFFICIENT OF FLUCTUATION__ OF ENERGY :- It may be defined as the ratio of the maximum fluctuation of energy, to the workdone per cycle.

Co-efficient of fluctuation of energy = C_E

$$C_E = \frac{\text{Maximum fluctuation of energy}}{\text{Workdone per cycle}}$$

Workdone per cycle may be obtained by,

i, $T_{mean} \times \theta$, where, T_{mean} = Mean torque

θ = Angle turned (in rad/s) in 1 revolution.

$= 2\pi$ for steam engine and 2 stroke I.C. engines

$= 4\pi$ for 4 stroke I.C. engines.

- The mean torque, $T_{mean} = \frac{P \times 60}{2\pi N} = P/\omega$

Where, P = power transmitted in watts.

N = Speed in rpm.

ω = Angular speed in rad/s.

ii, Workdone per cycle $= \frac{P \times 60}{n}$

Where,

n = Number of working strokes per minute

$= N$, for steam engines and 2 stroke I.C. engines

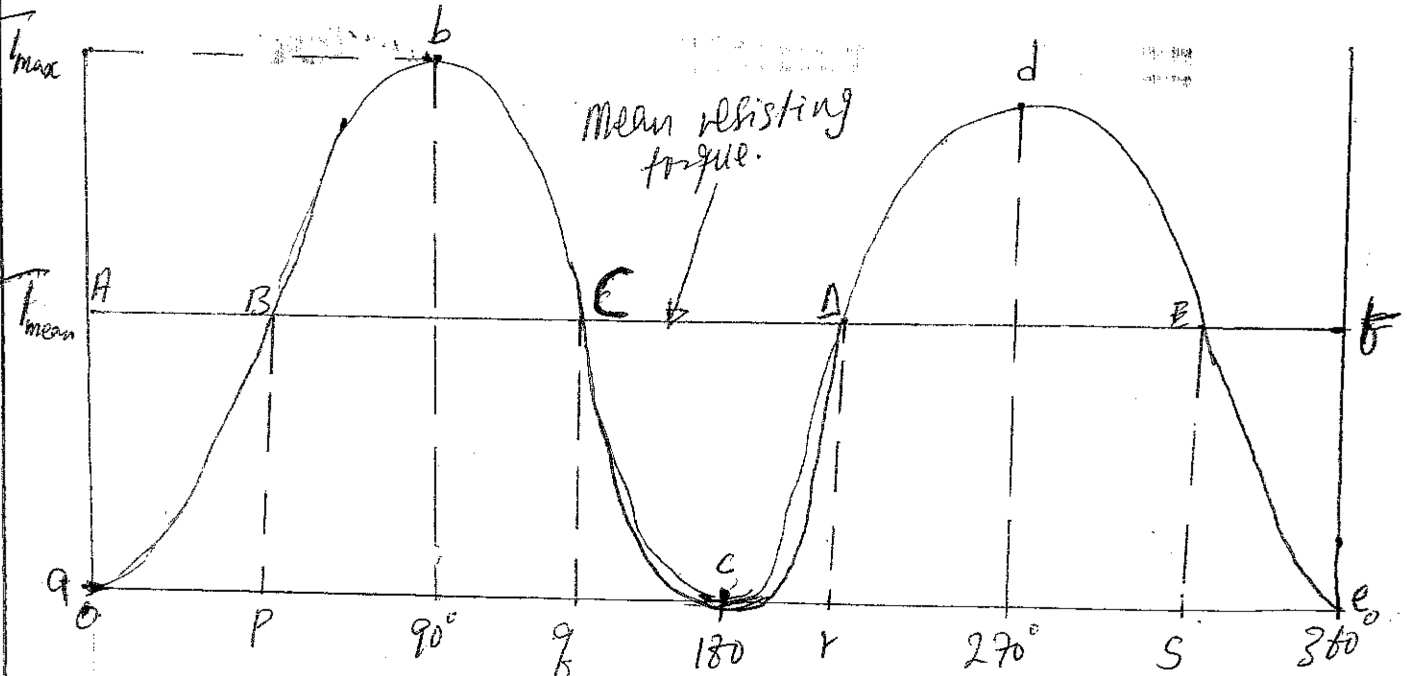
$= \frac{N}{2}$ for 4 stroke I.C. engines.

7, MINIMUM SPEED (N_2) :- This is the speed of the engine when flywheel releases energy during non-working strokes.

8, MAXIMUM SPEED [N_1] :- This is the speed of the engine when flywheel absorbs energy during the working stroke.

9. MEAN SPEED [N]: This is the average speed between the maximum and minimum speeds.

TURNING MOMENT DIAGRAM FOR A SINGLE CYLINDER DOUBLE ACTING STEAM ENGINE



The turning moment of the crankshaft "T" is given by

$$T = F_p \times r \left(\sin\theta + \frac{\sin 2\theta}{2\sqrt{n^2 - \sin^2\theta}} \right)$$

Where,

F_p = Piston effort

r = Radius of crank

n = Ratio of the ~~cran~~ connecting rod length and radius of crank.

θ = Angle turned by the crank from inner dead centre.

From the above expression, we see that the turning moment is zero, when the crank angle is 90°. This is shown by the curve above the area of the rectangle aAFe is proportional to the workdone against mean resisting torque. Note,

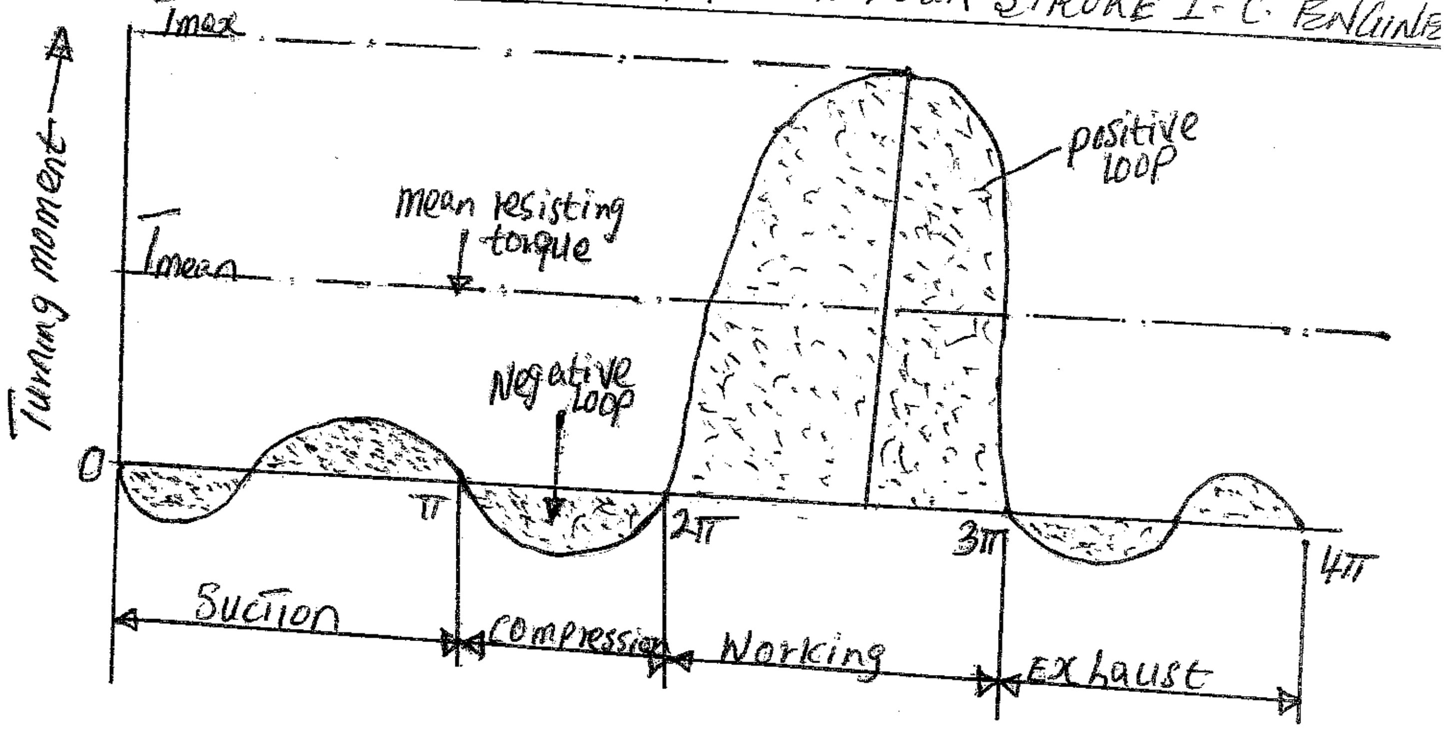
i) When the turning moment is positive (i.e B and C or D and E) the crankshaft accelerates and work is done by the steam.

ii) When the turning moment is negative (i.e C and D) the crankshaft retards and work is done on the steam.

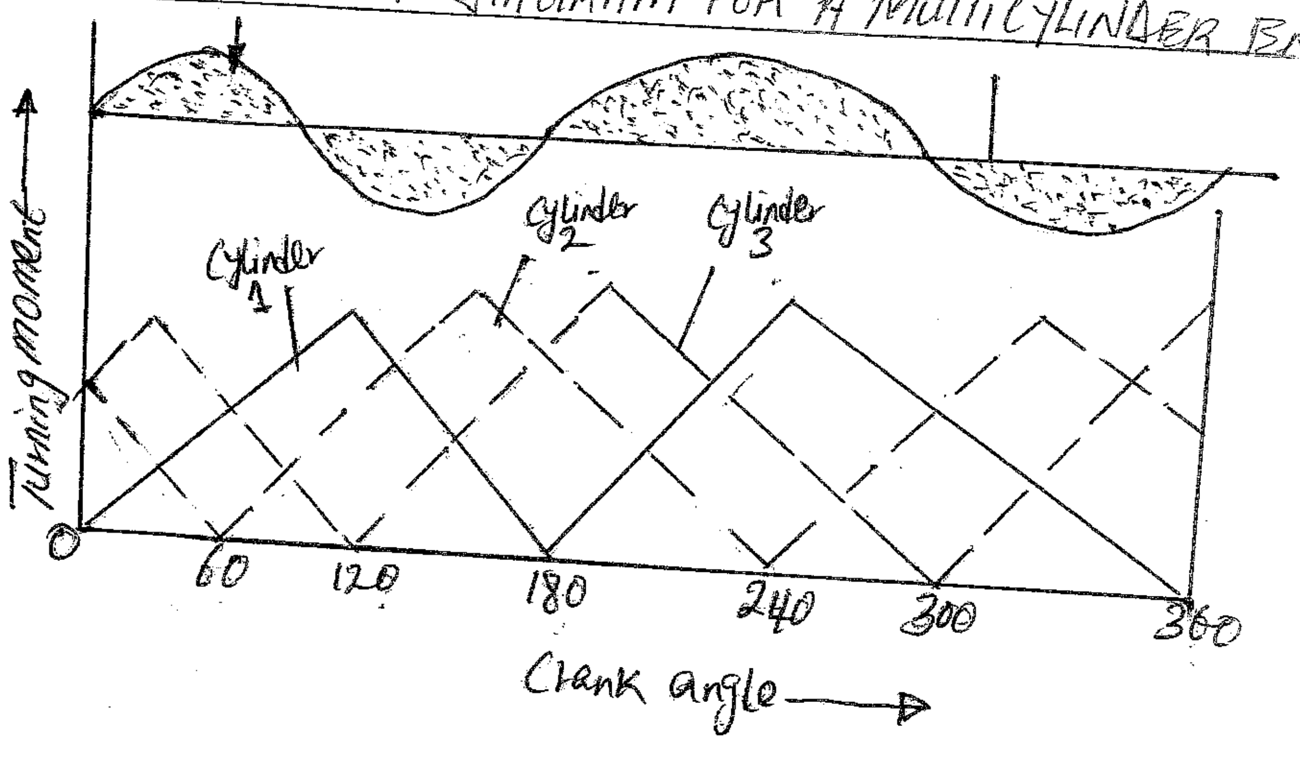
(iii) Accelerating torque on the rotating parts of the engine (4
$= T - T_{mean}$

(iv) If $T - T_{mean}$ is +ve, the flywheel accelerates
If $T - T_{mean}$ is −ve, the flywheel retards.

TURNING MOMENT DIAGRAM FOR A FOUR STROKE I.C. ENGINE

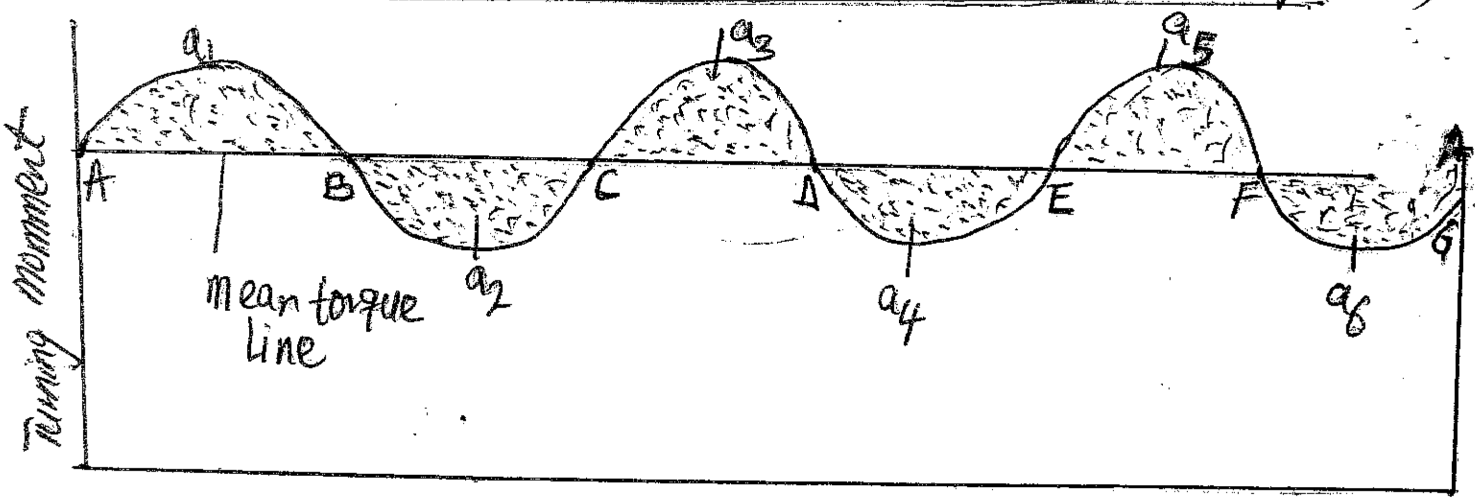


TURNING MOMENT DIAGRAM FOR A MULTICYLINDER ENGINE



P.T.O.

DETERMINATION OF MAXIMUM FLUCTUATION OF ENERGY (43)



Crank angle →

Let a_1, a_3, a_5 be the areas above the mean torque line and a_2, a_4 and a_6 be the areas below the mean torque line. These areas represent some quantity of energy which is either added or subtracted from the energy of the moving parts of the engine.

Let the energy in flywheel at A = E

then,

Energy at B = $E + a_1$

Energy at C = $E + a_1 - a_2$

Energy at D = $E + a_1 - a_2 + a_3$

Energy at E = $E + a_1 - a_2 + a_3 - a_4$

Energy at F = $E + a_1 - a_2 + a_3 - a_4 + a_5$

Energy at G = $E + a_1 - a_2 + a_3 - a_4 + a_5 - a_6$

= Energy at A (i.e cycle repeats after G)

Suppose the greatest of these energies is at B and least at E

∴ Maximum energy in flywheel = $E + a_1$

Minimum energy in flywheel = $E + a_1 - a_2 + a_3 - a_4$

∴ Maximum fluctuation of energy

ΔE = Maximum energy − Minimum energy

$= \left[(E + a_1) - (E + a_1 - a_2 + a_3 - a_4)\right]$

$= a_2 - a_3 + a_4$

D-ALEMBERT'S PRINCIPLE

Consider a rigid body acted upon by a system of forces. The system may be reduced to a single resultant force acting on the body whose magnitude is given by the product of the mass of the body and the linear acceleration of the centre of mass of the body. Remember, $F = ma$ (from Newton's second law of motion). It may be re-written as $F - ma = 0$. If the quantity $(-ma)$ is treated as a force, equal, opposite and with the same line of action as the resultant force F, and include this force with the system of forces of which "F" is the resultant, then, the complete system of forces will be in equilibrium. This principle is known as D-Alembert's principle. The equal and opposite force "$-ma$" is known as reversed effective force or the inertia force i.e F_1, then, the equation above becomes $F + F_1 = 0$. D-Alembert's principle states that, the resultant force acting on a body together with the reversed effective force (or inertia force) are in equilibrium.

- This principle is used to reduce a dynamic problem into an equivalent static problem.

Example I: The mass of flywheel of an engine is 6.5 tonnes and the radius of gyration is 1.8m. It is found from the turning moment diagram that the fluctuation of energy is 560KNM. If mean speed of the engine is 120rpm. Find the maximum and minimum speeds.

Given: $m = 6.5$ tonnes $= 6500kg$, $K = 1.8m$, $\Delta E = 560KNm$

$N = 120rpm$.

Let N_1 and N_2 = Max. and Min. speed respectively.

$$\Delta E = 560 \times 10^3 = \frac{\pi^2}{90} \times mK_1^2 N [N_1 - N_2].$$

$$= 27715 [N_1 - N_2]$$

$$\therefore N_1 - N_2 = 2rpm \quad ---------- (1)$$

$$N = \frac{N_1 + N_2}{2} \quad ; \quad 120 = \frac{N_1 + N_2}{2}$$

$\therefore N_1 + N_2 = 240\,rpm$ ---------- (2)

Solve eqn (1) & (2)

$N_1 = 121\,rpm, \quad N_2 = 119\,rpm.$

Example II

The flywheel of a Steam Engine has a radius of gyration of 1m and mass of 2500kg. The Starting torque of the Steam Engine is 1500Nm and may be assumed Constant. Determine.

(i) The Angular acceleration of the flywheel

(ii) The kinetic Energy of the flywheel after 10 Sec. from the Start.

Given

$K = 1m, \quad M = 2500kg, \quad T = 1500\,Nm,$

(i) The Angular acceleration of the flywheel, α

$I = MK^2 = 2500 \times 1^2 = 2500\,kg\,m^2$

Starting torque of the Engine, T

$$T = I\alpha, \quad = 1500 = 2500\,\alpha$$

$$\alpha = \frac{1500}{2500} = 0.6\,rad/s^2$$

(ii) The Kinetic Energy of flywheel after 10 sec. from the Start

$K.E = \frac{1}{2} I \omega_2^2$

$\omega_2 = \omega_1 + \alpha t = 0 + 0.6 \times 10 = 6\,rad/s.$

$K.E = \frac{1}{2} I \omega_2^2 = \frac{1}{2} \times 2500 \times 6^2 = 45\,kNm$

Example III A horizontal cross Compound Steam Engine develops 300kW at 90rpm. The Co-efficient of fluctuation of Energy as found from the turning moment diagram is to be 0.1 and the speed is to be kept within 0.5% of the mean speed. Find the ~~mean~~ mass of the flywheel required, if the radius of gyration is 2m.

Given: $P = 300kW$, $N = 90rpm$, $C_E = 0.1$, $k = 2m$,

$\omega = \frac{2\pi N}{60} = \frac{2\pi \times 90}{60} = 9.426$ rad/s.

Let ω_1 and ω_2 = Max. and Min. speed respectively, since the fluctuation of speed is 0.5% of mean speed

∴ Total fluctuation of speed, $\omega_1 - \omega_2 = 1\% \omega = 0.01\omega$

Co-efficient of fluctuation of speed, C_s.

$$C_s = \frac{\omega_1 - \omega_2}{\omega} = 0.01$$

Work done per cycle $= P \times \frac{60}{N} = 300 \times 10^3 \times \frac{60}{90} = 200 \times 10^3 Nm$

∴ Maximum fluctuation of energy, ΔE

ΔE = Work done per cycle $\times C_E$

$= 200 \times 10^3 \times 0.1$

$= 20 \times 10^3 Nm$.

Let m = Mass of flywheel

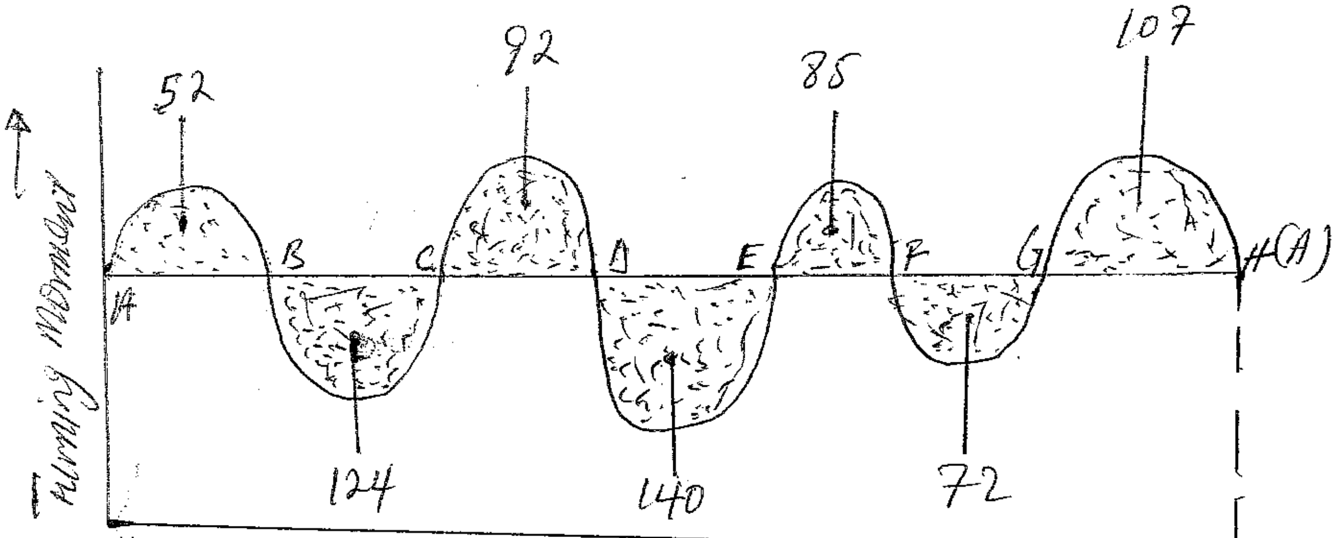
Also $\Delta E = 20 \times 10^3 = mk^2\omega^2 C_s$

$20 \times 10^3 = m \times 2^2 (9.426)^2 \times 0.01$

∴ $m = 563.0 kg$//

Example IV: The turning moment diagram for a multicylinder engine has been drawn to a scale 1mm = 600Nm vertically and 1mm = 3° horizontally. The intercepted areas between the output torque curve and mean resistance line taken in order from one end are as follows, +52, −124, +92, −140, +85, −72 and +107mm², when the engine is running at a speed of 600rpm. If the total fluctuation of speed is not to exceed ±1.5% of the mean, find the necessary mass of the flywheel of radius 0.5m.

Given: $N = 600$ rpm or $\omega = \frac{2\pi \times 600}{60} = 62.842$ rad/s
$R = 0.5$ m



Since the total fluctuation of speed is not to exceed $\pm 1.5\%$ of the mean speed.

$\therefore \omega_1 - \omega_2 = 3\% \omega = 0.03\omega$

Co-efficient of fluctuation of speed, $C_s = \frac{\omega_1 - \omega_2}{\omega} = 0.03$

The turning moment scale is 1mm = 600Nm

Crank angle scale is $1\text{mm} = 3^0 = 3 \times \pi/180 = \frac{\pi}{60}$ rad

1mm^2 on the turning moment diagram $= 600 \times \pi/60 = 31.42$ Nm

Let the total energy at $A = E$

Energy at $B = E + 52$

Energy at $C = E + 52 - 124 = E - 72$

Energy at $D = E - 72 + 92 = E + 20$

Energy at $E = E + 20 - 140 = E - 120$

Energy at $F = E - 120 + 85 = E - 35$

Energy at $G = E - 35 - 72 = E - 107$

Energy at $H = E - 107 + 107 = E =$ Energy at A

The maximum fluctuation of energy

$\Delta E =$ Max. Energy $-$ Minimum Energy

$= (E+52) - (E-120) = 172, \quad = 172 \times 31.42$
$= 5404\,NM$

m = Mass of the flywheel in kg

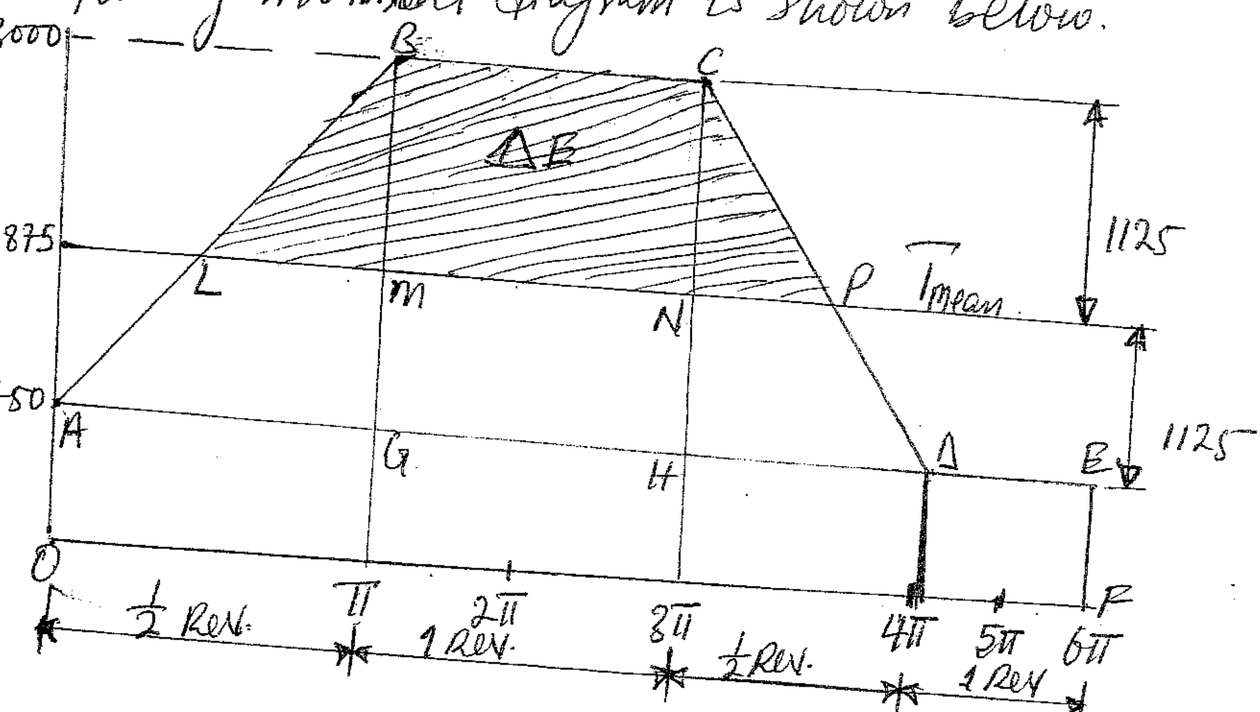
$\therefore \Delta E = 5404 = mR^2\omega^2 C_s = m \times (0.5)^2 \times (62.84)^2 \times 0.03$
$= 29.6m$

$\therefore \quad m = 183\,kg$

Example V A shaft is fitted with a flywheel rotates at 250rpm and drives a machine. The torque of machine varies in a cyclic manner over a period of 3 revolutions. The torque rises from 750 Nm to 3000 Nm uniformly during ½ revolution and remains constant for one revolution, the cycle being repeated thereafter. Determine the power required to drive the machine and percentage fluctuation in speed, if the driving torque applied to the shaft is constant and the mass of the flywheel is 500kg with radius of gyration of 600mm.

Given. $N = 250\,rpm$, or $\omega = \frac{2\pi \times 250}{60} = 26.2\,rad/s$, $m = 500\,kg$
$k = 600mm = 0.6m$

The turning moment diagram is shown below.



The torque required for one complete cycle
= Area of figure OABCDEF

$= \text{Area } OAEF + \text{Area } ABG + \text{Area } BCHG + \text{Area } CDH$

$= (OF \times OA) + (\frac{1}{2} AG \times BG) + (GH \times CH) + (\frac{1}{2} HD \times CH)$

$= (6\pi \times 750) + (\frac{1}{2} \times \pi (3000 - 750)) + [2\pi (3000 - 750)] + (\frac{1}{2} \pi (3000 - 750))$

$= 11250\pi$ Nm

If T_{mean} is the mean torque in Nm

then, torque required for one complete cycle

$= T_{mean} \times 6\pi$ Nm

$$T_{mean} = \frac{11250\pi}{6\pi} = 1875 \text{ Nm}$$

Power required to drive the machine

$P = T_{mean} \times \omega = 1875 \times 26.2 = 49125$ W

Co-efficient of fluctuation of speed, C_s.

lets find the values of LM and NP

from similar triangles ABG and BLM,

$$\frac{LM}{AG} = \frac{BM}{BG} \text{ or } \frac{LM}{\pi} = \frac{3000 - 1875}{3000 - 750} = 0.5$$

$\therefore LM = 0.5\pi$

From similar triangles CHD and CNP

$$\frac{NP}{HD} = \frac{CN}{CH} \text{ or } \frac{NP}{\pi} = \frac{3000 - 1875}{3000 - 750}, \quad NP = 0.5\pi$$

$BM = CN = 3000 - 1875 = 1125$ Nm.

Since, the area above the mean torque line represents the maximum fluctuation of energy,

$\therefore$ maximum fluctuation of energy.

$\Delta E = \text{Area } LBCP = \text{Area } LBM + \text{Area } MBCN + \text{Area } PNC$

$= (\frac{1}{2} \times LM \times BM) + (MN \times BM) + (\frac{1}{2} \times NP \times CN)$

$= (\frac{1}{2} \times 0.5\pi \times 1125) + (2\pi \times 1125) + (\frac{1}{2} \times 0.5\pi \times 1125)$

$= 8837$ Nm

Now, the co-efficient of fluctuation of speed.

$$\Delta E = 8837 = mk^2\omega^2 C_s$$
$$= 500(0.6)^2 \times (26.2)^2 \times C_s$$
$$C_s = 0.071$$

VIBRATION

When the forces on an individual part are such that, the displacement of the mass centre of the part is oscillating or periodically reversing in sense, it is said to be vibrating. All matter (solid, liquid and gaseous) are capable of vibration. Vibration of gases occurs in tail ducts of jet engines causing troublesome noise, sometimes fatigue cracks in the metal. Vibration in liquid is almost always longitudinal and can cause large forces because of the low compressibility of liquids. A typical machine has many moving parts each of which is a potential sources of vibration or shock-excitation. Designers face the problem of compromising between an acceptable amount of vibration, ~~and~~ noise and costs involved in reducing reduction.

Reciprocating inter = static

When elastic bodies such as a spring, a beam and a shaft are displaced from the equilibrium position by the application of external forces and then released, they execute a vibratory motion. This is due to the ~~release~~ reason that, when a body is displaced, the internal forces in the form of elastic or strain energy are present in the body. At release, these forces brings the body to its original position. When the body reaches the equilibrium position, the whole of the elastic or strain energy is converted into K.E due to which the body continues to move in the opposite direction. The whole of the K.E is converted into strain energy due to which the body again returns to the equilibrium position. In this way, the vibratory motion is repeated indefinitely.

Every object has a natural frequency of vibration in each of the three coordinate axes directions and also a rotational sense about each of these axes. There are a total of six degrees of freedom or modes of vibration.

DEFINATION OF TERMS.

1. AMPLITUDE: It is the maximum displacement of a body from its mean position. It is always equal to the radius of a circle. Or it is the maximum displacement of any part of the system from its static equilibrium position.

2. CYCLE: It is defined as motion completed in any one period of time or the sequence of values of a periodic quantity throughout a complete period.

3. PERIOD: It is the time taken for one complete revolution of the particle. $t_p = \frac{2\pi}{\omega}$, if $a = \omega^2 x$, or $\omega^2 = \frac{a}{x}$ or $\omega = \sqrt{\frac{a}{x}}$

$\therefore t_p = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{x}{a}} = 2\pi\sqrt{\frac{\text{Displacement}}{\text{Acceleration}}}$ Seconds.

4. FREQUENCY: It is the number of cycles per second and is the reciprocal of time period.

$$\text{Frequency, } n = \frac{\omega}{2\pi} = \frac{1}{t_p} = \frac{1}{2\pi}\sqrt{\frac{a}{x}} \text{ Hz}$$

Or, it is the number of cycles per unit of time.

5 PHASE ANGLE: This is the angle between two vectors representing two harmonically varying quantities which have the same frequency. It is the difference in phase measurement as an angle

6, DEGREE OF FREEDOM: This is the minimum number of co-ordinates which are required to specify the possible motion of a mechanical or other system. Or it is the number of independent co-ordinates necessary to describe the motion of a system. If the motion of a body is constrained to vibrate or oscillate in only one manner or mode, it is said to have a single degre of freedom. e.g pendulum, helical spring e.t.c whereas (S. a body moving in space e.g. a cricket ball, satellite e.t. has six degrees of freedom (3 translational and 3 rotational

7, DAMPING RATIO: Damping is the dissipation of energy from a vibrating system and thus prevent excessive response. Damping ratio is the ratio of the damping co-efficient to the critical damping co-efficient. Note that, critical damping co-efficient is the smallest value of the damping co-efficient required to prevent vibration while the damping co-efficient is the constant co-efficient of the velocity term $\dot{x}$ in a motion defined by the differential equation $m\ddot{x} + \dot{x} + kx = 0$ where m is the mass and k is the stiffness of the system.

8, LOG - DECREMENT: The logarithmic decrement is the natural logarithm of the ratio of the peak values of the amplitudes of a vibration in successive cycles when the amplitudes ~~of a vibration~~ decrease exponentially.

9, RESONANCE: If the frequency of excitation ~~coin~~ coincides with one of the natural frequencies of the system, a condition of resonance is encountered and dangerously large oscillations may result. The failure of major structures such as bridges, buildings, or airplane wings is an awesome possibility under resonance.

TYPES OF VIBRATORY MOTION

1. FREE OR NATURAL VIBRATIONS: Free vibration takes place when a system oscillates under the action of forces inherent in the system itself, and when external impressed forces are absent. The system under free vibration will vibrate at one ~~of~~ or more of its natural frequencies which are properties of the dynamic system established by its mass and stiffness distribution

2. FORCE VIBRATIONS: It takes place under the excitation of external forces. The external force applied to the body is

a periodic disturbing force created by unbalance. (53)
The vibrations have the same frequency as the applied force.

3. DAMPED VIBRATIONS: The motion is said to be damped vibration, when there is a reduction in amplitude over every cycle of vibration. This is due to the fact that a certain amount of energy possessed by the vibrating system is always dissipated in overcoming frictional resistances to the motion.

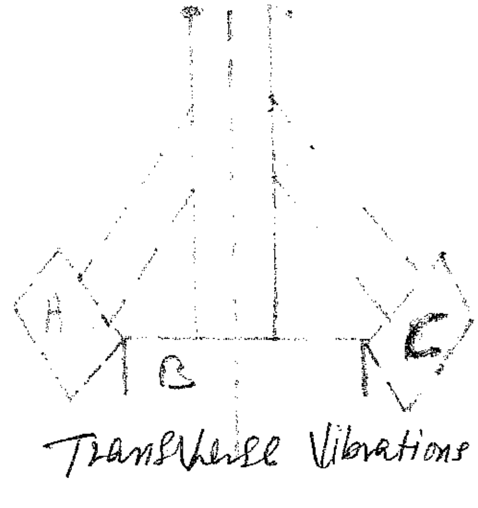
THREE TYPES OF FREE VIBRATION

i) LONGITUDINAL VIBRATION: This occurs when the particles of the shaft or disc moves parallel to the axis of the shaft. In this case, ~~case~~, the shaft is elongated and shortened alternately and thus the tensile and compressive stresses are induced alternately in the shaft.

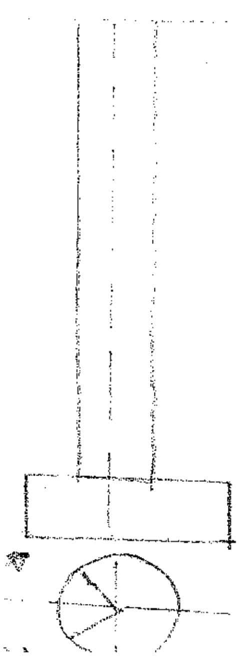
ii) TRANSVERSE VIBRATIONS: This occurs when the particles of the shaft or disc move approximately perpendicular to the axis of the shaft. In this case, the shaft is straight and bent alternately and bending stresses are induced in the shaft.

iii) TORSIONAL VIBRATIONS: This occurs when the particles of the shaft or disc moves in a circle about the axis of the shaft. In this case, the shaft is twisted and untwisted alternately and the torsional shear stresses are induced in the shaft.

Longitudinal Vibrations.

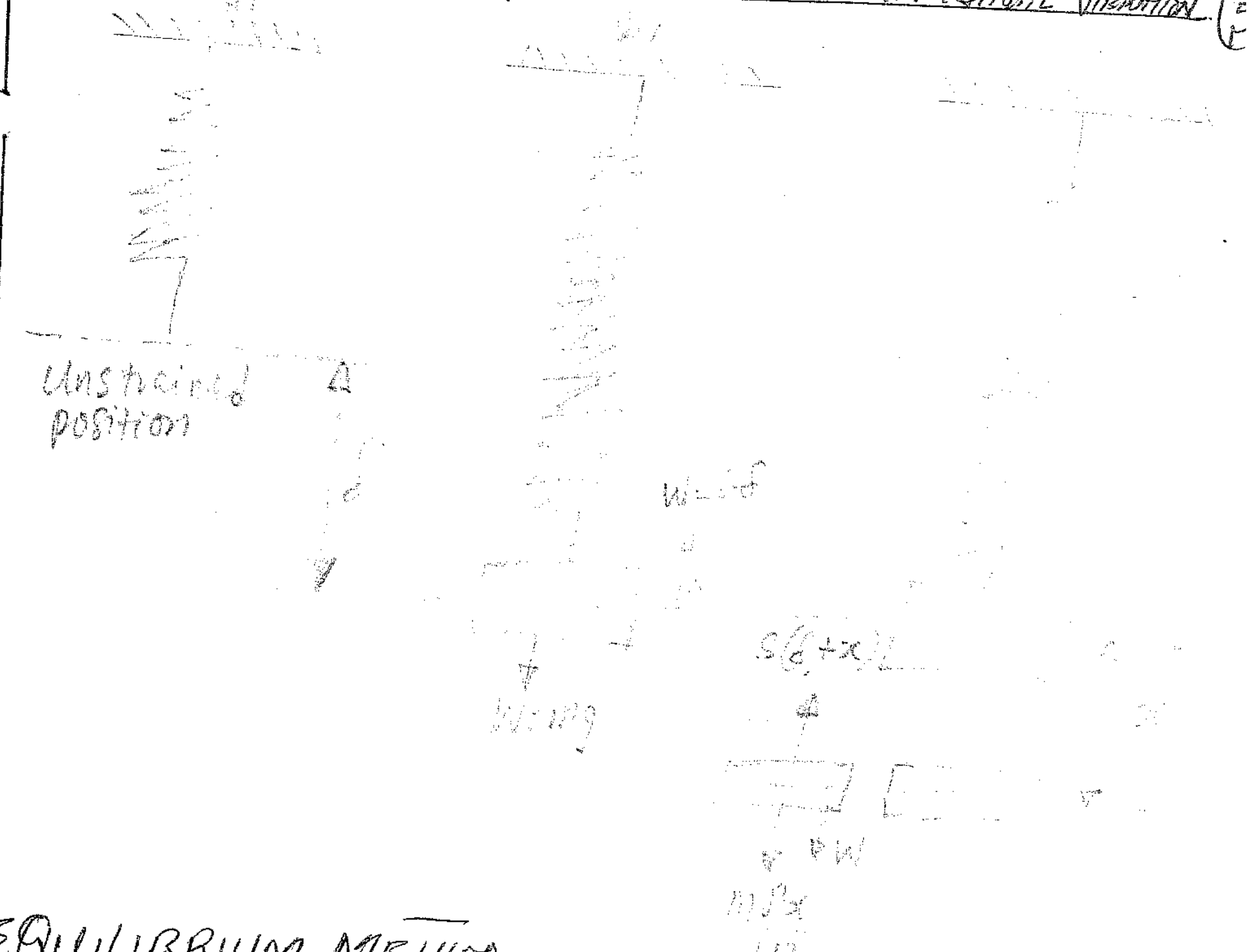


Transverse Vibrations



Torsional Vibrations

NATURAL FREQUENCY OF FREE LONGITUDINAL VIBRATION



EQUILIBRIUM METHOD

Consider a constraint (spring) of negligible mass in an unstrained position.

Let:

S = Stiffness of the constraint. It is the force required to produce unit displacement in the direction of vibration. (N/m)

m = Mass of the body suspended from the constraint in kg.

W = Weight of the body in Newtons $= m \cdot g$

δ = Static deflection of the spring in metres due to weight W Newtons.

x = Displacement given to the body by the external force in metres.

In the equilibrium position, the gravitational pull $W = mg$ is balanced by a force of spring such that $W = S \cdot \delta$ (as shown in (b)). In (c), since the mass is now displaced from its equilibrium position by a distance x and is then released after time t,

Restoring force $= W - s(\delta + x) = W - s\delta - sx$

$= s\delta - s\delta - sx = -sx \qquad W = s\delta$ ------(1)

(Taken upward force as $-ve$)

Accelerating force = Mass × Acceleration

$= m \times \frac{d^2x}{dt^2}$ ---------- (2) (Taking downwards force as $+ve$)

Equating eqn (1) and (2)

the eqn of motion of the body of mass m after time t

$m \times \frac{d^2x}{dt^2} = -sx$ or $m \times \frac{d^2x}{dt^2} + sx = 0$

or $\frac{d^2x}{dt^2} + \frac{sx}{m} = 0$ ---------- (3)

We know that the fundamental equation of simple harmonic motion.

$\frac{d^2x}{dt^2} + \omega^2 x = 0$ ---------- (4)

Comparing eqn (3) and (4)

$\omega = \sqrt{s/m}$

∴ Time period, $t_p = \frac{2\pi}{\omega} = 2\pi\sqrt{m/s}$

Natural frequency, $f_n = \frac{1}{t_p} = \frac{1}{2\pi}\sqrt{s/m} = \frac{1}{2\pi}\sqrt{g/\delta}$ $(mg = s\delta)$

Taking the value of g as $9.81\, m/s^2$ and δ in metres

$f_n = \frac{1}{2\pi}\sqrt{\frac{9.81}{\delta}} = \frac{0.4985}{\sqrt{\delta}}$ Hz

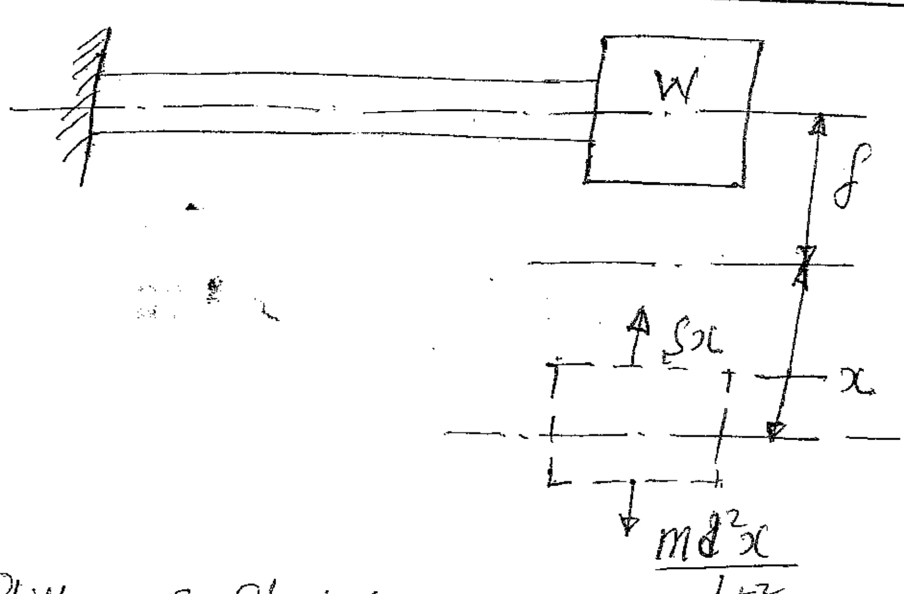
The value for deflection "δ" may be obtained, for longitudinal ~~For~~ Vibrations.

$\frac{\delta}{\varepsilon} = E$ or $\frac{W}{A} \times \frac{L}{\delta} = E$

strain — ε

∴ $\delta = \frac{WL}{EA}$

NATURAL FREQUENCY OF FREE TRANSVERSE VIBRATIONS



S = Stiffness of Shaft
δ = Static deflection due to weight of the body
x = Displacement of body from mean position after time t
m = Mass of the body = W/g.

Consider a shaft of negligible mass, whose one end is fixed and the other end carries a body of weight W.

Restoring force $= -Sx$ ---(1)

Accelerating force $= \frac{md^2x}{dt^2}$ ---(2)

Equating eqn (1) & (2)

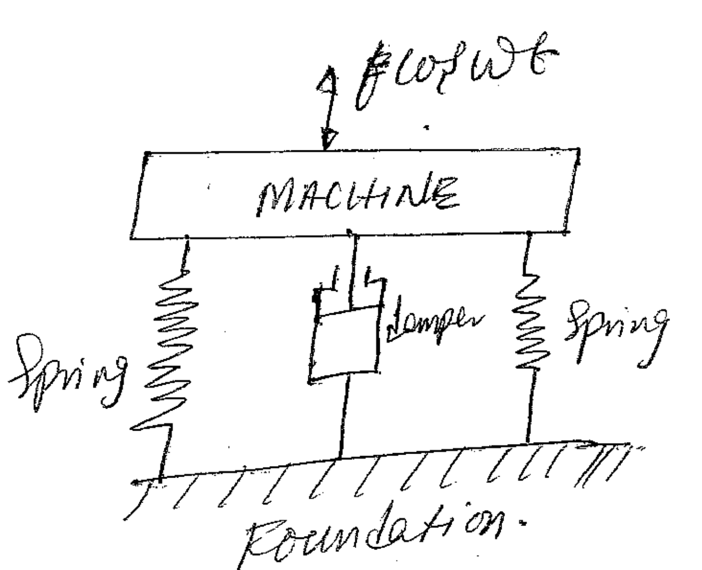
$$\frac{md^2x}{dt^2} = -Sx$$

$$\frac{d^2x}{dt^2} + \frac{S}{m}x = 0$$

Time period $t_p = 2\pi\sqrt{\frac{m}{S}}$

Natural frequency $f_n = \frac{1}{t_p} = \frac{1}{2\pi}\sqrt{\frac{S}{m}} = \frac{1}{2\pi}\sqrt{g/\delta}$

VIBRATION ISOLATION AND TRANSMISSIBILITY



A little consideration will show that when an unbalanced machine is installed on the foundation, it produces vibration in the foundation. In order to prevent these vibrations or to minimise the transmission of forces to the foundation, the ~~mass~~ machines are mounted on a springs and dampers or some vibration isolating material. The arrangement is assumed to have one degree of freedom i.e it can move up and down only.

It may be noted that, when a periodic (i.e simple harmonic) disturbing force $F\cos\omega t$ is applied to a machine of mass m supported by a spring of stiffness S, then the force is transmitted by means of the spring and the damper or dashpot to the fixed support or foundation. The ratio of the force transmitted (F_T) to the force applied (F) is known as isolation factor or transmissibility ratio of the spring support.

The force transmitted to the foundation consists of the following two forces.

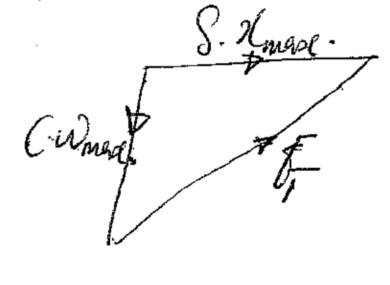
(i) Spring force or elastic force which is equal to $S \cdot x_{max}$

(ii) Damping force which is equal to $C \cdot \omega \cdot x_{max}$.

Since these two forces are perpendicular to one another, therefore the force transmitted F_T

$$F_T = \sqrt{(S \cdot x_{max})^2 + (C \cdot \omega \cdot x_{max})^2}$$

$$= x_{max}\sqrt{S^2 + C^2\omega^2}$$



Transmissibility ratio $\mathcal{E} = \frac{F_T}{F} = \frac{x_{max}\sqrt{S^2 + C^2\omega^2}}{F}$

We know that

$x_{max} = x_0 \times \Delta = \frac{F}{S} \times \Delta$ $\boxed{x_0 = F/S}$

$$\mathcal{E} = \frac{\Delta}{S}\sqrt{S^2 + C^2\omega^2} = \Delta\sqrt{1 + \frac{C^2\omega^2}{S^2}}$$

$$= \Delta\sqrt{1 + \left[\frac{2C}{C_c} \times \frac{\omega}{\omega_n}\right]^2}$$ $\boxed{\frac{C \times \omega}{S} = \frac{2C}{C_c} \times \frac{\omega}{\omega_n}}$

We know that, the magnification factor Δ

$$\Delta = \frac{1}{\sqrt{\left(\frac{2C\omega}{C_c\omega_n}\right)^2 + \left[1 - \frac{\omega^2}{(\omega_n)^2}\right]^2}}$$

C = Damping coefficient
C_c = Critical damping coefficient
ω_n = Natural circular frequency Vibration of the machine.
x_{max} = Maximum amplitude
ω = Circular frequency.

$$\mathcal{E} = \frac{\sqrt{1 + \left(\frac{2C\omega}{C_c\omega_n}\right)^2}}{\sqrt{\left(\frac{2C\omega}{C_c\omega_n}\right)^2 + \left(1 - \frac{\omega^2}{\omega_n^2}\right)^2}}$$

When the damper is not provided, then $C = 0$ and

$$\mathcal{E} = \frac{1}{\left(\frac{\omega}{\omega_n}\right)^2 - 1}$$

Example I A Cantilever Shaft 50mm diameter and 300mm long has a disc of mass 100kg at its free end. The young modulus for the shaft material is $200\ GN/m^2$. Determine the frequency of longitudinal and transverse vibrations of the shaft.

Given: $d = 50mm = 0.05m$, $L = 300mm = 0.3m$, $M = 100kg$,

$E = 200\ GN/m^2 = 200 \times 10^9\ N/m^2$

$A = \frac{\pi d^2}{4} = \frac{\pi}{4}(0.05)^2 = 1.96 \times 10^{-3}\ m^2$

P.T.O.

Moment of inertia of the shaft I

$$I = \frac{\pi d^4}{64} = \frac{\pi}{64}(0.05)^4 = 0.3 \times 10^{-6}\, m$$

Static deflection of the shaft, δ

$$\delta = \frac{WL}{AE} = \frac{100 \times 9.81 \times 0.3}{1.96 \times 10^{-3} \times 200 \times 10^9}$$

$$\delta = 0.751 \times 10^{-6}\, m$$

∴ Frequency of longitudinal vibration, f_n

$$f_n = \frac{0.4985}{\sqrt{\delta}} = \frac{0.4985}{\sqrt{0.751 \times 10^{-6}}} = 575\, KHz.$$

To find the frequency of transverse vibration f_n

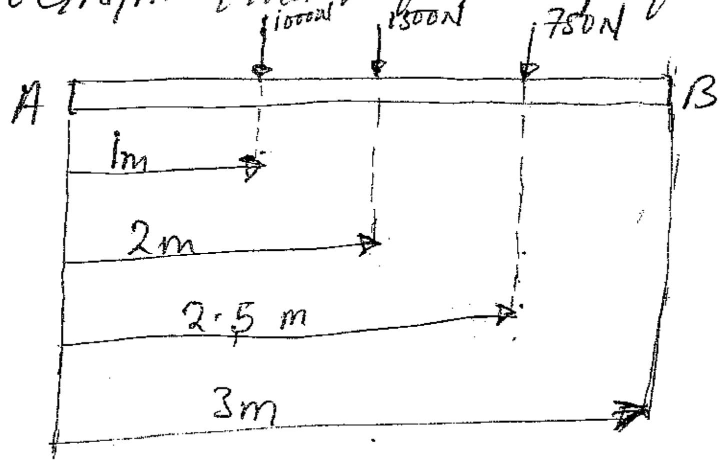
Static deflection of the shaft, $\delta = \frac{WL^3}{3EI}$

$$= \frac{100 \times 9.81 \times (0.3)^3}{3 \times 200 \times 10^9 \times 0.3 \times 10^{-6}} = 0.147 \times 10^{-3}\, m$$

Frequency of transverse vibration, f_n

$$f_n = \frac{0.4985}{\sqrt{\delta}} = \frac{0.4985}{\sqrt{0.147 \times 10^{-3}}} = 41 Hz$$

Example II A shaft 50mm diameter and 3 metres long is simply supported at the ends and carries three loads of 1000N, 1500N and 750N at 1m, 2m and 2.5m from the left support. Take $E = 200\, GN/m^2$. Find the frequency of transverse vibration.



$$I = \frac{\pi d^4}{64} = \frac{\pi (0.05)^4}{64} = 0.307\times10^{-6}\,m$$

Static deflection due to point Load, $\delta = \frac{Wa^2b^2}{3EIL}$

Static deflection due to load 1000N, δ_1

$$\delta_1 = \frac{1000\times1^2\times2^2}{3\times200\times10^9\times0.307\times10^{-6}\times3} = 7.24\times10^{-3}\,m$$

Static deflection due to load of 1500N, δ_2

$$\delta_2 = \frac{1500\times2^2\times1^2}{3\times200\times10^9\times0.307\times10^{-6}\times3} = 1.086\times10^{-2}\,m$$

Static deflection due to load of 750N, δ_3

$$\delta_3 = \frac{750(2.5)^2\times(0.5)^2}{3\times200\times10^9\times0.307\times10^{-6}\times3} = 2.12\times10^{-3}\,m$$

Frequency of transverse Vibration, f_n

$$f_n = \frac{0.4985}{\sqrt{\delta_1+\delta_2+\delta_3}} = \frac{0.4985}{\sqrt{7.24\times10^{-3}+1.086\times10^{-2}+2.12\times10^{-3}}} = 21.8\,Hz$$

Example III A Vertical Shaft 5mm diameter is 200mm long and is Supported in long bearings at its ends. A disc of mass 50kg is attached to the Centre of the Shaft. Neglecting any increase in Stiffness due to attachment of the disc to the Shaft. Find the Critical Speed of rotation and the Maximum bending Stress when the Shaft is rotating at 75% of the Critical Speed. The Centre of the disc is 0.25mm from the geometric axis of the Shaft. $E = 200\,GN/m^2$

$I = \frac{\pi d^4}{64} = \frac{\pi}{64}(0.005)^4 = \cancel{30.7\times10^{-2}\ m^4} = 3.07\times10^{-11}\,m^4$

Since the shaft is supported in long bearings, it is assumed to be fixed at both ends.

$\delta = \frac{WL^3}{192EI} = \frac{50\times9.81\times(0.2)^3}{192\times200\times10^9\times30.7\times10^{-12}} = 3.33\times10^{-3}\,m$

Critical speed of rotation or natural frequency of transverse vibrations $N_c = \frac{0.4985}{\sqrt{3.33\times10^{-3}}} = \cancel{8.64}$ rps (rotation per sec.) 273 rps

N = Speed of the shaft = 75% of critical speed = $0.75 N_c$

When the shaft starts rotating, the additional dynamic load (W_1) to which the shaft is subjected may be obtained by using the bending equation $\frac{M}{I} = \frac{\sigma}{y_1}$ or $M = \frac{\sigma I}{y_1}$

For a shaft fixed at both ends and carrying a point load (W_1) at the centre, the max. bending moment, $M = W_1 L/8$

$\therefore \frac{W_1 L}{8} = \frac{\sigma I}{d/2}$

$W_1 = \frac{\sigma I\times8}{d/2\times L} = \frac{\sigma\times\overset{3.07\times10^{-11}}{\cancel{30.7\times10^{-12}}}\times8}{0.005/2\times0.2} = \cancel{0.49\times10^{-6}\,\sigma\,N}\ 4.9\times10^{-7}\,\sigma\,N$

$\therefore$ Additional deflection due to load W_1

$y = \frac{W_1}{W}\times\delta = \frac{4.9\times10^{-7}\,\sigma}{50\times9.81}\times3.33\times10^{-3} = 3.327\times10^{-12}\,\sigma$

Also $y = \frac{\pm e}{\left(\frac{\omega_c}{\omega}\right)^2-1} = \frac{\pm e}{\left(\frac{N_c}{N}\right)^2-1}$ equate y to each other

$3.327\times10^{-12}\,\sigma = \frac{\pm0.25\times10^{-3}}{\left(\frac{N_c}{0.75N_c}\right)^2-1}$, $\therefore\ \sigma = \frac{0.32\times10^{-3}}{3.327\times10^{-12}}$

$= 96.1\ MN/m^2$

Substituting $\omega_c = N_c$ & $\omega = N$

Example IV The following data are given for a vibratory system with viscous damping, mass = 2.5 kg, Spring constant = 3×10^3 N/m and the amplitude decreases to 0.25 of the initial value after five consecutive cycles. Determine the damping co-efficient of the damper in the system. 6

Given, $m = 2.5$ kg, $S = 3 \times 10^3$ N/m $= 3000$ N/m, $x_6 = 0.25 x_1$

Natural circular frequency of vibration ω_n

$$\omega_n = \sqrt{S/m} = \sqrt{\frac{3000}{2.5}} = 34.64 \text{ rad/s}.$$

let C = damping co-efficient of the damper in N/m/s

x_1 = Initial amplitude

x_6 = Final amplitude after five consecutive cycles $= 0.25 x_1$

We know that

$$\frac{x_1}{x_2} = \frac{x_2}{x_3} = \frac{x_3}{x_4} = \frac{x_4}{x_5} = \frac{x_5}{x_6}$$

$$\frac{x_1}{x_6} = \frac{x_1}{x_2} \times \frac{x_2}{x_3} \times \frac{x_3}{x_4} \times \frac{x_4}{x_5} \times \frac{x_5}{x_6} = \left(\frac{x_1}{x_2}\right)^5$$

$$\therefore \frac{x_1}{x_2} = \left(\frac{x_1}{x_6}\right)^{\frac{1}{5}} = \left(\frac{x_1}{0.25 x_1}\right)^{\frac{1}{5}} = (4)^{\frac{1}{5}} = 1.32$$

Known that,

$$\log_e\left(\frac{x_1}{x_2}\right) = a \times \frac{2\pi}{\sqrt{(\omega_n)^2 - a^2}}$$

$$\log_e(1.32) = a \times \frac{2\pi}{\sqrt{(34.64)^2 - a^2}}$$

$$0.2776 = \frac{a \times 2\pi}{\sqrt{1200 - a^2}}$$ Squaring both sides

$$0.077 = \frac{39.5 a^2}{1200 - a^2} \quad \text{or} \quad a = 1.53$$

Since $a = \frac{C}{2m}$ or $C = a \times 2m = 1.53 \times 2 \times 2.5 = 7.65$ N/m/s.

Example V

An instrument vibrates with a frequency of 1 Hz when there is no damping. When the damping is provided, the frequency of damped vibrations was observed to be 0.9 Hz. Find the damping factor and logarithmic decrement.

Let m = mass of the instrument in kg.

c = Damping co-efficient or damping force/unit velocity in N/m/s.

c_c = Critical damping co-efficient in N/m/s.

Natural frequency of undamped vibrations, ω_n

$\omega_n = 2\pi f_n = 2\pi \times 1 = 6.28\cancel{3}$ rad/s.

Circular frequency of damped vibrations, ω_d

$\omega_d = 2\pi f_d = 2\pi \times 0.9 = 5.6\cancel{5}$ rad/s.

Also $\omega_d = \sqrt{\omega_n^2 - a^2}$, $\quad 5.66 = \sqrt{(6.284)^2 - a^2}$

Square both sides,

$$(5.66)^2 = \left(\sqrt{(6.284)^2 - a^2}\right)^2$$

$$(5.66)^2 = (6.284)^2 - a^2, \quad a = 2.7\cancel{6}$$

$$a = \frac{c}{2m} \text{ or } c = a \times 2m = 2.7\cancel{6} \times 2m = 5.48m \text{ N/m/s.}$$

$$c_c = 2m\omega_n = 2m \times 6.284 = 12.568m$$

$$\therefore \text{Damping factor} = \frac{c}{c_c} = \frac{5.48m}{12.568m} = 0.4\cancel{48}$$

Logarithmic decrement, δ

$$\delta = \frac{2\pi c}{\sqrt{(c_c)^2 - c^2}} = \frac{2\pi \times 5.48m}{\sqrt{(12.568m)^2 - (5.48m)^2}} = 3.04$$

Example VI A machine has a mass of 100kg and unbalanced reciprocating parts of mass 2kg which moves through a vertical stroke of 80mm with simple harmonic motion. The machine is mounted on four springs symmetrically arranged with respect to centre of mass, in such a way that the machine has one degree of freedom and can undergo vertical displacement only. Neglecting damping calculate the combined stiffness of the spring in order that the force transmitted to the foundation is $\frac{1}{25}$th of the applied force, when the speed of rotation of machine crankshaft is 1000rpm. When the machine is actually supported on the springs, it is found that the damping reduces the amplitude of the successive free vibrations by 25%. Find the force transmitted to the foundation at 1000rpm, the force transmitted to the foundation at resonance and amplitude of the forced vibration of the machine at resonance.

Given, $M_1 = 100kg$, $M_2 = 2kg$, $L = 80mm = 0.08M$, $\varepsilon = \frac{1}{25}$

$N = 1000rpm$ or $\omega = 2\pi N/60 = 104.7 rad/s$.

Let, S = Combined stiffness of springs in N/m

ω_n = Natural circular frequency of vibration of the machine in rad/s.

Transmissibility ratio, ε

$$\varepsilon = \frac{1}{25} = \frac{1}{\left(\frac{\omega}{\omega_n}\right)^2 - 1} = \frac{(\omega_n)^2}{\omega^2 - \omega_n^2} = \frac{\omega_n^2}{(104.7)^2 - (\omega_n)^2}$$

$$(104.7)^2 - (\omega_n)^2 = 25(\omega_n)^2, \quad \omega_n = 20.5 rad/s.$$

$$\text{Also } \omega_n = \sqrt{\frac{S}{M_1}} \quad \text{or } S = M_1(\omega_n)^2 = 100 \times 421.6 = 42160 N/m$$

$$\therefore S = 42160 N/m \text{ — combine stiffness}$$

Let, F_T = Force transmitted to the foundation at 1000rpm

x_1 = Initial amplitude of vibration.

Since, the damping reduces the amplitude of successive free vibrations by 25%, $\therefore$ final amplitude of vibration, $x_2 = 0.75x_1$

$$\log_e\left(\frac{x_1}{x_2}\right) = \frac{Q \times 2\pi}{\sqrt{(\omega_n)^2 - Q^2}}, \quad \log_e\left(\frac{x_1}{0.75x_1}\right) = \frac{Q \times 2\pi}{\sqrt{421.6 - Q^2}}$$ (65-66)

Square both side

$$(0.2877)^2 = \frac{Q^2 \times 4\pi^2}{421.6 - Q^2}$$

$$0.083 = \frac{39.5Q^2}{421.6 - Q^2} \qquad \left(\log_e \frac{1}{0.75} = \log_e 1.333 = 0.2877\right)$$

$$Q = 0.94$$

Damping Co-efficient or damping force per unit Velocity

$$C = Q \times 2m_1 = 0.94 \times 2 \times 100 = 188\,N/m/s$$

Critical damping co-efficient, C_c

$$C_c = 2m\omega_n = 2 \times 100 \times 20.5 = 4100\,N/m/s.$$

∴ Actual Value of transmissibility ratio.

$$\varepsilon = \frac{\sqrt{1 + \left(\frac{2c\omega}{c_c\omega_n}\right)^2}}{\sqrt{\left(\frac{2c\omega}{c_c\omega_n}\right)^2 + \left(1 - \frac{\omega^2}{(\omega_n)^2}\right)^2}} = \frac{\sqrt{1 + \left(\frac{2 \times 188 \times 104.7}{4100 \times 20.5}\right)^2}}{\sqrt{\left(\frac{2 \times 188 \times 104.7}{4100 \times 20.5}\right)^2 + \left[1 - \left(\frac{104.7}{20.5}\right)^2\right]^2}}$$

$$= \frac{\sqrt{1 + 0.22}}{\sqrt{0.22 + 0.629}} = \frac{1.104}{25.08} = 0.044$$

The maximum unbalanced force on the machine due to reciprocating parts.

$$F = m_2\omega^2 r = 2(104.7)^2 \times (0.08)/2 = 877\,N$$

∴ Force transmitted to foundation, F_T

$$F_T = \varepsilon F = 0.044 \times 877 = 38.6\,N.$$

To calculate the force transmitted to the foundation at resonance F_T. Since at resonance, $\omega = \omega_n$

∴ The transmissibility ratio ε

$$\varepsilon = \frac{\sqrt{1+\left(\frac{2c}{c_c}\right)^2}}{\sqrt{\left(\frac{2c}{c_c}\right)^2}} = \frac{\sqrt{1+\left(\frac{2\times188}{4100}\right)^2}}{\sqrt{\left(\frac{2\times188}{4100}\right)^2}} = 10.92$$

The maximum unbalanced force on the machine due to reciprocat parts at resonance speed ω_n, F

$$F = m_2(\omega_n)^2 r = 2(20.5)^2\left(\frac{0.08}{2}\right) = 33.6N$$

∴ Force transmitted to the foundation at resonance F_T

$$F_T = \varepsilon F = 10.92 \times 33.6 = 367N$$

Amplitude of the forced vibration at resonance $= \frac{\text{force transmitted at res}}{\text{combined stiffness}}$

$$= \frac{367}{42160} = 8.7mm \quad \text{i.e } \frac{F_T}{S_1}$$

POWER TRANSMISSION – BELTS OR ROPES.

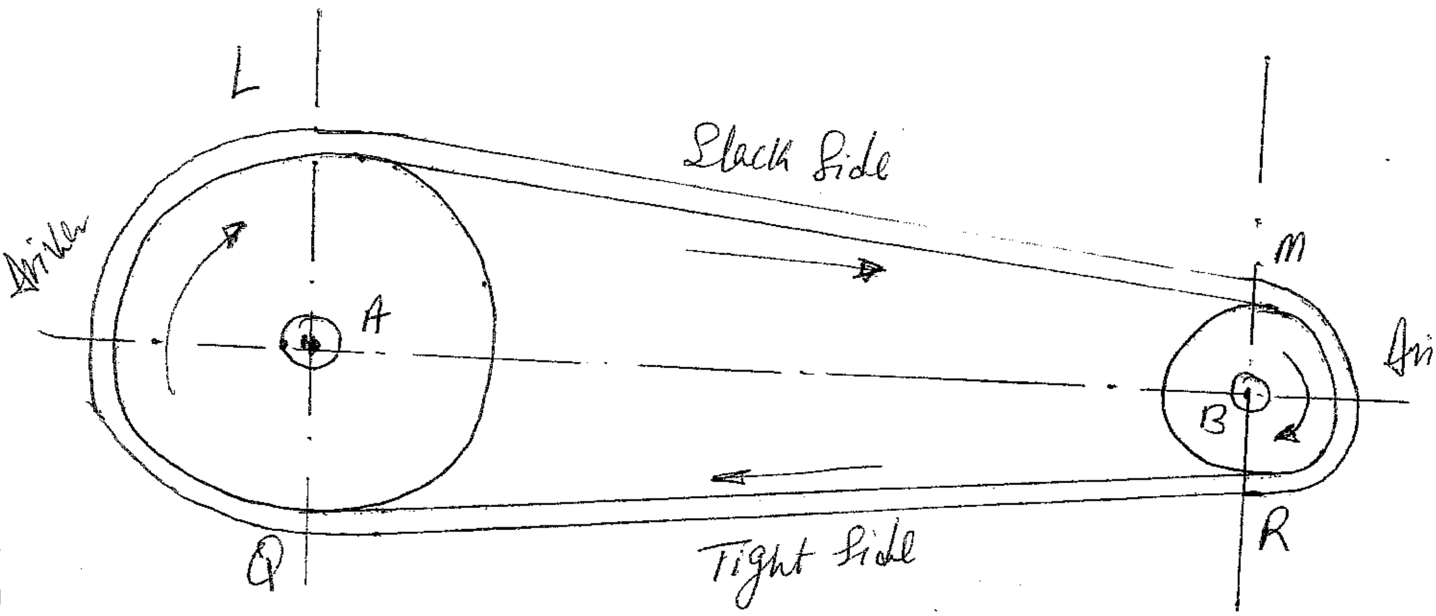
The belt or ropes are used to transmit power from one shaf to another by means of pulleys which rotate at the same spee or different speeds. The amount of power transmitted depends upon the following factors.

i) The velocity of the ~~shaft~~ belt.
ii) The tension under which the belt is placed on the pulleys.
iii) The arc of contact between the belt and the smaller pulley.
iv) The condition under which the belt is used.

Types of belts are flat, vee and circular belts. The material used for belts are leather, cotton or fabrics and rubber belts.

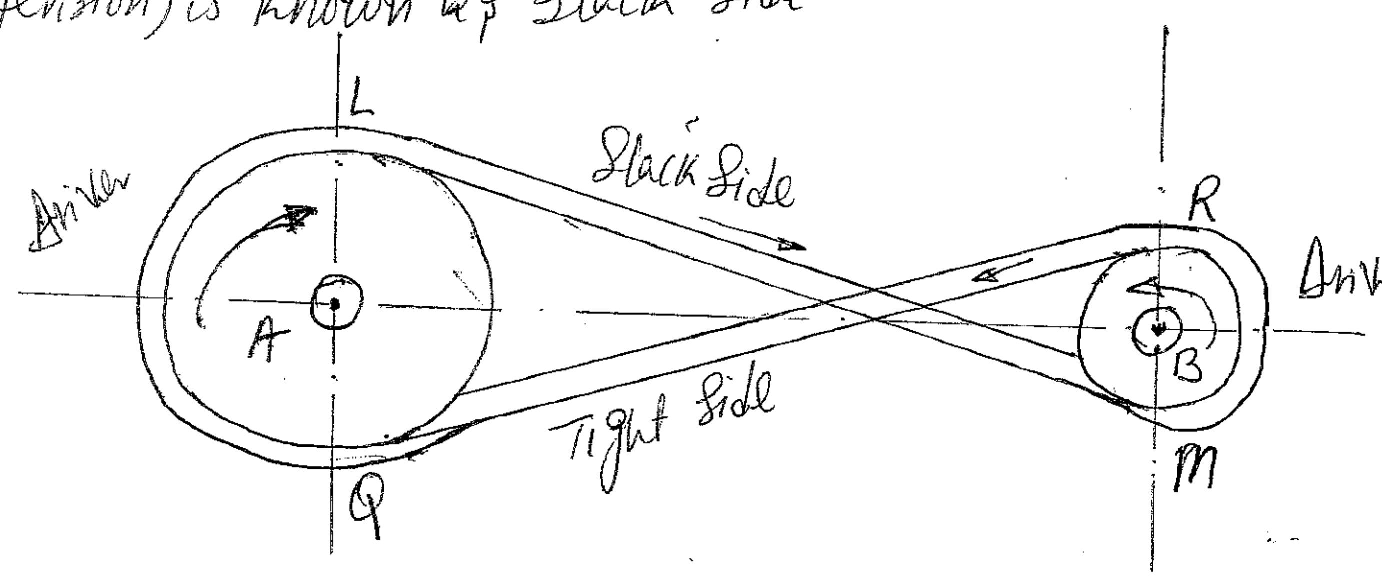
OPEN BELT DRIVE

The open belt drive shown below is used with shafts arranged parallel and rotating in the same direction. In this case the driver "A" pulls the belt from one side i.e side RQ and deliver it to the other side (i.e upper side LM). Thus the tension in the lower side belt will be more than that in the upper side belt. The lower side belt (because of more tension) is known as tight side whereas the upper side belt (because of less tension) is known as slack side.

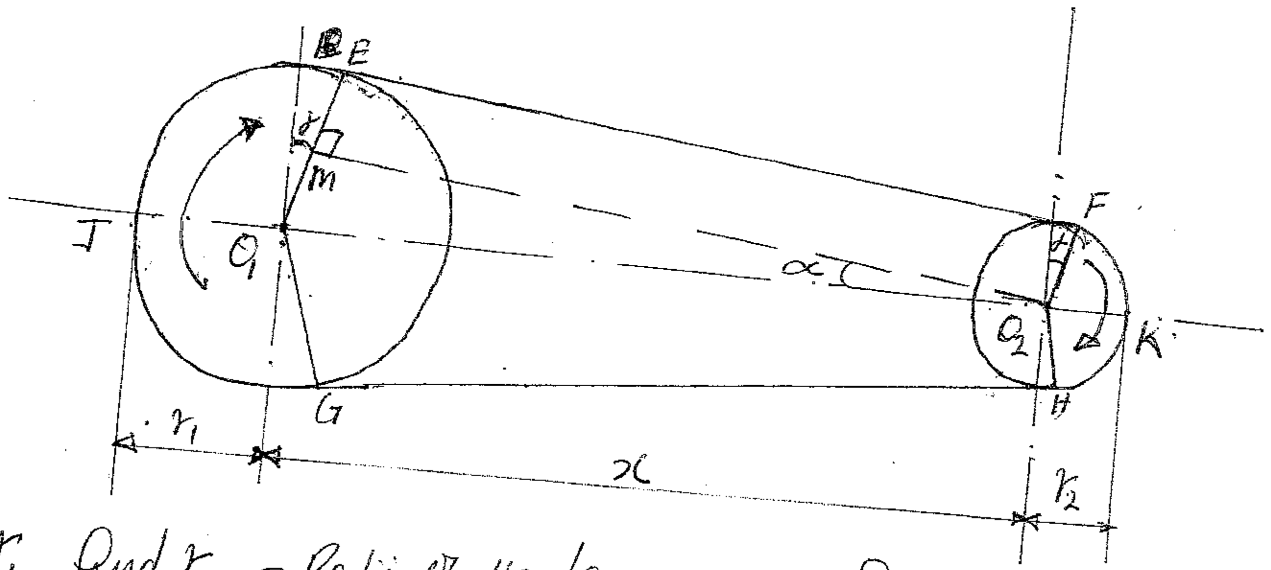


CROSSED OR TWIST BELT DRIVE

The Crossed belt drive shown below is used with shafts arranged parallel and rotating in the opposite directions. In this case, the ~~driv~~ driver pulls the belt from one side (RQ) and deliver it to the other side (LM). Thus, the tension of the ~~shaft~~ belt RQ will be more than that in the belt LM. The belt RQ (because of more tension) is known as tight side, whereas the belt LM (because of less tension) is known as slack side.



LENGTH OF AN OPEN BELT DRIVE (6.



Let r_1 and r_2 = Radii of the larger and smaller pulleys.

x = Distance between the centres of two pulleys ($O_1 O_2$)

l = Total length of the belt

Let the belt leaves the larger pulley at E and G and the smaller pulley at F and H. Through O_2 draw O_2M parallel to FE.

From geometry of figure, we find that O_2M will be perpendicular to O_1E.

Let the angle MO_2O_1 be α

The length of the belt = Arc GJE + EF + Arc FKH + HG

$$= 2\,(\text{Arc JE} + EF + \text{Arc FK}) \quad \text{------- (1)}$$

From the geometry of the figures.

$$\sin\alpha = \frac{O_1M}{O_1O_2} = \frac{O_1E - EM}{O_1O_2} = \frac{r_1 - r_2}{x}$$

Since α is very small

$\therefore$ $\sin\alpha = \alpha$ in radians $= \dfrac{r_1 - r_2}{x}$ ------- (2)

$$\text{Arc JE} = r_1\left(\frac{\pi}{2} + \alpha\right) \quad \text{------- (3)}$$

$$\text{Arc FK} = r_2\left(\frac{\pi}{2} - \alpha\right) \quad \text{------- (4)}$$

$$EF = MO_2 = \sqrt{(O_1O_2)^2 - (O_1M)^2} = \sqrt{x^2 - (r_1 - r_2)^2}$$

$$= x\sqrt{1 - \left(\frac{r_1 - r_2}{x}\right)^2}$$

Expanding this equation by binomial theorem

$$BF = x\left[1-\frac{1}{2}\left(\frac{r_1-r_2}{x}\right)^2+\cdots\cdots\right] = x-\frac{(r_1-r_2)^2}{2x} \quad \text{------} (5)$$

Substitute the values of arc JE from eqn 3, arc FK from eqn (4) and EF from eqn (5) in eqn (1)

$$L = 2\left[\left(r_1\left(\frac{\pi}{2}+\alpha\right)\right)+\left(x-\frac{(r_1-r_2)^2}{2x}\right)+\left(r_2\left(\frac{\pi}{2}-\alpha\right)\right)\right]$$

$$= 2\left[r_1\times\frac{\pi}{2}+r_1\alpha+x-\frac{(r_1-r_2)^2}{2x}+r_2\times\frac{\pi}{2}-r_2\times\alpha\right]$$

$$= 2\left[\frac{\pi}{2}(r_1+r_2)+\alpha(r_1-r_2)+x-\frac{(r_1-r_2)^2}{2x}\right]$$

$$= \pi\left[(r_1+r_2)+2\alpha\left[r_1-r_2\right)+\left(2x-\frac{(r_1-r_2)^2}{x}\right)\right]$$

Substituting the value of $\alpha = \frac{r_1-r_2}{x}$ from eqn (2)

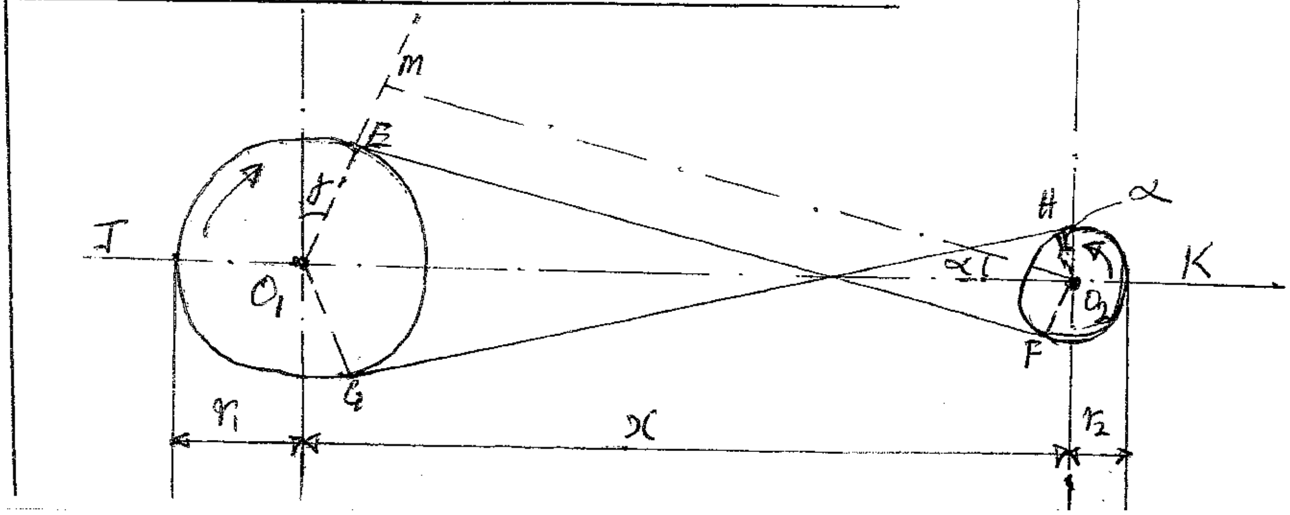
$$L = \left[\pi\left[r_1+r_2\right]+\left(2\frac{(r_1-r_2)}{x}\right)\times(r_1-r_2)+\left(2x-\frac{(r_1-r_2)^2}{x}\right)\right]$$

$$= \pi(r_1+r_2)+\frac{2(r_1-r_2)^2}{x}+2x-\frac{(r_1-r_2)^2}{x}$$

$$= \pi\left[r_1+r_2\right]+2x+\frac{(r_1-r_2)^2}{x} \quad \text{--------} \text{[in terms of pulley radii]}$$

$$= \frac{\pi}{2}(d_1+d_2)+2x+\frac{(d_1-d_2)^2}{4x} \quad \text{------} \text{[in terms of pulley diameter]}$$

LENGTH OF A CROSS BELT DRIVE



Let,

r_1 and r_2 = Radii of the larger and smaller pulleys. (7)

x = Distance between the centres of two pulleys ($O_1 O_2$)

L = Total length of the belt.

Let the belt leaves the larger pulley at E and G and the smaller pulley at F and H. Through O_2 draw O_2M parallel to FE. From the geometry of figure, we find that O_2M will be perpendicular to O_1E. Let the angle $MO_2O_1 = \alpha$ radius.

The length of the belt, L = Arc GJE + EF + Arc FKH + HG.

$$= 2\,[\text{Arc } JE + EF + \text{Arc } FK] \quad \text{------} \quad (1)$$

From the geometry of figure, we find that,

$$\sin\alpha = \frac{O_1M}{O_1O_2} = \frac{O_1E + EM}{O_1O_2} = \frac{r_1 + r_2}{x}$$

Since α is very small, ∴ putting $\sin\alpha = \alpha(\text{rad}) = \frac{r_1+r_2}{2}$ -(2)

$$\text{Arc } JE = r_1\left(\frac{\pi}{2} + \alpha\right) \quad \text{------} \quad (3)$$

Similarly, $\text{Arc } FK = r_2\left(\frac{\pi}{2} + \alpha\right)$ ------ (4)

$$EF = MO_2 = \sqrt{(O_1O_2)^2 - (O_1M)^2} = \sqrt{x^2 - (r_1 + r_2)^2}$$

$$= x\sqrt{1 - \left(\frac{r_1+r_2}{x}\right)^2}$$

Expanding this equation by binomial theorem

$$EF = x\left[1 - \frac{1}{2}\left(\frac{r_1+r_2}{x}\right)^2 + \cdots\cdots\right] = x - \left[\frac{(r_1+r_2)^2}{2x}\right]$$

$$= x - \frac{(r_1 - r_2)^2}{2x} \quad \cancel{= x\sqrt{1 - \frac{(r_1+r_2)^2}{2x}}} \quad \text{---} \quad (5)$$

Substituting the value of arc JE from equation (iii) arc FK from equation (iv) and EF from equation (V) in eqn (i) we get:

$$L = 2\left[r_1\left(\frac{\pi}{2} + \alpha\right) + x - \frac{(r_1+r_2)^2}{2x} + r_2\left(\frac{\pi}{2} + \alpha\right)\right]$$

$$= 2\left[r_1 \times \frac{\pi}{2} + r_1\alpha + x - \frac{(r_1+r_2)^2}{2x} + r_2 \times \frac{\pi}{2} + r_2\alpha\right]$$

$$= 2\left[\frac{\pi}{2}(r_1+r_2) + \alpha(r_1+r_2) + x - \frac{(r_1+r_2)^2}{2x}\right]$$

$$= \pi(r_1+r_2) + 2\alpha(r_1+r_2) + 2x - \frac{(r_1+r_2)^2}{x}$$

Substituting the value of $\alpha = \frac{r_1+r_2}{x}$ from eqn. (ii)

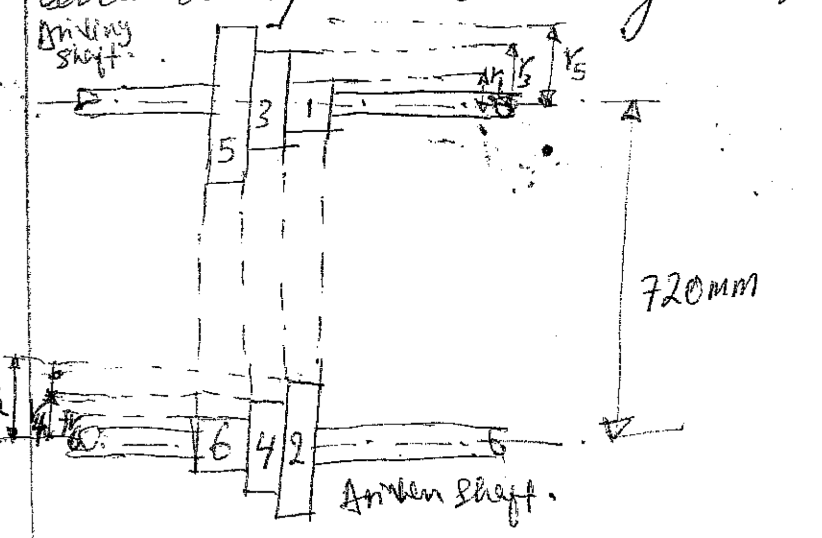
$$L = \pi(r_1+r_2) + \left(\frac{2(r_1+r_2)}{x}\right) \times (r_1+r_2) + \left(2x - \frac{(r_1+r_2)^2}{x}\right)$$

$$= \pi(r_1+r_2) + \frac{2(r_1+r_2)^2}{x} + \left(2x - \frac{(r_1+r_2)^2}{x}\right)$$

$$= \pi(r_1+r_2) + 2x + \frac{(r_1+r_2)^2}{x} \quad \text{(in terms of pulley radii)}$$

$$= \frac{\pi}{2}(d_1+d_2) + 2x + \frac{(d_1+d_2)^2}{4x} \quad \text{(in terms of pulley diameter)}$$

Example I A shaft rotates at a constant speed of 160 rpm is connected by belting to a parallel shaft 720mm apart which has to run at 60, 80 and 100 rpm. The smallest pulley on the driving shaft is 40mm in radius. Determine the remaining radii of the two stepped pulleys for a crossed belt and an open belt. Neglect the belt thickness and slip.



Given:

$N_1 = N_3 = N_5 = 160$ rpm

$x = 720$ mm

$N_2 = 60$ rpm

$N_4 = 80$ rpm

$N_6 = 100$ rpm

$r_1 = 40$ mm.

Let r_2, r_3, r_4, r_5 and r_6 be the radii of the pulleys 2, 3, 4, (7 5, and 6 respectively.

For ~~A~~ Crossed Belt.

For pulleys 1 and 2

$$\frac{N_2}{N_1} = \frac{r_1}{r_2} \quad \therefore\ r_2 = r_1 \times \frac{N_1}{N_2} = 40 \times \frac{160}{60} = 106.7\,mm$$

For pulleys 3 and 4, $\frac{N_4}{N_3} = \frac{r_3}{r_4} \quad \therefore\ r_4 = r_3 \times \frac{N_3}{N_4} = \frac{160}{80} \times r_3 = 2r_3$

For crossed belt drive, $r_1 + r_2 = r_3 + r_4 = r_5 + r_6 = 40 + 106.7 = 146.7\,m$ ...(i)

$\therefore\ r_3 + r_4 = r_3 + 2r_3 = 146.7 \quad \therefore\ r_3 = 146.7 \div 3 = 48.9\,mm$

$r_4 = 2r_3 = 2 \times 48.9 = 97.8\,mm$.

Now for pulleys 5 and 6, $\frac{N_6}{N_5} = \frac{r_5}{r_6}$ or $r_6 = r_5 \times \frac{N_5}{N_6} = r_5 \times \frac{160}{100} = 1.6r_5$

From eqn (i) $r_5 + 1.6r_5 = 146.7 \quad \therefore\ r_5 = 146.7 \div 2.6 = 56.4\,mm$

$r_6 = 1.6r_5 = 1.6 \times 56.4 = 90.2\,mm$.

For ~~an~~ an open belt.

$r_2 = 106.7\,mm$ (as above)

Also pulleys 3 and 4, $r_4 = 2r_3$ (as above)

Length of belt for an open belt drive

$$L = \pi(r_1 + r_2) + \frac{(r_2 - r_1)^2}{x} + 2x$$

$$= \pi(40 + 106.7) + \frac{(106.7 - 40)^2}{720} + 2 \times 720 = 1907\,mm$$

Since the length of belt in an open belt drive is constant $\therefore$ pulleys 3 and 4 length of the belt $L = 1907 = \pi(r_3 + r_4) + \frac{(r_4 - r_3)^2}{x} + 2x$

$$= \pi(r_3 + 2r_3) + \frac{(2r_3 - r_3)^2}{720} + 2 \times 720$$

$$= 9.426r_3 + 0.00014r_3^2 + 1440$$

$\underline{r_3}$ $\quad 0.0014r_3^2 + 9.426r_3 - 467 = 0$

$$r_3 = \frac{-9.426 \pm \sqrt{(9.426)^2 + 4 \times 0.0014 \times 467}}{2 \times 0.0014} = 49.3\,mm$$

$\therefore r_4 = 2r_3 = 2 \times 49.3 = 98.6\,mm$

Now for pulleys 5 and 6

$\frac{N_6}{N_5} = \frac{r_5}{r_6}$ or $r_6 = \frac{N_5}{N_6} \times r_5 = \frac{160}{100} r_5 = 1.6r_5$

length of the belt L, $1907 = \pi(r_5 + r_6) + \frac{(r_6 - r_5)^2}{x} + 2x$

$$= \pi(r_5 + 1.6r_5) + \frac{(1.6r_5 - r_5)^2}{720} + 2 \times 720$$

$$= 8.17r_5 + 0.0005(r_5)^2 + 1440$$

or $0.0005r_5^2 + 8.17r_5 - 467 = 0$

$$\therefore r_5 = \frac{-8.17 \pm \sqrt{(8.17)^2 + 4 \times 0.0005 \times 467}}{2 \times 0.0005} = 60\,mm$$

$r_6 = 1.6r_5 = 1.6 \times 60 = 96\,mm$

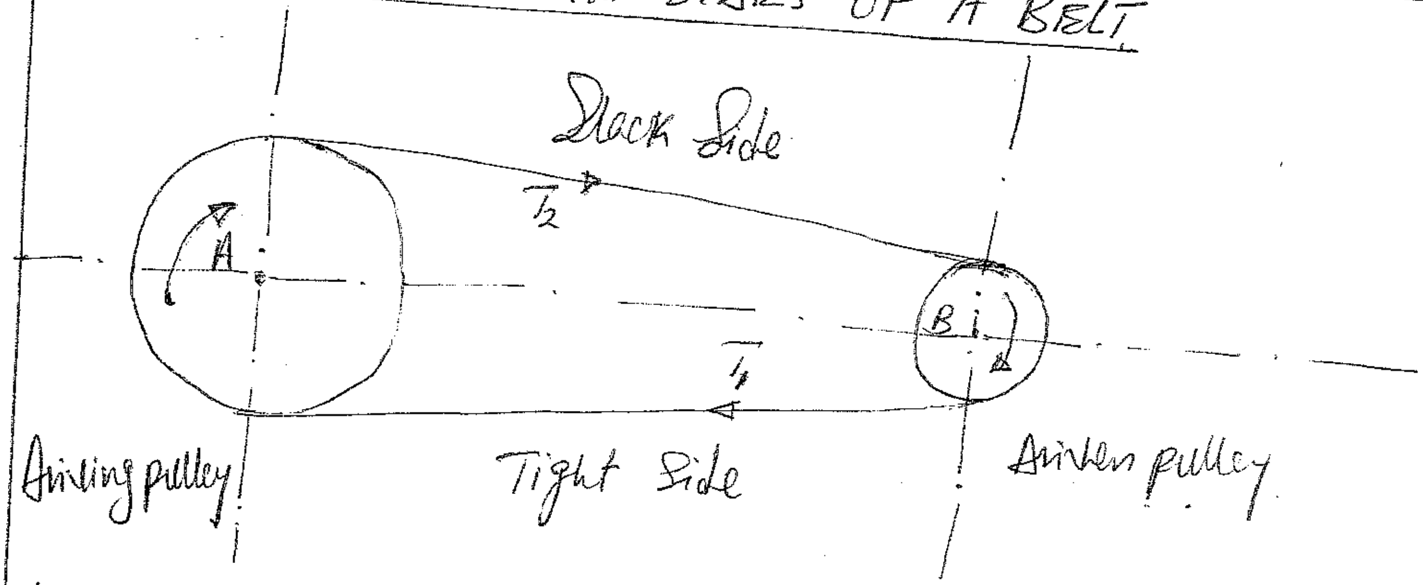
IDLER PULLEY: It is used with shafts arranged parallel and when an open belt drive cannot be used due to small angle of contact on the smaller pulley. This type of drive is provided to obtain high velocity ratio and when the required belt tension cannot be obtained by other means.

STEPPED PULLEY: It is used for changing the speed of the driven shaft while the main or driving shaft runs at constant speed. This is accomplished by shifting the belt from one part of the steps to the other.

COUNTERSHAFT PULLEY: It is used as an intermediate shaft in a line of shafting placed between a driving and a driven shaft

... a greater Speed-ratio or where direct (7)
Connection is difficult.

EXPRESSION RELATING INITIAL TENSION TO TENSIONS IN THE SLACK AND TIGHT SIDES OF A BELT



Let
T_1 and T_2 = Tension in the tight and Slack Sides of the belt respec... in Newtons.
r_1 and r_2 = Radii of the driver and follower respectively.
V = Velocity of the belt in m/s.

The effective turning (driving) force at the Circumference of the follow... is the difference between the two tensions $(T_1 - T_2)$

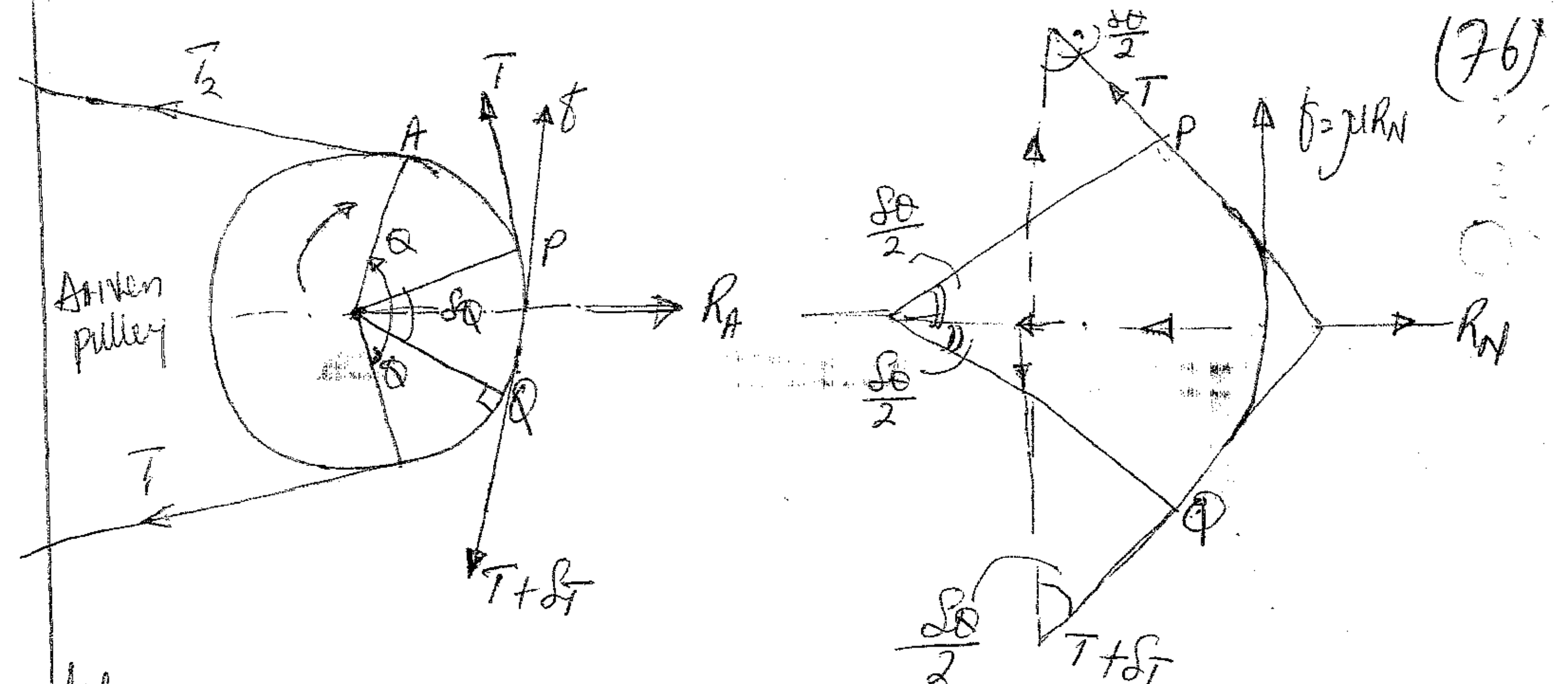
$\therefore$ Workdone per Second $= (T_1 - T_2)V$ N-m/s

power transmitted $P = (T_1 - T_2)V$ watts.

Torque exerted $= (T_1 - T_2)r_1$ and $(T_1 - T_2)r_2$.

RATIO OF DRIVING TENSIONS FOR FLAT BELT DRIVE

Consider a driven pulley rotating in the clockwise direction as shown below.



Let,
θ - Angle of Contact in radians
Now Consider small portion of the belt PQ, subtending an angle $\delta\theta$ at the Centre of pulley. The belt PQ is in equilibrium under the following forces.
i, Tension T in the belt P
ii, Tension $(T+\delta T)$ in the belt at Q
(iii) Normal reaction R_N
(IV) Frictional force $f = \mu R_N$ where, μ is the Co-efficient of friction between the belt and pulley.
Resolving all the forces horizontally and equating the same.

$$R_N = (T+\delta T)\sin\frac{\delta\theta}{2} + T\sin\frac{\delta\theta}{2} \quad \text{------------ (1)}$$

Since angle $\delta\theta$ is very small, $\therefore$ putting $\sin\frac{\delta\theta}{2} = \frac{\delta\theta}{2}$

$$R_N = (T+\delta T)\frac{\delta\theta}{2} + T\times\frac{\delta\theta}{2} = \frac{T\delta\theta}{2} + \frac{\delta T\delta\theta}{2} * \frac{T\delta\theta}{2} = T\delta\theta \quad \text{------ (2)}$$

(Neglecting $\frac{\delta T\delta\theta}{2}$)

Now resolving the forces vertically, we have

$$\mu R_N = (T+\delta T)\cos\frac{\delta\theta}{2} - T\cos\frac{\delta\theta}{2} \quad \text{------------ (3)}$$

Since $\delta\theta$ is very small, $\therefore$ putting $\cos\frac{\delta\theta}{2} = 1$

$\therefore \mu R_N = T + \delta T - T = \delta T$ or $R_N = \frac{\delta T}{\mu}$ ------ (4) (7)

Equating eqn (1) and (4)

$T\delta\theta = \frac{\delta T}{\mu}$ or $\frac{\delta T}{T} = \mu\delta\theta$

Integrating both sides between limits T_2 and T_1 and from 0 to θ respectively.

$$\int_{T_2}^{T_1} \frac{\delta T}{T} = \mu \int_0^{\theta} \delta\theta \text{ or } \log_e\left(\frac{T_1}{T_2}\right) = \mu\theta \text{ or } \frac{T_1}{T_2} = e^{\mu\theta} \text{ ---- (5)}$$

Equation (5), can be expressed in terms of corresponding logarithm base 10 i.e $2.3 \log\left(\frac{T_1}{T_2}\right) = \mu\theta$.

Example 1 Find the power transmitted by a belt running over a pulley of 600mm diameter at 200rpm. The co-efficient of friction between the belt and the pulley is 0.25. Angle of lap 160° and maximum tension in the belt is 2500N.

Given: $d = 600mm = 0.6m$, $N = 200rpm$, $\mu = 0.25$, $\theta = 160 = 160 \times \frac{\pi}{180}$

$T_1 = 2500N$. $= 2.793 rad.$

Velocity of the belt, $V = \frac{\pi d N}{60} = \frac{\pi \times 0.6 \times 200}{60} = 6.284 m/s.$

T_2 = Tension in the slack side of the belt

$2.3 \log\left(\frac{T_1}{T_2}\right) = \mu\theta = 0.25 \times 2.793 = 0.6982.$

$\log\left(\frac{T_1}{T_2}\right) = \frac{0.6982}{2.3} = 0.3036$

$\frac{T_1}{T_2} = 2.01$, $T_2 = \frac{T_1}{2.01} = \frac{2500}{2.01} = 1244N$

Power transmitted by the belt, P $P = (T_1 - T_2)V = (2500 - 1244)6.284$

$= 7890W = 7.89KW$

Example II Two pulleys, one 450mm diameter and the other 200mm diameter are on parallel shaft 1.95m apart and are connected by a crossed belt. Find the length of the belt required and the angle of contact between the belt and each pulley. What power can be transmitted by the belt when the larger pulley rotates at 200 rev/min, if the maximum permissible tension in the belt is 1KN and the co-efficient of friction between the belt and pulley is 0.25

Given: $d_1 = 450mm = 0.45m$ or $r_1 = 0.225m$, $d_2 = 200mm = 0.2m$ or $r_2 = 0.1m$, $x = 1.95m$, $N_1 = 200rpm$, $T_1 = 1KN = 1000N$

$\mu = 0.25$

$$V = \frac{\pi d_1 N_1}{60} = \frac{\pi \times 0.45 \times 200}{60} = 4.714 \, m/s.$$

Length of the crossed belt $L = \pi(r_1 + r_2) + 2x + \frac{(r_1+r_2)^2}{x}$

$$= \pi(0.225 + 0.1) + (2 \times 1.95) + \frac{(0.225+0.1)^2}{1.95} = 4.975m$$

Angle of contact between the belt and each pulley θ for a crossed belt drive.

$$\sin\alpha = \frac{r_1 + r_2}{x} = \frac{0.225 + 0.1}{1.95} = 0.1667, \quad \alpha = 9.6^\circ$$

$$\therefore \theta = 180 + 2\alpha = 180 + (2 \times 9.6^\circ) = 199.2^\circ$$

$$= 199.2 \times \frac{\pi}{180} = 3.477 \, rad$$

Power transmitted P

Let T_2 = Tension in the slack side of the belt.

$$2.3 \log\left(\frac{T_1}{T_2}\right) = \mu\theta = 0.25 \times 3.477 = 0.8692$$

$$\log\left(\frac{T_1}{T_2}\right) = \frac{0.8692}{2.3} = 0.378 \text{ or } \frac{T_1}{T_2} = 2.387 \quad \text{(Taking antilog of 0.378)}$$

$$T_2 = \frac{T_1}{2.387} = \frac{1000}{2.387} = 419N; \quad \therefore P = (T_1 - T_2)V$$

$$= (1000 - 419)4.714$$

$$= 2740W$$

$$= 2.74KW$$

MAXIMUM TENSION IN THE BELT

A little consideration will show that the maximum tension in the belt "T" is equal to the total tension in the tight side of the belt T_{t_1}, let

σ = Maximum safe stress in N/mm^2

b = Width of the belt in mm

t = Thickness of the belt in mm

T = Maximum tension in the belt

= Maximum stress × cross-sectional area of the belt

$= \sigma \times b \times t$.

When centrifugal tension is neglected, then

T or $T_{t_1} = T_1$ i.e. tension in the tight side of the belt.

When centrifugal tension is considered

T or $T_{t_1} = T_1 + T_c$, $\quad T = T_1 + T_c$

~~Cond~~ CONDITION FOR THE TRANSMISSION OF MAXIMUM POWER

Note that, $P = (T_1 - T_2)V$ ------------ (1)

Also $\frac{T_1}{T_2} = e^{\mu\theta}$ or $T_2 = \frac{T_1}{e^{\mu\theta}}$ ------------ (2)

Substituting the value of T_2 in eqn (1)

$$P = \left(T_1 - \frac{T_1}{e^{\mu\theta}}\right)V = T_1\left(1 - \frac{1}{e^{\mu\theta}}\right)V = T_1 V C \text{ ------ (3)}$$

Where, $C = 1 - \frac{1}{e^{\mu\theta}}$

Also known that, $T_1 = T - T_c$

Substituting the value of T_1 in eqn (3)

$P = (T - T_c)V \cdot C = (T - mV^2)VC = (TV - mV^3)C$

(i.e, $T_c = mV^2$)

For Maximum power, differentiate the above expression with respect to V and equate to zero.

$$\frac{\delta P}{\delta V} = 0, \quad \frac{\delta}{\delta V}(TV - mV^3)C = 0$$

$$\therefore T - 3mV^2 = 0, \quad T - 3T_c = 0, \quad T = 3T_c$$

It shows that, when the power transmitted is maximum, $\frac{1}{3}$rd of the maximum tension is absorbed as centrifugal tension.

Note, $T_1 = T - T_c$ (For maximum power, $T_c = \frac{T}{3}$)

$$T_1 = T - \frac{T}{3} = \frac{2T}{3}$$

Also

$$T - 3mV^2 = 0$$

$$T = 3mV^2$$

$$V = \sqrt{\frac{T}{3m}}$$

Example III A shaft rotating at 200 rpm drives another shaft at 300 rpm and transmits 6 kW through a belt. The pulley is 100 mm wide and 10 mm thick. The distance between the shafts is 4 m. The smaller pulley is 0.5 m in diameter. Calculate the stress in the belt, if it is an open belt drive and a cross belt drive. Take $\mu = 0.3$.

Given: $N_1 = 200$ rpm, $N_2 = 300$ rpm, $P = 6$ kW $= 6 \times 10^3$ W

$b = 100$ mm, $t = 10$ mm, $x = 4$ m, $d_2 = 0.5$ m, $\mu = 0.3$

To find the stress in the belt for an open belt drive. Find the diameter of the larger pulley d_1

$\frac{N_2}{N_1} = \frac{d_1}{d_2}$ $\therefore$ $d_1 = \frac{N_2 d_2}{N_1} = \frac{300 \times 0.5}{200} = 0.75 m$

Also $V = \frac{\pi d_2 N_2}{60} = \frac{\pi \times 0.5 \times 300}{60} = 7.855 m/s$

Find the angle of contact on the smaller pulley for open belt drive

$\sin\alpha = \frac{r_1 - r_2}{x} = \frac{d_1 - d_2}{2x} = \frac{0.75 - 0.5}{2 \times 4} = 0.03125$

$\alpha = 1.8^\circ$

$\therefore$ Angle of contact, $\theta = 180^\circ - 2\alpha = 180 - (2 \times 1.8) = 176.4^\circ$

$= 176.4^\circ \times \pi/180 = 3.08$ rad.

Known that, $2.3 \log\left(\frac{T_1}{T_2}\right) = \mu\theta = 0.3 \times 3.08 = 0.924$

$\log\left(\frac{T_1}{T_2}\right) = \frac{0.924}{2.3} = 0.4017$ (Taking antilog of 0.4017)

$\frac{T_1}{T_2} = 2.52$

$P = (T_1 - T_2) V = 6 \times 10^3 = (T_1 - T_2) 7.855$

$T_1 - T_2 = \frac{6 \times 10^3}{7.855} = 764 N$

From above, $2.52 T_2 - T_2 = 764 N$

$\therefore T_2 = 503 N$ and $T_1 = 1267 N$

Maximum Tension in belt T_1 and the stress can be obtain from.

$T_1 = \sigma b t$, $1267 = \sigma \times 100 \times 10 = 1000\sigma$

$\sigma = \frac{1267}{1000} = 1.267 N/mm^2 = 1.267 N/mm^2$ or Pa.

Stress in the belt for a cross belt drive.

$\sin\alpha = \frac{r_1 + r_2}{x} = \frac{d_1 + d_2}{2x} = \frac{0.75 + 0.5}{2 \times 4} = 0.1562$

$\alpha = 9^\circ$ (from $\sin^{-1}$ ~~of~~ 0.1562)

Angle of Contact, $\theta = 180 + 2\alpha = 180 + (2\times 9) = 198^\circ$

$= 198 \times \pi/180 = 3.456\,\text{rad}.$

$$2.3 \log\left(\frac{T_1}{T_2}\right) = \mu\theta = 0.3 \times 3.456 = 1.0368$$

$$\log\left(\frac{T_1}{T_2}\right) = \frac{1.0368}{2.3} = 0.4508, \quad \therefore \frac{T_1}{T_2} = 2.82$$

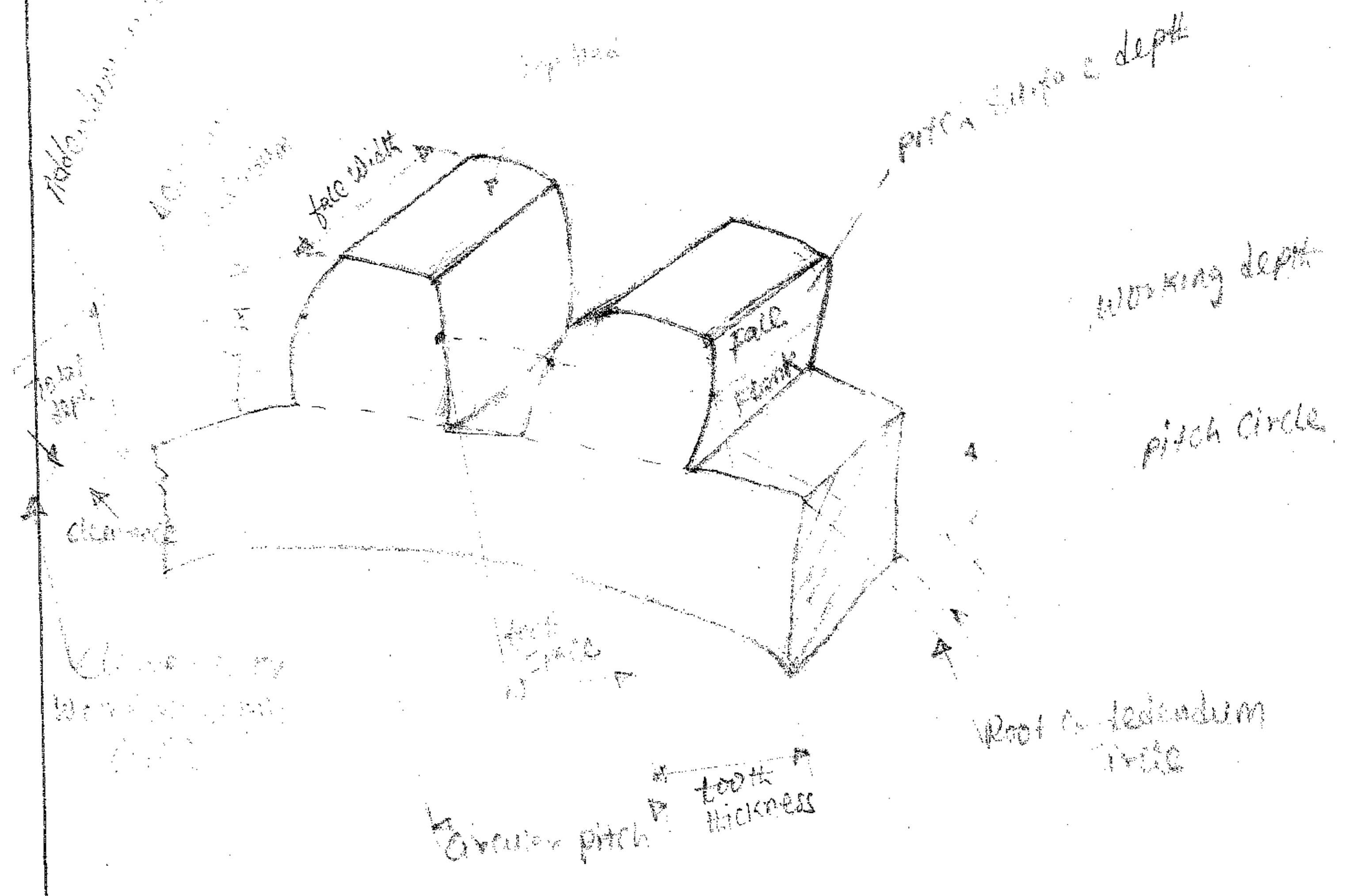
From above, $T_1 - T_2 = 764$, $\therefore T_1 = 1184N$ and $T_2 = 420N$

Maximum tension in the belt T_1 and the stress can be obtain by

$T_1 = \sigma bt$, $1184 = \sigma \times 100 \times 10 = 1000\sigma$

$\sigma = \dfrac{1184}{1000} = 1.184\,N/mm^2$ or $1.184\,mpa$ — ?

TOOTHED GEARING



A gear drive provide a definite velocity ratio and
when the distance between the driver and the follower is sma

TERMS USED IN GEARS.

(1) PITCH CIRCLE: It is an imaginary or theoretical circle which by p rolling action would give the same motion as the actual gear.

(2) PITCH CIRCLE DIAMETER: It is the diameter of the pitch circle. The size of the gear is usually specified by the pitch circle diameter.

3. PRESSURE ANGLE OR ANGLE OF OBLIQUITY: It is the angle between the common normal to two gear teeth at the point of contact an the common tangent at the pitch point. It is ~~used~~ usually denote by ϕ. The standard pressure angles are $14\frac{1}{2}°$ and $20°$.

4. ADDENDUM: It is the radial distance of a tooth from the pitch circle to the top of the tooth.

(5) DEDENDUM: It is the radial distance of a tooth from the pitch circle t the bottom of the tooth

(6) CIRCULAR PITCH: It is the distance measured on the circumference of the pitch circle from a point of one tooth to the corresponding point on the next tooth. It is usually denoted by $P_c = \pi D/T$

(7) DIAMETRAL PITCH: It is the ratio of number of teeth to the pitch circle diameter in millimetres. It is denoted by $P_d = T/D = \pi/P_c$

(8) MODULE: It is the ratio of the pitch circle diameter in millimeters to the number of teeth. It is usually denoted by $m = D/T$

(9) CLEARANCE: It is the radial distance from the top of the tooth to the bottom of the tooth in a meshing gear.

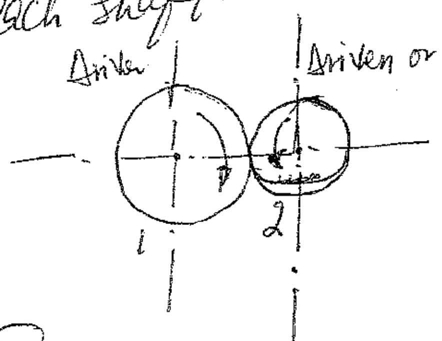
ADVANTAGES OF GEAR DRIVE

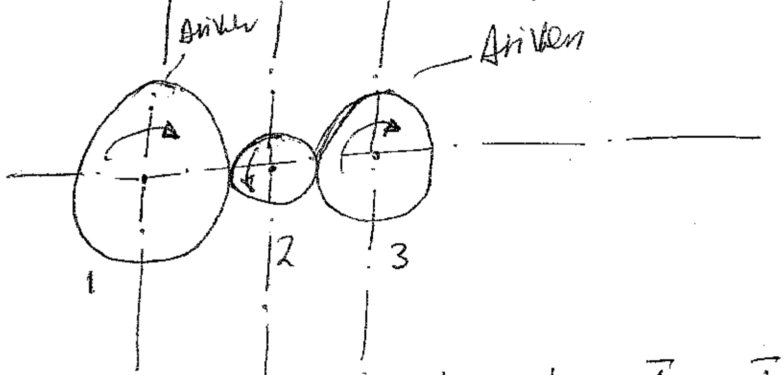
i) It transmits exact velocity ratio.
ii) It may be used to transmit large power
iii) It has high efficiency.
iv) It has reliable service
v) It has compact layout.

DISADVANTAGES OF GEAR DRIVE

i) The manufacture of gears require special tools and equipment.
ii) The error in cutting teeth may cause vibrations and noise during operation.

SIMPLE GEAR TRAIN :- This is when there is only one gear on each shaft



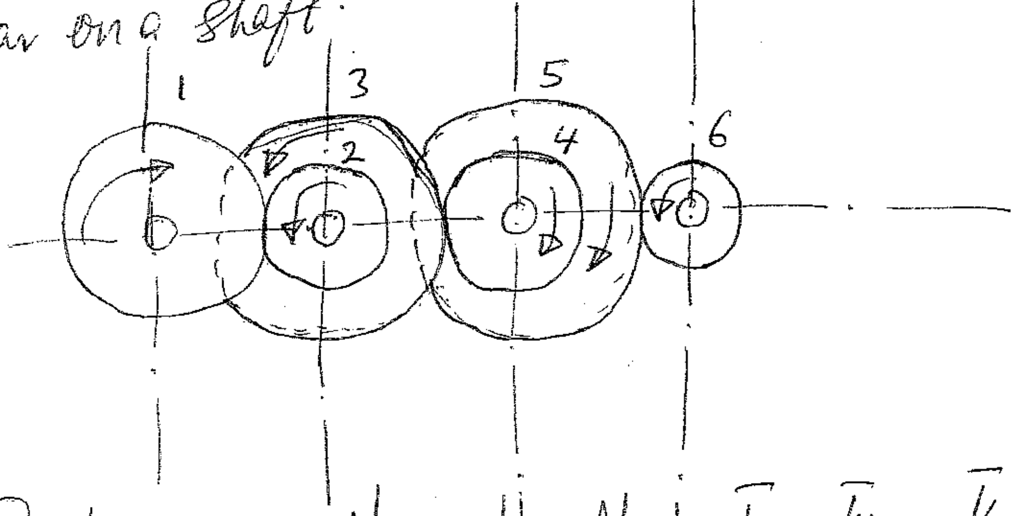


Speed ratio $= \frac{N_1}{N_2} = \frac{T_2}{T_1}$

Speed ratio $= \frac{N_1}{N_2} \times \frac{N_2}{N_3} = \frac{T_2}{T_1} \times \frac{T_3}{T_2}$

$= \frac{N_1}{N_3} = \frac{T_3}{T_1}$

COMPOUND GEAR TRAIN :- This is when there are more than one gear on a shaft.



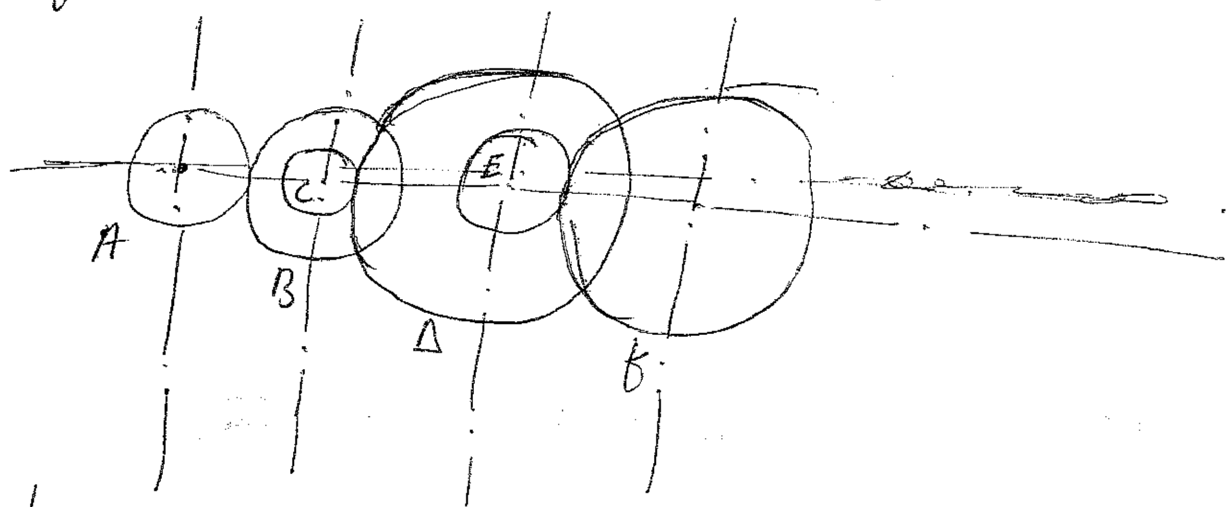
Speed ratio $= \frac{N_1}{N_2} \times \frac{N_3}{N_4} \times \frac{N_5}{N_6} = \frac{T_2}{T_1} \times \frac{T_4}{T_3} \times \frac{T_6}{T_5}$

$\frac{N_1}{N_6} = \frac{T_2 \times T_4 \times T_6}{T_1 \times T_3 \times T_5}$, $\frac{\text{Speed of the first driver}}{\text{Speed of the last driven or follower}} =$

$= \frac{\text{product of the number of teeth on the drivens}}{\text{product of the number of teeth on the drivers}}$

Example A gearing of a machine tool is shown below. The shaft is connected to gear A and rotates at 975 rpm. The gear wheels B, C, D and E are fixed to parallel shaft rotates together. The final gear F is fixed on the output shaft. What is the speed of gear F? The number of teeth on each gear are given below.

Gear	A	B	C	D	E	F
No. of teeth	20	50	25	75	26	65



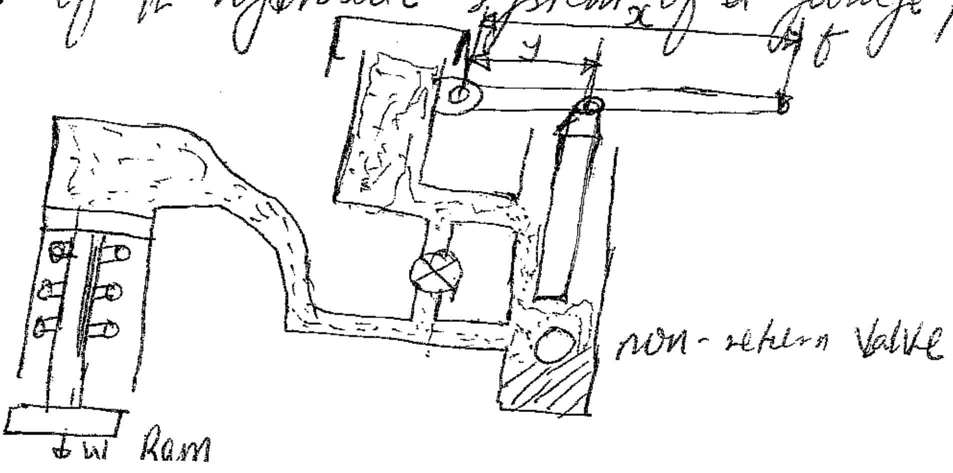
$$\frac{N_A}{N_F} = \frac{T_B \times T_D \times T_F}{T_A \times T_C \times T_E} = \frac{50 \times 75 \times 65}{20 \times 25 \times 26} = 18.75$$

$$\frac{N_A}{N_F} = 18.75, \quad N_F = \frac{975}{18.75} = 52 \text{ rpm}$$

POWER TRANSMISSION BY HYDRAULIC SYSTEMS

This form of power is often used for operating units of the motor vehicle and machines in the repair workshop. Examples are the operation of brakes, clutches, steering, ignition timing, doors, load tipping gear, lifting jack, and power tools.

The layout of the hydraulic system of a garage press is shown below.



The plunger which exerts the load or working force is known as the ram, to distinguish it from the pumping plunger.

$$M.A = \frac{\text{Load}}{\text{Effort}}, \quad V.R = \frac{\text{Distance moved by effort}}{\text{Distance moved by ram}}$$

When the fluid is used, the displacement of the fluid by the plunger equals the displacement of fluid by the ram.

∴ Quantity of fluid = Cross-Sectional Area of plunger × length of Stroke

= Cross-Sectional Area of ram × Distance moved.

$$\text{Hydrolic pressure} = \frac{\text{force applied to end of lever} \times \text{leverage}}{\text{Cross-Sectional Area of plunger}}$$

$$= \frac{\text{Force } \cancel{P} \times \frac{x}{y}}{\text{Area of plunger}}$$

The load or force exerted by the ram.

= Hydraulic pressure × ~~cs~~ Cross Sectional Area of ram.

Example I

A hydraulic garage press is operated by a force of 265 N applied perpendicular to the end of a lever, the lever ratio is 15 : 1. The pumping plunger is 25 mm diameter and the stroke is 50 mm. ~~If~~ the ram is 200 mm diameter. Neglecting friction, find

(a) The load on press

(b) The number of pumping strokes required to move the ram

80mm:

a) Hydraulic pressure $= \dfrac{\text{Force on lever} \times \text{leverage}}{\text{Cross Sectional Area of plunger}}$

$$= \frac{265 \times 15}{\pi/4 \times (0.025)^2} = \frac{265 \times 15}{0.7854 \times (0.025)^2}$$

Load on press = Hydraulic pressure × Ram Area.

$$= \frac{265 \times 15}{0.7854\,(0.025)^2} \times 0.7854 \times 0.2 \times 0.2$$

$$= 254.4\,\text{kN}$$

b) Number of pumping strokes required.

$$= \frac{\text{Volume of fluid pumped into ram Cylinder}}{\text{Volume of fluid pumped per Stroke of Plunger}}$$

$$= \frac{\frac{\pi}{4} D^2 \times \text{distance moved}}{\frac{\pi d^2}{4} \times \text{length of Stroke}} = \frac{\cancel{\frac{\pi}{4}} D^2 \times \cancel{80}}{\cancel{\frac{\pi}{4}} d^2 \times \cancel{80}}$$

$$= \frac{D^2}{d^2} = \frac{0.2^2}{0.025^2} = 64 \text{ Strokes of pumps.}$$

Example II A force of 310N is applied perpendicular to the foot brake pedal of a Car fitted with hydraulic brakes. The pedal leverage is 6:1 and the diameter of the Master Cylinder is 25mm. The Wheel Cylinder (4 double acting) are 32mm diameter and each Wheel Cylinder Plunger moves 0.8mm when the brakes are applied. Allowing 25 mm of free pedal movement, find, (a) distance moved by the pedal when the brakes are applied (b) force exerted on each brake shoe.

(a) To find the pedal movement, the stroke of the master cylinder must be known. This is found from the volume of fluid entering the wheel cylinders when the brakes are applied. Volume of fluid entering wheel cylinders.

= Area × Stroke × number of cylinders

$= \frac{\pi d^2}{4} \times$ number of cylinders

$= \cancel{\frac{\pi d^2}{4} \times 0.8}\ \frac{\pi}{4} \times 32^2 \times 0.8 \times 4 \times 2$

The volume is equal to the volume displaced by the master cylinder which $= \pi/4 \times 25^2 \times$ Stroke

$\therefore \frac{\pi}{4} \times 25^2 \times \text{Stroke} = \pi/4 \times 32^2 \times 0.8 \times 4 \times 2$

Stroke = 10.49 mm.

pedal movement = (Stroke of piston × pedal leverage) + free movement

$= (10.49 \times 6) + 25 = 87.94\,mm$

(b) To find the force exerted on each brake shoe, the hydraulic pressure must be known, then the force exerted on each brake shoe can be found.

$$\text{Hydraulic pressure} = \frac{\text{Applied force} \times \text{leverage}}{\text{Area of master cylinder piston}}$$

$$= \frac{310 \times 6}{\pi/4 \times (0.025)^2} = 3789160\ N/m^2$$

$$= 3789\ kN/m^2$$

$$\text{Force on brake shoe plunger} = \text{Hydraulic pressure} \times \text{Area of wheel cylinder.}$$

$$= 3789 \times \pi/4 \times (0.032)^2$$

$$= 3.047\ kN$$

POWER TRANSMISSION BY COUPLINGS

Flexible coupling must couple two pieces of rotating equipment, equipment with shaft, flanges or both. It must transmit power efficiently. Coupling are divided into two major classes: The rigid and flexible coupling.

Four basic categories of flexible industrial couplings are

1. Mechanical flexible couplings
2. Elastomeric couplings
3. Metallic membrane couplings
4. Miscellaneous couplings.

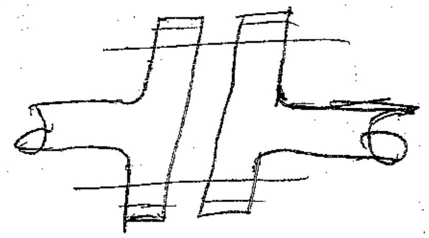
1) MECHANICAL FLEXIBLE COUPLINGS: These couplings obtain their flexibility from loose-fitting parts and or rolling or sliding of mating parts. They usually require lubrication unless one moving part is made of a material that supplies its own lubrication.

needs e.g Nylon gear coupling. E.g Axle, Universal Joints for Vehicles, Chain coupling e.t.c.

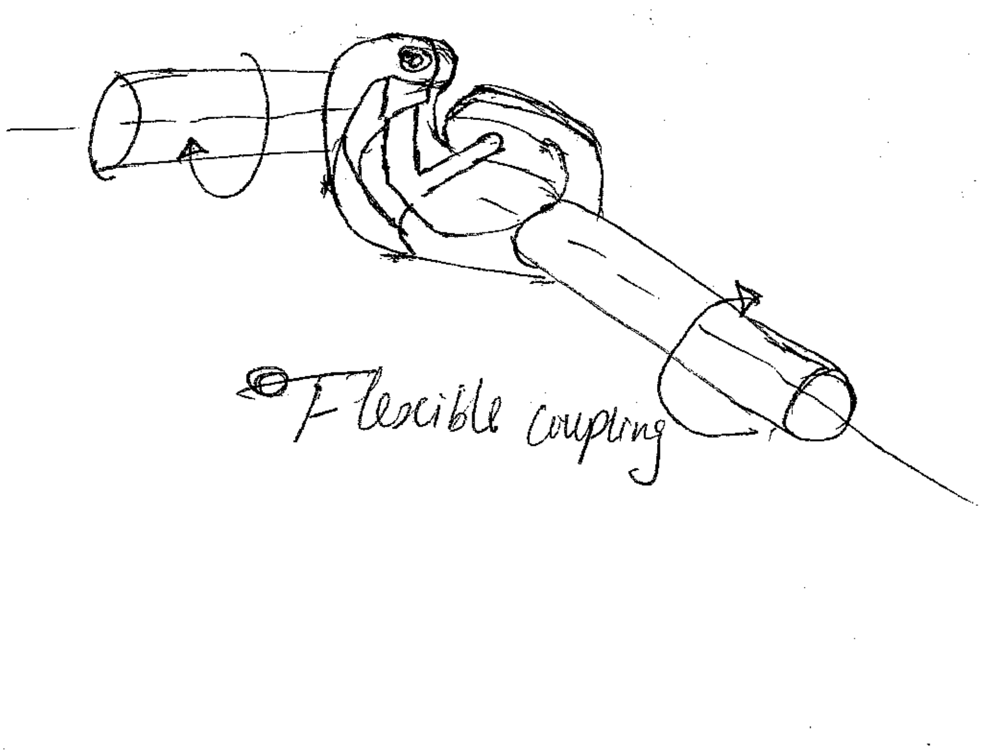
2, ELASTOMETRIC COUPLINGS:- These coupling obtained their flexibility from stretching or compressing a resilient material (rubber, plastic e.t.c). Some sliding or rolling may take place, but it is usually minimal. e.g Rubber tyre coupling.

3, METALLIC MEMBRANE COUPLINGS:- The flexibility of this couplings is obtained from the flexing of thin metallic discs or diaphragms. e.g Disc coupling, tapered contour diaphram.

4, MISCELLANEOUS COUPLING:- These couplings obtain their flexibility from a combination of the mechanisms described above through a unique mechanism. e.g pin and bushing coupling, spring coupling e.t.c.



Rigid flange connection.



Flexible coupling